Industry Contribution to U.S. Wage Inequality

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Abstract

Industry dimension is increasingly dominant to investigate the upward trend of inequality. This paper examines the key drivers of U.S. wage inequality, emphasising the role of heterogeneous capital-labour substitution elasticities across industries in shaping wage dispersion. Key is the distinction of a *quantity effect* (changes in the composition of capital and labour inputs) and a *structural effect* (reflecting technological transformations in inputs substitutability) from Skill-Biased Technological Change (SBTC). Findings suggest that industry-level transformations on the labour side – differentials in job tasks substitutability and workforce composition – constitute the principal drivers of real wage inequality, overshadowing the contribution of capital-side adjustments. A structural estimation of the model reveals that trend-asymmetries in sectoral elasticities of substitution between ICT capital, routine and non-routine workers account for 94% of observed wage variance, while stronger sorting and segregation effects further exacerbate such dispersion. Upon neutralising structural differences between industries, quantity adjustments reckon merely 6-15% of the observed wage inequality.

JEL: E24, J31, J82, L16

KEYWORDS: wage inequality, structural transformations, industry, tasks, labour force composition, elasticity of substitution

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Introduction

Wage inequality has been on the rise in United States over the last decades, and understanding the cause behind such turn is a secular challenge in economics. A large body of research has identified key drivers behind rising wage inequality. One prominent explanation focuses on employment polarization, largely attributed to new technologies and job tasks (*e.g.*, Autor, Katz, et al. 2006, Acemoglu and Autor 2011).[1] Complementing this view, recent evidence shows that more than 60% of the growth in U.S. wage dispersion over the past three decades reflects systematic differences across industries (*e.g.*, Haltiwanger et al. 2024). Thus, a thorough understanding of wage inequality requires considering both labour market characteristics and the broader organization of economic activity across sectors.

Addressing the issue, this paper argues that what matters for inequality is not the amount of capital or labour, but the structural transformation of substitution elasticities: some industries increasingly substitute routine for non-routine workers while complementing ICT capital, generating major wage dispersion. In fact, inequality in wages can be influenced along two dimensions. If elasticities of substitution among factors of production are uniform across industries, inequality arises primarily from differences in capital and labour accumulation, in consistency with the Skill Biased Technological Change (SBTC) framework.[2] If industries differ in their substitution elasticities, even similar production inputs growth lead to diverging wage outcomes: inequality is driven less by the *quantity* of inputs and more by the *structure of technology* and its interaction with factors accumulation.

Given these overlapping explanations, the analysis aims to quantify the relative importance of each factor contributing to wage inequality. While the "quantity" channel has been examined at length, largely unexplored remains the "structural" realm, whose effects are driven by *structural transformations* – hereafter interpreted as sectoral heterogeneous trends in the capital-labour substitution elasticities as in Alvarez-Cuadrado et al. (2018). Key result is that inequality emerges from industry-level technological heterogeneity rather than from factor accumulation alone.

The initial empirical analysis shows that technological change has uneven effects on the composition of the U.S. labour force across industries, and thus on real wages. As ICT technologies have become pervasive, changes in capital deepening are accompanied by systematic shifts in job tasks: the rising share of non-routine workers reflects the replacement of routine workers rather than an expansion in sectoral employment size. Wage differentials move with this reorganization of production, as industries contributing more to wage inequality adopt ICT capital more intensively

[1] Providing a sectoral perspective, Cerina et al. (2021) assert that the downward polarization of employment is characterized by the prevalence of routine tasks but it is fundamentally shaped by the type of sector (services and non-services sectors) rather than the performed job task.

[2] SBTC posits that changes in technological endowments disproportionately benefits high-skilled workers while displacing lower-skilled ones, contributing to wage inequality. Refer, for example, to Tinbergen (1974), Katz and Murphy (1992), Card and DiNardo (2002), and Acemoglu and Autor (2011).

and shift their task composition toward non-routine workers. Inequality thus arises from how industries deploy technology and reallocate tasks, linking capital structure directly to labour market outcomes.

To elucidate how differences in the industrial composition of capital and labour types affect wage dispersion, I build a general equilibrium model of industries differing in degrees of substitution among inputs of production. The market structure equips the economy with a block of industries populated by monopolistically competitive firms employing two types of both capital and labour: besides physical capital, non-routine workers are complementary to ICT capital while routine workers are substitutes, in the spirit of Krusell et al. (2000). As for heterogeneity, an important phenomena of the labour market is that workers are themselves of varying capacities depending on where they are employed in; accordingly, the structure of the labour market is designed around the endogenous Roy (1951)-style sorting of households into the firm-industry pair where it is most productive. This mechanism results in upward-sloping, firm-specific labour supply curves, linking industry wage premia to sorting and segregation effects in a competitive labour market.[3]

Quantitatively, I estimate the model on U.S. industry-level data. The key structural parameters include two substitution elasticities (between ICT capital and non-routine labour, and between routine and non-routine tasks) as well as the degree of labour market concentration (strength of workers' sorting and segregation effects). Despite its convoluted structure, the model's general equilibrium structure enables a clear identification of production technology parameters. Following Karabarbounis and Neiman (2014), industry-level elasticities are estimated using sectoral labour shares: estimates suggest "gross complementarity", with capital-labour elasticities of substitution ranging from 0.25 to 0.8.

Given the estimated set of parameters, the model successfully captures key features of the inter-industry wage structure and prevailing wage inequality levels. In the baseline cross-sectional calibration, substantial heterogeneity emerges in the estimated substitution elasticities: industries with the largest wage growth show the strongest impact of technological change, followed by those with minimal and moderate wage variation. A re-estimation over two distinct sub-periods confirms how these technological parameters evolved unevenly across industries both in direction and, more critically, in magnitude.[4]

Having provided reduced-form empirical support for the model's theoretical implications, I use the observed shifts in structural parameters to conduct counterfactual simulations. These exercises decompose the industry component of U.S. wage in-

[3] This framework aligns with insights from the class of "new monopsony models" (*e.g.*, Manning 2021). Appendix D shows that incorporating wage-setting power does not materially affect the model's predictions on industry wage inequality, consistent with the empirical findings of Card, Rothstein, et al. (2024a).

[4] Industries where wages increased the most (less) have experienced a decrease in the complementarity between technological capital and non-routine workers of about 25% (131%), while slightly increasing the complementarity between job tasks of 3% (11%). Differently, intermediate industries undergone a reduction (20%) in substitutability of ICT capital-non-routine tuple, along a pale increase (3%) in job tasks substitution.

equality since the early 2000s, accounting for the interaction between technological change, labour force composition, and labour market concentration. To isolate the key mechanisms, I examine how period-variations in individual or combined technological parameters affect key outcomes of the model related to observed moments. I also estimate sectoral productivity via industry-level production functions and assess its contribution – alongside capital and labour input changes – once structural heterogeneity in technological parameters is excluded.

My main findings highlight that structural transformations are the primary determinants of U.S. between-industry wage inequality, far outweighing the role of changes in factor inputs typically emphasized by SBTC dimension. The analysis allows to pin down four main takeaways. Firstly, *structural effects dominate quantity effects*: when allowing for cross-industry trend-differences in substitution elasticities, the model accounts for 94% of the observed between-industry wage dispersion. By contrast, when holding these elasticities uniform across industries, changes in production input quantities explain only 6% to 15% of the observed inequality. The "structural effect" (heterogeneous shifts in capital-labour elasticities of substitution across industries) thus accounts for the bulk of observed between-industry wage dispersion, while the "quantity effect" (industry-specific changes in ICT capital, labour force, and productivity) plays a minor role.

Secondly, *substitution across job tasks is the dominant channel*. The "indirect effect" of SBTC – namely, the reorganization of tasks in response to technological change – is way more impactful than its "direct effect" – changes in non-routine workers due to changes in technological capital. Rising inequality is therefore tightly linked to the *worker side* than the *capital side* of production across industries.

Thirdly, *labour market concentration magnifies structural effects*. Incorporating heterogeneous sorting and segregation effects into the model further amplifies inequality, explaining up to 98% of the total between-industry wage dispersion when interacted with the evolving technological structure, since they are reinforcing the role of structural transformations through workforce allocation dynamics.

These findings indicate that while Skill-Biased Technological Change is necessary in explaining U.S. wage inequality, it is not sufficient on its own. What matters most is not the scale of factor input quantities accumulation, but the sector-specific evolution of technology and production structures. As such, explaining wage inequality requires moving beyond quantity-based narratives toward a clearer focus on structural heterogeneity and the dynamic nature of technological change across industries.

Related literature.– This research contributes to several strands of the literature. A primary connection is with the study of wage inequality in the U.S., traditionally emphasizing *worker-level* (*e.g.*, Katz and Autor 1999, Autor, Katz, et al. 2008, Lemieux 2010) or *between-firm* (*e.g.*, Leonardi 2007, Card, Heining, et al. 2013, Song et al. 2019, Bingley and Cappellari 2022) differences. Anyhow, recent evidence unveils how differentials are primarily clustered at the industry level. While earlier studies had already shown that inter-industry wage differentials were substantial but stable throughout much of the 20th century (*e.g.*, Slichter 1950, Cullen 1956, Krueger and Summers 1988, Allen 1995),[5,6] the industry component of wage inequality has grown markedly over the past three decades. Central is the work of Haltiwanger et al. (2024): wage differentials across industries drive the rise in U.S. wage inequality from 1998 to 2018, with a small number of industries playing a disproportionate role.[7] Existing investigations overlook the key determinants of industry-driven wage dispersion; my contribution is a comprehensive framework that incorporates structural variations in capital-labour substitutability across U.S. industries to explain observed trends in between-industry wage inequality.

Second, my analysis directly dives into the task-based literature.[8] The role of tasks in shaping wage inequality is emphasized by Autor, Katz, et al. (2006), who highlight how technological change alters the distribution of job task demands, thus driving job polarization. Building on this, I examine how industry-driven technological change reshapes labour force composition through tasks division and reallocation, contributing to U.S. wage inequality.

Third, the paper builds on structural transformations studies (reallocation of economic activity across broadly defined sectors), in particular on the role of elasticity of substitution between capital and labour (*e.g.*, Buera and Kaboski 2012, Herrendorf, Rogerson, et al. 2014, Eden and Gaggl 2018). Sectoral differences in capital-labour elasticities of substitution, as in Herrendorf, Herrington, et al. (2015), shape factor reallocation across broadly defined sectors. This perspective intersects my results: ample differences in industry-level elasticities of substitution and their asymmetric shifts majorly explain wage inequality across industries.

Roadmap.– This introduction is succeeded by Section 1, which provides the motivating evidence at which the model presented in Section 2 refers to. Section 3 shows and assess the performance of the calibration strategy, while Section 4 quantifies the relevance of given parameters in accounting for the observed level of U.S. wage inequality. Section 5 determines the importance of capital and labour quantities and of an estimated measure of sectoral productivity. Last section concludes.

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[5] Refer to Krueger and Summers (1987) and Dickens and Katz (1987) for a discussion of the findings. A complementary literature explores the role of inter-industry dispersion in capital-labour ratios on wage inequality (*e.g.*, Montgomery 1991, Caselli 1999), while sectoral differences are important also to interpret both the racial (*e.g.*, Card, Rothstein, et al. 2024b) and the gender (*e.g.*, Fields and Wolff 1995) pay gaps, and wage differentials under a behavioural economics perspective (*e.g.*, Thaler 1989).

[6] Allen (2001) notices how the "stability" argument may be misleading since important within-industry factors changes over time, while the constancy depends on a strong autocorrelation over a long time period. Under a cross-country comparisons, substantial but stable inter-industry wage differentials exist among also E.U. (*e.g.*, Genre, Kohn, et al. 2011, Genre, Momferatou, et al. 2005) and OECD (*e.g.*, Gittleman and Wolff 1993) countries.

[7] Rising between-industry wage inequality is a well documented fact (*e.g.*, Davidson and Reich 1988, Bell and Freeman 1991, Howell and Wolff 1991, Allen 2001). Haltiwanger et al. (2024)'s contribution is to show its importance on the increase in total U.S. wage inequality relative to both individual and firm components. Similar results for Italy are in Briskar et al. (2022).

[8] Strong emphasis in measuring employment trends by job task content: rapid employment growth for non-routine tasks from early 1990s relative to routines (*e.g.*, Goos and Manning 2007, Autor and Dorn 2013, Cortes et al. 2017, Jaimovich and Siu 2020, Cerina et al. 2021, Siena and Zago 2024, Cossu et al. 2024).

1. Motivating Evidence

To account for trends in wage dispersion I employ annual U.S. data, and the 3-digit U.S. 2017 North American Industrial Classification System (NAICS) is adopted for the industry level definition. The (balanced) panel data built is a combination of two different data sources: first, information on the stocks of physical and technological capital types are recovered from the Bureau of Economic Analysis (BEA), in the *Detailed Data for Fixed Assets* tables; the second source is the *Occupational Employment and Wage Statistics* (OEWS) program of the Bureau of Labor Statistics (BLS), which contains information on occupational employment and wage rates.

Capital inputs are grouped into four categories using BEA data for 1998-2022: physical capital, digital equipment, intangible capital; the sum of the last two measurers ICT capital. Occupational data from the BLS cover routine and non-routine tasks for private U.S. industries from 2003 onward. BEA and BLS data are harmonized at the 3-digit NAICS level, yielding a balanced panel of 62 private industries observed annually from 2003 to 2022. For additional details refer to Appendix A.

1.1. Validating the sample

From Haltiwanger et al. (2024), total wage variance growth across industries can be decomposed as each industry-$s \in \mathcal{S}$ contribution to wage inequality

$$\underbrace{\Delta\, var\Big(w(s) - \overline{w}\Big)}_{\substack{\textit{between-industry}\\ \textit{wage variance growth}}} = \sum_{s} \overbrace{\Delta\, \underbrace{\left(\frac{\ell(s)}{\ell}\right)}_{\substack{\textit{employment}\\ \textit{share}}} \underbrace{\Big(w(s) - \overline{w}\Big)^2}_{\substack{\textit{relative}\\ \textit{wage}}}}^{\substack{\textit{industry-s contribution to}\\ \textit{between-industry wage variance}}} \tag{1}$$

in which the employment share component allows to attach the proper weight to each industry. In each year, $w(s)$ is the average real log-wage of industry-s, $\overline{w}$ the associated economy-wide period mean, while $\ell(s)$ and ℓ represent sectoral and aggregate employment, respectively. A shift-share analysis provides further insights on industry factors accounting for such variance growth, inspecting the importance of relative wage changes versus employment share changes:

$$\underbrace{\Delta \left(\frac{\ell(s)}{\ell}\right) \Big(w(s) - \overline{w}\Big)^2}_{\substack{\textit{industry s contribution to}\\ \textit{between-industry wage variance}}} = \underbrace{\overline{\Big(w(s) - \overline{w}\Big)^2}\, \Delta \left(\frac{\ell(s)}{\ell}\right)}_{\textit{shift share: employment}} + \underbrace{\overline{\left(\frac{\ell(s)}{\ell}\right)}\, \Delta \Big(w(s) - \overline{w}\Big)^2}_{\textit{shift share: wage}} \tag{2}$$

The two decompositions are reported in Table 1. Wage inequality growth is highly concentrated across industries: 15 out of 62 sectors account for 93% of the total increase in wage dispersion, despite employing less than 40% of the workforce. Concentration is even sharper at the top, with just three industries explaining more than half of the total growth in real log-wage variance while representing only 8% of total employment. Contributions to wage dispersion are further examined through a shift-share analysis. An expansion in industries' labour force plays a limited role,

TABLE 1: CONTRIBUTION TO BETWEEN-INDUSTRY WAGE VARIANCE

			share		***shift-share***	
contribution	*industries*	*variance*		*employment*	*wage*	*employment*
> 5%	3	.26	54%	8%	90%	10%
1% to 5%	7	.14	29%	18%	76%	24%
.05% to 1%	6	.05	10%	13%	124%	-24%
-.05% to .05%	46	.03	7%	61%		·
quantiles						
0-25	15	.24	50%	31%	91%	9%
25-50	16	.04	8%	28%	77%	23%
50-75	16	.02	4%	15%		·
75-100	15	.18	38%	26%	98%	2%

Estimates are referred to eq. (1) for 3-digit U.S. 2017 NAICS industries. The last two columns report a quantification of the components in eq. (2); not reported estimates '·' imply that the shift-share for employment is highly less than zero. Operator Δ in the equations is $x_t - x_{t-1}$, and not a percentage change. Industries are grouped according to their own contribution to between-industry wage inequality in the first part of the table while, in the second part, grouping follows the overall percentage change in real log-wage per capita of each industry. Source: BEA and own calculations.

whereas relative wage dynamics dominate. As shown in the last two columns of Table 1, the bulk of wage variance growth stems from shifts in relative industry wages rather than from changes in employment composition.

FACT 1 (Contribution) *A small subset of industries drives the rise in wage inequality; these are in the tails of the industry-level wage growth distribution.*

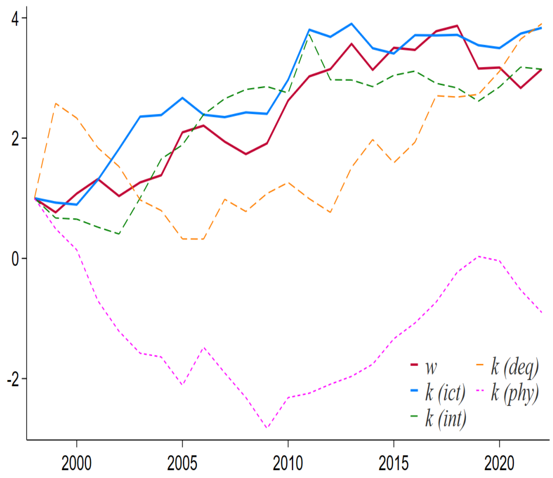
A similar exercise groups industries by changes in real log-wages (second panel of Table 1). Top and bottom 25% together account for 88% of the growth in wage inequality while employing nearly 60% of the workforce. As before, relative wage movements significantly dominate: changes in industry employment shares contribute little to wage dispersion, thus reconciling with Fact n. 3. These results align with Haltiwanger et al. (2024).[9]

What intrinsic characteristics underpin the dispersion of wages across industries? Why does the employment size of industries seem to play a minimal role? These questions are explored in the following subsections.

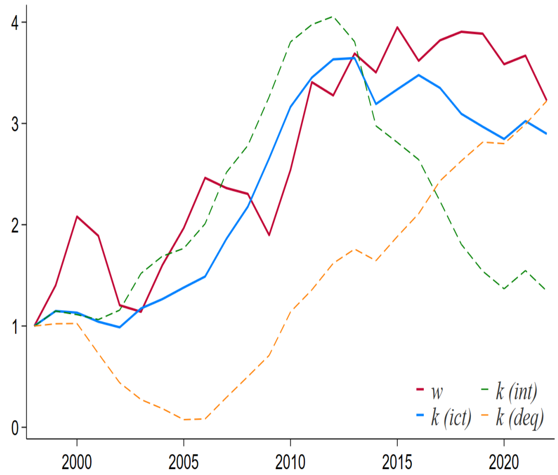
1.2. U.S. INDUSTRIES DIGITALIZATION

As a first step, the analysis shall examine whether cross-industry differences in capital stocks are reflected in wage differentials. To capture the rise in dispersion, Figure 1 traces the evolution of the differentials in capital intensities and real wages

[9] According to their estimates, 10% of 4-digit U.S. 2017 NAICS industries account for nearly 98% of the total increase in between-industry wage dispersion from 1996 to 2018, as well located at the tails of the industry-level earnings distribution.



(A) INTERQUARTILE RANGE, 90^{th}-10^{th}



(B) LOG-VARIANCE

FIGURE 1: CROSS-INDUSTRY DISPERSIONS OF CAPITAL TYPES

Note: the figure depicts dispersion across industries of average real log-wage, physical, ICT, intangible capital types and digital equipment per capita. Solid red and blue lines are related to wages and ICT capital, while dashed green, orange and purple lines are intangible capital, digital structures, and physical capital, respectively, all taken in per capita log-terms. Panel 1a plots the yearly difference between top and bottom 10% of each component, while Panel 1b plots the associated log-variance. Series are standardized and indexed to 1 in 1998, so that both y-axis indexes the respective measure given the initial value at unity. Plots are referred to 3-digit U.S. 2017 NAICS industries. *Source*: BEA and own calculations.

across industries. Panel 1a measures industry-level dispersion using the interquartile range – the gap between the top and bottom 10% of industries in each log-variable. Dispersion in ICT-related capital ratios increases steadily over time, indicating a widening gap between industries with high and low ICT capital per worker. Intangible assets and digital equipment display similar, though less uniform, patterns. By contrast, dispersion in physical capital declines, suggesting a convergence in physical capital stocks across industries.

FACT 2 (Industry capital gaps) *Dispersion in physical capital-labour ratio is decreasing across industries, while it is not that of ICT capital-labour ratio.*

Comparing inter-industry differentials in capital intensities with those in labour income reveals that real wage gaps are primarily associated with disparities in ICT capital, rather than with other forms of capital. This pattern highlights the role of a macro-complementarity:[10] as shown in Panel 1b, dispersion in intangibles or digital equipment alone fails to match the evolution of wage dispersion unless the two are combined into a unified ICT capital measure. The interaction between intangible capital and digital equipment is therefore central: taken separately, intangibles loosely track wage dispersion until around 2012, while digital equipment becomes relevant only thereafter.

[10] A growing empirical and methodological debate emphasizes the importance of such complementarities. For instance, Corrado et al. (2017) show that returns to ICT depend critically on otherwise "unmeasured" intangible assets, documenting ICT-intangible complementarity in both U.S. and EU data. A similar argument appears in Crouzet et al. (2022), who note that intangible capital, lacking physical embodiment, requires digital infrastructure or storage medium to operate.

Further evidence on the role of ICT capital on real wages is provided in Table A.1. Fixed-effects panel regressions largely confirm the preceding patterns. While increases in physical capital per worker are associated with higher real wages, the effect of ICT capital alone is weaker (column 1), suggesting substantial heterogeneity across industries. Intangible capital plays a central role, accounting for a large share of wage variation (column 2), whereas digital equipment on its own appears less influential. Crucially, the interaction between intangibles and digital equipment (column 3) remains significant, and continues to do so when all capital types are included jointly (column 4), underscoring the importance of ICT complementarities in explaining industry-level real log-wage differences.

1.3. Accounting for workforce composition

Since their advent, advances in technical and technological processes have raised the issue of their underlying effects on employees' qualification requirements at both aggregate and industry layers (*e.g.*, Horowitz and Herrnstadt 1966, Rumberger 1981, Autor 2022), and that the mechanism through which these impact workers does not transmit uniquely on the absolute level of occupational employment, but especially in its relative term.[11] A natural intuition to analyse is whether changes in the relative workforce of each task can be attributed to changing sizes of industries or to the reallocation of worker-types within those industries. Denoting the set of tasks as $\{a, a'\} \in \mathcal{A}$ and with $s \in \mathcal{S}$ that of industries, the following labour force decomposition provides the answer to this inquiry:

$$\Delta\left(\frac{\ell(a)}{\ell(a')}\right) = \underbrace{\sum_s \overline{\left(\frac{\ell(a,s)}{\ell(s)}\right)} \Delta\left(\frac{\ell(a,s)}{\ell(a',s)}\right)}_{\textit{within-industry: substitution effect}} + \underbrace{\sum_s \overline{\left(\frac{\ell(a,s)}{\ell(a',s)}\right)} \Delta\left(\frac{\ell(a,s)}{\ell(s)}\right)}_{\textit{between-industry: size effect}} \tag{3}$$

Shifts in the economy-wide ratio of task-a to task-a' can be decomposed into within- and between-industry components. The within component captures changes in task intensity within industries – thereby capturing the substitution and reallocation of tasks –, while the between component reflects changes in the relative size of task-specific employment across industries. Figure A.3 illustrates the dynamics of fitted routine and non-routine labour shares obtained from Fixed Effects (FE) regressions with industry and year effects, used to implement equation (3). The share of non-routine workers rises steadily over time as routine employment declines, raising the question of whether this shift reflects adjustments in industry size or substitution in task composition within industries. This structural change is exactly the underlying logic behind eq. (3).

Table 2 reports the results of the decomposition. The increase in the non-routine-to-routine task ratio is driven almost entirely by within-industry adjustments, indicating that industries are replacing routine tasks with non-routine ones rather than

[11] Refer to Melman (1951) and Braverman (1974) for an analysis over the period 1820-1970.

TABLE 2: LABOUR FORCE COMPOSITION

	routine		*non-routine*	
interval	*within*	*between*	*within*	*between*
2003-2008	83%	17%	38%	62%
2009-2015	81%	19%	58%	42%
2016-2022	78%	22%	69%	31%

Quantification of eq. (3); changing components (Δ) are linear trends predicted by a Fixed Effects (FE) regression of the form $(y_t \mid \mathcal{X}_t) = \beta_c + \beta_t \mathcal{X}_t + u_t$, with y representing the task ratio, and $\mathcal{X}$ a vector of industry and year fixed effects, at 3-digit U.S. 2017 NAICS industries. Source: BLS and own calculations.

expanding their workforce. Conversely, changes in the routine-to-non-routine ratio are associated with modest between-industry effects, reflecting limited employment expansion rather than increased reliance on routine tasks.[12]

FACT 3 (Labour force composition) *Increases in non-routine relative share are determined by a substitution effect rather than by the employment size of industries.*

The analysis next combines the evidence on capital accumulation with changes in labour force composition. The key question is whether industries experiencing faster real wage growth are also those that expand ICT capital and increase their reliance on non-routine labour. In this framework, non-routine workers are those whose tasks are more ICT-intensive, whereas routine workers primarily perform manual tasks, as formalized later in the model. To investigate this relationship, a set of industry-level fixed-effects panel regressions is estimated, grouping industries by their position in the real log-wage growth distribution

$$y_\omega \left(\Delta \, log \, (w_t(s)) \mid \mathcal{X}_{i,\omega t}, \mathcal{Z}_{j,\omega t} \right) = \beta_{c,\omega} + \beta_{i,\omega} \mathcal{X}_{i,\omega t} + \delta_{j,\omega} \mathcal{Z}_{j,\omega t} + u_{\omega t} \tag{4}$$

with ω identifying each quantile in Table 1, $\mathcal{X}_i$ bundles percentage changes in both ICT-to-physical capital (k) and non-routine-over-routine workers (ℓ) ratios, and $\mathcal{Z}_j$ serves as controls. The interaction between ICT capital accumulation and task reallocation is particularly pronounced in industries at both ends of the wage growth distribution – those at the top and those at the bottom –, as shown in Table 3, thereby aligning with stylized Fact n. 1. High-growth industries exhibit large joint increases in ICT capital and non-routine employment, translating into substantial wage gains (1.56), whereas low-growth industries show milder adoption of these factors, resulting in weaker wage performance (0.876). No significant – and in some cases negative – effects are observed for industries in the middle, which contribute relatively little to overall wage inequality. Table A.2 reinforces this pattern. Column 3 shows that the interaction term $\beta_{\Delta k \times \Delta \ell}$, capturing joint changes in ICT capital and task composition, has a large and statistically significant effect on real log-wage growth (.919), far

[12] A link between industry affiliation and occupational distribution is also documented in Haltiwanger et al. (2024), although that analysis does not address task substitution within industries.

TABLE 3: COMBINED REGRESSIONS BY GROUPS, PERCENTAGE CHANGES

	$\Delta\, log(w(s))$			
	0-25 quantile	*25-50 quantile*	*50-75 quantile*	*75-100 quantile*
$\beta_{\Delta k}$	-.693	.302***	.078	-.082*
	(.618)	(.072)	(.179)	(.042)
$\beta_{\Delta \ell}$	.041***	-.297	.015***	.052*
	(.004)	(.247)	(.003)	(.025)
$\beta_{\Delta k \times \Delta \ell}$	.876***	-1.304	-.643	1.560*
	(.100)	(2.945)	(.470)	(.789)

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). Standard error in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries in 2003-2022. Each Fixed Effects (FE) regression – performed on groups of industries clustered in quantiles (ω) according to their overall growth in real wage – , is of the form of eq. (4), with $\mathcal{X}_i$ representing the percentage change in both ICT-to-physical capital and non-routine-over-routine workers ratios, and $\mathcal{Z}_j$ being a set of time-varying controls. Variables are all in log format. Constant not reported to save space. Source: BEA, BLS and own calculations.*

exceeding the impact of isolated changes in either ICT capital (.009) or labour composition (.035). Therefore, wage inequality is driven primarily by the concentrated performance of industries at the extremes of the wage growth distribution, where the combined evolution of ICT capital and non-routine labour is most pronounced. The evidence underscores that the interaction of these two factors, rather than isolated changes, plays a central role in shaping wage dispersion.

An argument strongly substantiated by a bulk of quantile regressions of the form $\mathcal{Q}_\omega\left(log\left(w_t(s)\right) \mid \mathcal{X}_{i,\omega t}, \mathcal{Z}_{j,\omega t}\right) = \beta_{i,\omega}\mathcal{X}_{i,\omega t} + \delta_{j,\omega}\mathcal{Z}_{j,\omega t} + u_{\omega t}$, where ω represents each quantile (defined on the independent variable), and $\mathcal{X}_i$ and $\mathcal{Z}_j$ as before. The results point to a positive effect of both the task ratio and the ICT-to-physical capital ratio, with the impact strengthening from the lower to the higher end of the industry wage growth distribution. Table A.3 shows that the task ratio (β_ℓ) has a substantially larger effect on real log-wage growth than the ICT ratio (β_k, column 1). Effects are uneven across industries: the magnitude of both ratios increases along the distribution, with industries farther from the top experiencing smaller impacts (columns 2-5). Importantly, the interaction term reaches its largest values in regressions based on changes ($\beta_{\Delta k \times \Delta \ell}$) as opposed to those in levels ($\beta_{k \times \ell}$), suggesting that the joint evolution of ICT capital and task composition is most informative for explaining over-time variation in real log-wages, rather than cross-sectional differences.

FACT 4 (Structural transformations) *Industries marked by highest changes in real wages experience a substantial rise in their non-routine workers relative share along an increasing ICT capital ratio dynamics.*

When jointly considered, the presented stylized facts reveal some important characteristics of the U.S. inter-industry wage structure: the dispersed increase in industries' real average wage is influenced by a growing dispersion in technology (Fact n. 2); in turn, such technological differences reflect variations in the labour force composition within each industry (Fact n. 3). In other words, industries that have experienced a larger and more significant increase in the stock of ICT capital relative to physical capital tend to exhibit a higher rate of substitution of routine tasks with non-routine tasks (Fact n. 4). Reminiscent from the introduction, these findings are opened to two possible interpretations: wage inequality across industries could be the result of *(i)* a *quantity* effect, under different trajectories in factor of productions; or *(ii)* a *structural* effect, stemming from changes in the technological parameters that govern the relationship between capital and labour types. These general equilibrium considerations provide the motivating background to build a model with industry-specific capital-labour substitutability and heterogeneous labour force composition to account for trends in wage inequality in the United States.

2. Structural Model

To rationalize the presented empirical regularities, I build and simulate a structural general equilibrium model. The households block features a skill-related heterogeneity: households are *ex-ante* divided in routine and non-routine workers, sorting into firms and industries given heterogeneous productivities. Final and sectoral outputs are competitively aggregated, while firms in each industry compete monopolistically; they all use ICT and non-ICT capital types, non-routine labour is complementary to ICT capital, while routine workers are substitutable. Hence, the economy identifies the structure of production functions to be analogous in all industries except for their degrees of substitutability across factors of production. To deepen the role of substitution parameters, productivity differences across industries are excluded.[13]

2.1. Households

There is a unit mass of households, each labelled as $i \in \mathcal{I}$, split in tasks $a = \{rt, nrt\}$. Household-i of type-a gets utility from consumption, $\mathcal{C}_i$, and it is increasing in a known idiosyncratic factor, $\wp^i_h\,(a,s)$, drawn once from a specified distribution, measuring its efficiency level when working with firm-$h \in [1,H]$ in industry-$h \in [1,S]$:

$$\mathcal{U}^i_h\,(a,s) := log\;\mathcal{C}^i + log\;\mathcal{B}_{h,t}\,(a,s) + \wp^i_h\,(a,s) \tag{5}$$

Increasing utility in $\wp^i_h\,(a,s)$ suggests that household-i is optimally allocated (*i.e.*, endogenously sort) to a workplace that maximizes its productivity.[14] This outcome directly follows from assuming a self-selection based labour market, involving indi-

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[13] Sectoral productivity and structural changes are theoretically analysed by Ngai and Pissarides (2007), linking the elasticity of substitution among intermediate inputs to the sectoral reallocation of employment. In Section 5 I consider the effect of estimated industry-level productivity series on wage dispersion.

[14] An high value for $\wp^i_h\,(a,s)$ indicates that household-i is utilitarianism better off, as it reflects the household's ability to achieve its maximum productivity when working as type-a within the given firm and industry (h,s) tuple. This way of designing the utility function is common in the discrete choice literature (*e.g.*, Card, Cardoso, Heining, et al. 2018, Hsieh et al. 2019). Due to the absence of a non-employment option and unemployment, the terms household(s) and worker(s) are then used interchangeably.

viduals choosing the firm-industry combination that maximizes their utility based on their spectrum of productivities across different sectors. In other words, individuals self-select into workplaces by comparing the potential outcomes of various firm-industry pairs. Moreover, borrowing from the spatial economics literature, I include a firm-h, industry-s amenity for household type-a, entering multiplicatively in the utility function (*e.g.*, Kleinman et al. 2023). It is defined as $\mathcal{B}_{h,t}(a,s) = \left[g_{h,t}(a/a',s)\right]^{-\varsigma}$, where $g_{h,t}(a/a',s)$ reflects the relative share of task-a in task-a' (keeping the same notation of eq. 3), negatively scaled by the elasticity parameter ς (an indicator of the degree of substitutability among job tasks).[15]

An household can invest either in bonds (b^i, which are in zero net supply) at the rate r_t, and in capital types $k^i(j), \forall j = \{phy, ict\}$, while receiving dividends $\mathcal{D}^i$ from firms, so that its expected utility problem displays the budget constraint to be

$$C_t^i + I_t^i(phy) + I_t^i(ict) + b_{t+1}^i - (1+r_t)b_t^i =$$
$$w_{h,t}(a,s)\,\ell_h^i(a,s) + R_t(phy)\,k_t^i(phy) + R_t(ict)\,k_t^i(ict) + \mathcal{D}_t^i$$

where the wage rate associated to type-a in firm-industry pair (h,s) is $w_h(a,s)$, with labour $\ell_h^i(a,s)$ inelastically supplied and set to unity. Any capital stock depreciates at a rate δ and accumulates over time according to a quantity-adjusted law of motion: it is a negative function of ζ_t^i (the relative ICT capital share) that scales new capital investment, $I_t^i(j)$: as the capital stock owned by household-i becomes more technologically advanced (higher ζ_t^i), a higher rate of capital investment is required to maintain the future stock of that particular capital type.[16] ζ_t^i thus represents the technology level of the total capital stock owned by household-i.

Inter-temporal utility maximization implies a standard Euler condition for future path of marginal utility from consumption, and an equation displaying the evolution of the capital rental rate

$$R_t = (1+r_{t+1})\,\zeta_t - (1-\delta)\,\zeta_{t+1} \tag{6}$$

in which changes across states of R_t are determined by changes in the (aggregate) relative quantities across capital types, $\zeta_t = \int_i \zeta_t^i\,\mathrm{d}i$. In the spirit of Karabarbounis and Neiman (2014), eq. (6) determines that investing in capital types is profitable as long as the marginal benefit of investment is at least lower than its marginal cost.

In a market economy workers do not randomly sort across workplaces. Rather, in a Roy (1951)-style model of self-selection, households locate based on how their idiosyncratic productivity varies across firms and industries. Dropping time-t subscript, the idiosyncratic *productivity* of household-i, task-a working in firm-industry (h,s) tuple is drawn once from a multivariate Frechét-type cumulative distribution

[15] Below, the component $\mathcal{B}_{h,t}(a,s)$ allows to characterize the equilibrium wages as a function of the task ratio too, as in Section 1. Its relevance in determining the endogenous labour supply of eq. (7) – *i.e.*, the outcome of the sorting choice by the part of households – will be empirically tested and validated in Subsection 3.1.

[16] Household-i's capital dynamics follows a law of motion as $k_{t+1}^i(j) = \frac{I_t^i(j)}{\zeta_t^i} + (1-\delta_j)k_t^i(j)$, $\forall j = \{phy, ict\}$. A higher ζ_t^i is an improvement in the technological sophistication of the capital bundle, symptom that any additional unit of invested capital faces diminishing marginal returns, as described by Inada (1963)'s conditions.

$$F_i\left(\wp^i_{h,\dots,H}(a,1),\dots,\wp^i_{h,\dots,H}(a,s),\dots,\wp^i_{h,\dots,H}(a,s)\right)=exp\left[-\sum_s\left(\int_h \wp^i_h(a,s)\,\mathrm{d}h\right)^{-\theta}\right]$$

Its shape parameter, $\theta > 1$, governs the degree of dispersion in idiosyncratic productivities' draws (*i.e.*, the labour supply elasticity), and lower values of θ imply major degrees of dispersion (and thus a more elastic labour supply response to wage differences);[17] without loss of generality, location and scale parameters are normalized to 1. Such Type-I Extreme Value Distribution identifies a discrete choice on how household of type-a sorts into firm (h,s) given heterogeneous productivity levels across different firm-industry pairs.[18]

Central to this modelling framework is that workers are potentially employable in any firm within any industry, but their productivity varies across firm-industry pairs. This structure captures both absolute and comparative advantage. Absolute advantage implies that any worker performing a task can, in principle, work in any firm-industry combination. Comparative advantage, however, determines actual employment: workers choose to be employed in the firm and industry where their relative productivity in that task is highest. Consequently, labour allocation reflects not only wage differences but also task-specific efficiency across different environments. This mechanism generates a "segregation" effect, whereby firms increasingly hire workers whose comparative productivity aligns closely with their operational and technological requirements.

The labour market, while competitive and frictionless in a traditional sense, is thus structurally imperfect due to this idiosyncratic productivity heterogeneity. The distribution of households' productivities – and thus the self-decision around employment choices – makes the labour supply to become endogenous, shaped by both firm-industry level wages and resulting task-specific comparative advantages. Described sorting preferences allows to derive analytical form for the measure of each worker-type in firms and industries given the respective wage level: in Appendix B I show that the labour supply curve of each (a,h,s) takes the form of

$$\ell_h(a,s)=\left(\frac{w_h(a,s)}{\mathcal{W}_{\mathcal{H}}(a,\mathcal{S})}\,\frac{\mathcal{B}_h(a,s)}{\mathcal{B}_{\mathcal{H}}(a,\mathcal{S})}\right)^{\theta} \tag{7}$$

[17] It measures the centrality of a change in labour supply of task-a induced by a change in its firm-h, industry-s wage level. Rogerson (2024) stresses the importance of the (aggregate) labour supply elasticity in macroeconomics, pointing to both intensive (average hours worked) and extensive (employed people in the total population) margins. In my model only the latter margin (relative task-a share of total employment) is considered, since households are assumed to supply inelastically one unit of work (so that $\ell^i_h(a,s)=1$), linking this elasticity to a certain degree of workers' concentration in the labour market; refer to Figure 2.

[18] Self-selection concept dates back to Roy (1951)'s verbal-model, whose scope is to describe how the subjective choice for a given occupation would affect the distribution of wages: it is not only a function of gaps in potential wages, but rather it includes also occupational sorting choices. Roy's idea has been substantially tested in the literature (right after Heckman and Sedlacek 1985 and Borjas 1987), and the macroeconomic implications of such microeconomics sorting choice have been firstly introduced by Hsieh et al. (2019).

with $\mathcal{W}_{\mathcal{H}}(a,\mathcal{S}) = \sum_{h,s} w_h(a,s)$ and $\mathcal{B}_{\mathcal{H}}(a,\mathcal{S}) = \sum_{h,s} \mathcal{B}_h(a,s)$. Labour supply is modelled as depending not only on the absolute wage offered to task-specific workers but also on its relative attractiveness, captured by a "pay premia" effect – that is, how the wage compares with those offered in other firms and industries. Additionally, to attract a larger share of workers with similar productivity levels (the "sorting" effect), a firm must offer higher wages.[19]

Such labour supply allows to summarize the role of wage premium, sorting and segregation, all effects of pivotal importance in shaping wage inequality. Haltiwanger et al. (2024) highlight how wage dispersion across industries is increasingly driven by sorting (referred to as the covariance of industry wage premium and the segregation effect across industries) associated to wage premium.[20] Card, Rothstein, et al. (2024a) find how the correlation between workers sorting across industries and the associated wage premium is high.[21] In other words, this constitutive self-selection directly enhances a mechanism on the critical interaction of the workplaces' labour force composition with sorting and segregation effects, both associated to the firm-industry wage premium from not perfectly elastic labour supplies.

Remark 1 (Labour market elasticity, sorting and segregation) *Selected specification of households' productivities allows to characterize firms and industries workforce composition of tasks as determined by a not fully elastic labour supply elasticity, shaped by comparative advantage, interacting with equilibrium wage premium designed through perfect competition in the labour market.*

Labour supply curves in eq. (7) are upward-sloping not due to frictions, but due to the structure of comparative advantage and the imperfect substitutability of workers across jobs.[22] In this sense, workers in the same task are "highly rival factors" (Hicks 1932) since they can be freely substituted for one another, yet ensuring that high-wage workers do sort in high-wage firms and industries, and also that more

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[19] As shown in Appendix B, the endogenous sorting decision is determined by a comparison of the combination of wages and idiosyncratic productivities (thus the total earnings) of working in a given workplace compared to other ones, *i.e.*, $Pr\left[w(a,s)\,\mathcal{B}(a,s)\,\wp^i(a,s) > w(a,s')\,\mathcal{B}(a,s')\,\wp^i(a,s')\right]$.

[20] In continuity with Card, Heining, et al. (2013) for Germany, several empirical works (*e.g.*, Alvarez et al. (2018) for Brazil, Morin (2023) for Denmark, Briskar et al. (2022) for Italy, Card, Cardoso, and Kline (2016) for Portugal, Håkanson et al. (2020) for Sweden, Barth et al. (2016), Song et al. (2019) and Haltiwanger et al. (2023) for US) detect how these major effects are increasingly impacting the distribution of wages.

[21] Using a cross-sectional approach at sectoral level, Gibbons et al. (2005) show how the high-skill occupational sorting is determined by differentials in return to skills, and the comparative advantage motive, rather than learning, seems to affect the resulting wage premiums.

[22] An important observation is that the presented model traces these effects back to a deeper micro-structural foundation: endogenous heterogeneity in worker productivity and endogenous allocation of labour in a competitive but non-random labour market in the sense of "new classical monopsony" (Manning 2021). The not-fully-flexible labour supply (due to parameter $\theta \neq 1$) is central to the understanding of how wage premia emerge and interact with sorting and segregation forces. In this perspective monopsony is not a dynamic search-and-matching friction; rather, its static nature ensures the labour supply elasticity a notable role, affecting labour in a fully competitive labour market. Even if I will not allow firms to take their respective labour supply of each task-a as given (not encompassing a proper monopsony power), the equilibrium wage level of firm (h,s) directly determines the number and the type of workers working there through sorting and segregation effects.

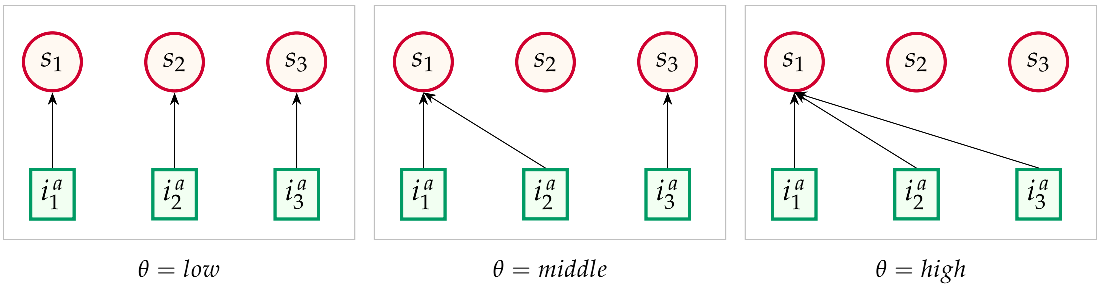


FIGURE 2: PRODUCTIVITY DISPERSION, SORTING AND SEGREGATION

Note: these graphs display some stylized examples on how productivity dispersion of households relates to labour market concentration at industry level. Each i^a represents an household of task-a, while s is a specific industry. The higher is the value of θ, the lower is the dispersion of households' productivities, so that the larger are the effects of sorting and segregation, thus the higher is the degree of labour market concentration of workers across industries.

efficient people would be preferred to get a job in high-wage workplaces than ones who are less productive.

ASSUMPTION 1 *Since households' productivities are Frechét-distributed, workers are perfectly mobile across firms within an industry, and immobile across industries.*

Parameter θ governs the elasticity of labour supply and, since wage differentials primarily drive employment choices, it is interpreted as a structural measure of labour market concentration. As idiosyncratic productivities become more dispersed, the strength of these mechanisms diminishes, reducing labour market polarization. This concentration arises from the distribution of household productivities through sorting and segregation mechanisms. Figure 2 provides a conceptual illustration of how variations in productivity dispersion shape employment allocations.

EXAMPLE 1 (Labour market concentration) *Consider a stylized economy with* $\{s_1, s_2, s_3\} \in \mathcal{S}$ *industries, each represented by a single firm, and three households of unique type-a. Households' productivities may acquire only three values,* $\wp^i\left(a, s\right) = \left\{low, middle, high\right\}$ *depending where employed, with parameter* θ *determining even dispersion. When* $\theta = low$*, productivity levels are highly dispersed across households and industries, leading each household to sort into a different industry, and resulting in low labour market concentration. Conversely when* $\theta = high$*: productivities result to be very clustered, with all households concentrating in the same industry, thereby generating strong polarization. An intermediate case occurs when* $\theta = middle$*, yielding to partial sorting pattern and reflecting moderate labour market concentration.*

Under a broader perspective, I am interpreting the degree of type-a workers' concentration at industry level to be directly proportional to the degree of variability in households' productivities (and therefore to the labour supply elasticity).

2.2. INDUSTRIES

The supply side is made of a countable set of industries $s \in \mathcal{S}$, each producing a specific variety $y\left(s\right)$. A perfectly competitive final good producer combines intermediate outputs from industries using a Constant Elasticity of Substitution (CES) technology

$$Y = \left(\sum_{s} y\left(s\right)^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}$$

where $\eta > 1$ is the elasticity of substitution among varieties of differentiated goods/services from industries. Final producer's optimizing behaviour gives birth to a relative demand of each industry-s output, given by $\frac{y(s)}{Y} = \left(\frac{p(s)}{P}\right)^{-\eta}$, with aggregate price index $P = \left(\sum_s p\left(s\right)^{1-\eta} \right)^{\frac{1}{1-\eta}}$ since final good is competitively supplied.

A unitary mass of monopolistically competitive homogeneous firms, indexed with $h \in \mathcal{H}$, is comprised in all industries. Given market power of firms, aggregation for industry-s bundles together different varieties by a new CES aggregating function

$$y\left(s\right) = \left(\int_{h} y_{h}\left(s\right)^{\frac{\epsilon-1}{\epsilon}} \, \mathrm{d}h \right)^{\frac{\epsilon}{\epsilon-1}}$$

where $\epsilon > 1$ is the sectoral elasticity of substitution among firms' varieties. In a way analogous to final producer, the conditional demand of firm (h, s) arises from competitive profit maximization at industry level:

$$y_{h}\left(s\right) = \left(\frac{p_{h}\left(s\right)}{p\left(s\right)} \right)^{-\epsilon} y\left(s\right) \tag{8}$$

with the sectoral price index as the *numeraire*, $p\left(s\right) = \left(\int_{0}^{1} p_{h}\left(s\right)^{1-\epsilon} \, \mathrm{d}h \right)^{\frac{1}{1-\epsilon}} = 1$. Firm-$h$ in industry-s comprises two types of workers, $a = \{rt,\ nrt\}$: non-routine (nrt) workers are complementary to ICT capital, while routine (rt) labour force is substitutable with the ICT composite good produced under the non-routine-ICT capital complementarity. The production function $y = f\left(k^{phy}, k^{ict}, \ell^{rt}, \ell^{nrt}\right)$ of firm (h, s) exploits that in Krusell et al. (2000), that is

$$y_{h}\left(s\right) = \left(k_{h}\left(phy, s\right)\right)^{\alpha} \left[\mu \left(\ell_{h}\left(rt, s\right)\right)^{\varsigma} + \left(1 - \mu\right) \left(q_{h}\left(s\right)\right)^{\varsigma} \right]^{\frac{1-\alpha}{\varsigma}}$$

with

$$q_{h}\left(s\right) = \left[\lambda \left(k_{h}\left(ict, s\right)\right)^{\varrho} + \left(1 - \lambda\right) \left(\ell_{h}\left(nrt, s\right)\right)^{\varrho} \right]^{\frac{1}{\varrho}} \tag{9}$$

where $k_h\left(phy, s\right)$ and $k_h\left(ict, s\right)$ are non-ICT and ICT capital, $\ell_h\left(rt, s\right)$ and $\ell_h\left(nrt, s\right)$ are measures of routine and non-routine workers, while μ and λ are just weighting parameters that govern output share of each firm (h, s); parameter α is the physical capital share of output. Elasticity indicators (ϱ, ς) are $\varrho = \frac{\rho-1}{\rho}$ and $\varsigma = \frac{\sigma-1}{\sigma}$.

ASSUMPTION 2 *Substitution elasticities, $\rho_s, \sigma_s \in \left[0, \infty\right)$, physical capital and distributive shares, $\alpha_s, \mu_s, \lambda_s \in \left(0, 1\right)$, are industry-specific. Since elasticities are finite, each industry adopts strictly positive levels for capital and labour quantities.*

Parameter ρ is the elasticity of substitution between ICT capital and non-routine labour input (*i.e.*, the CES composite), while σ identifies the elasticity of substitution between routine labour input and the CES composite. The CES structure imposed above yields a symmetry constraint in the interpretation of σ: the elasticity of substitution between routine and non-routine workers is the same as that between routine labour force and the ICT composite. Capital-task complementarity requires $\sigma > \rho$: this condition resorts the idea that technological process (namely an increase in ICT capital relative to non-ICT one) widens the relative demand of non-routine labour force, thus lessening that of routine workers.

Given the market structure, the formation of wages is decided at firm level considering monopolistic competition among firms in the same industry. Profit maximizing behaviour thus implies the Cobb Douglas-nested CES production function in eq. (9) to be subject on the conditional firm demand in eq. (8):

$$\max_{p_h(s),k_h(j,s),\ell_h(a,s)} \left[\mathcal{D}_h\left(s\right) \;\middle|\; y_h\left(s\right)\right]$$

with $\mathcal{D}_h\left(s\right)$ being profits. Within an industry, firms are all homogeneous; thus, any wage decision by firm (h, s) is not altering the sectoral wage level.[23]

Assumption 3 *Firms are atomistic, so that any wage level offered by a firm (h, s) only does not affect the sectoral one:* $\frac{\partial w(a,s)}{\partial w_h(a,s)} = 0\ \forall a, h, s.$

Under this assumption, equilibrium wages paid by firm (h, s) are set, for routine and non-routine workers, respectively, to

$$\begin{aligned} w\left(rt,s\right) &= \left[\Lambda\left(s\right)\,\chi\left(rt,s\right)\,\left(k\left(phy,s\right)\right)^{\alpha}\,\mathcal{V}\left(s\right)^{\frac{1-\alpha-\varsigma}{\varsigma}}\,\left(\frac{\mathcal{B}\left(rt,s\right)}{\mathcal{WB}\left(rt,\mathcal{S}\right)}\right)^{\theta(\varsigma-1)}\right]^{\frac{1}{1+\theta-\theta\varsigma}} \\ w\left(nrt,s\right) &= \left[\Lambda\left(s\right)\,\chi\left(nrt,s\right)\,\left(k\left(phy,s\right)\right)^{\alpha}\,\mathcal{V}\left(s\right)^{\frac{1-\alpha-\varsigma}{\varsigma}}\,\mathcal{Q}\left(s\right)^{\frac{\varsigma-\varrho}{\varrho}}\,\left(\frac{\mathcal{B}\left(nrt,s\right)}{\mathcal{WB}\left(nrt,\mathcal{S}\right)}\right)^{\theta(\varrho-1)}\right]^{\frac{1}{1+\theta-\theta\varrho}} \end{aligned} \tag{10}$$

where industry-specific composite parameters are given by $\chi\left(rt,s\right) = \left(1-\alpha\right)\mu$ and $\chi\left(nrt,s\right) = \left(1-\alpha\right)\left(1-\mu\right)\left(1-\lambda\right)$, while $\Lambda\left(s\right) = p\left(s\right)\,\mathcal{M}^{-1}$ is an indicator combining the industry-specific price level multiplied by the price mark-up, $\mathcal{M} = \frac{\epsilon}{\epsilon-1}$, and the aggregate element $\mathcal{WB}\left(a,\mathcal{S}\right) = \sum_s \mathcal{W}_{\mathcal{H}}\left(a,s\right)\mathcal{B}_{\mathcal{H}}\left(a,s\right)$; refer to Appendix B for the definition of the other components. Aggregate industry wage is thus found by averaged aggregation among job tasks: $w\left(s\right) = \left(\mathcal{A}\right)^{-1}\sum_a w\left(a,s\right)$.

Before delving into the intuition behind the equations, it is important to establish a clear understanding of why wages are at industry level: gaining this foundational insight will help to better contextualize eq. (10).

Proposition 1 (Firm and industry layers) *In an economy characterized by sorting and segregation effects and a not perfectly elastic labour supply where the measure of workers in a given firm is determined by its wage relative to the others, as long as*

(a) firms within an industry have the same size; or

(b) workers are perfectly mobile across firms within an industry,

[23] This assumption is a natural outcome in the proof of Proposition 1 in Appendix B.

$$\frac{\partial \ell_h(a,s)}{\partial w_h(a,s)} = -\frac{\partial \ell_{h'}(a,s)}{\partial w_h(a,s)} \quad and \quad \frac{\partial w_h(a,s)}{\partial \ell_h(a,s)} = \frac{\partial w_{h'}(a,s)}{\partial \ell_{h'}(a,s)} \quad ,$$

profit-maximizing wages set by firms in a specific industry are equal, and thus the unique optimal wage level can be directly written under industry notation. Moreover,

(c) workers are immobile across firms between industries.

Proof in Appendix B.

Besides formal proofs, the movement of workers across firms and industries is directly influenced by the assumed distribution of households' idiosyncratic productivity parameter $\wp_h^i(a,s)$. In fact, the *max-stability property* of the Frechét distribution ensures that a worker, once choosing a workplace, will not move nor across firms neither among industries: since households' productivities are so distributed, such property guarantees that its maximum is as well distributed (Mc Fadden 1974), and each household always picks its utility-maximizer productivity level, $\wp_h^i(a,s)_{max}$. Note that the imposed within-industry firm-homogeneity implies that each worker performs its job task content with the same productivity among firms in the same industry, *i.e.*, $\wp_h^i(a,s) \equiv \wp_{h'}^i(a,s)$ with $h' = \{1,\dots,h,\dots,H\} \in s$.

REMARK 2 (Max stability and workers' movement) *As from eq. (5), a worker has no incentive to choose a workplace where performing worse since*

$$\mathcal{U}_h^i(a,s) \mid \wp_h^i(a,s)_{max} \gg \mathcal{U}_h^i(a,s) \mid \forall\, \wp_h^i(a,s) \in \left[\wp_h^i(a,s)_{min}\,,\, \wp_h^i(a,s)_{max}\right)$$

is ensured by each worker's selection of its maximal productivity for the (h,s) tuple. In addition, since firms are symmetric within an industry, the sorting choice is uniquely driven by the dispersion of households' productivities between industries so that workers are free to move across firms within an industry while getting the same utility.

Interpreting eq. (10), both wage rise with non-ICT capital and the price level, and each wage is a function of the economy-wide wage for the corresponding task. Due to the production structure, the wage of non-routine workers is directly influenced by the ICT composite – the joint evolution of ICT capital and non-routine employment –, while routine wages are not. Task differences also manifest through the distinct roles of the two elasticities of substitution, ρ and σ: routine wages are primarily affected by substitution among themselves and non-routine tasks, whereas non-routine wages respond mainly to the elasticity governing the ICT composite, *i.e.*, the substitution between ICT capital and non-routine labour. Both elasticities further affect wages indirectly through total industry output, $\mathcal{V}(s)$. Consistently, an increase in the industry-specific amenity of a task, $\mathcal{B}(a,s)$, driven by substitution effects (ς) following a higher relative share of the other task, raises the wage of that task: for instance, a rise in $\mathcal{B}(rt,s)$ increases the ratio of non-routine to routine workers, thereby increasing routine wages. Finally, wages are non-decreasing in their aggregate task wage $\mathcal{W}(a,\mathcal{S})$, but rather increasing if $(\varrho,\varsigma) < 1$ (as estimated in Subsection 3.2), since $\left[\sum_s \mathcal{W}_{\mathcal{H}}(a,s)\right]^{-\theta([\varrho,\varsigma]-1)}$ with $-\theta\big([\varrho,\varsigma]-1\big) > 0$.

Cross-sectional differences in ρ and σ reflect not only divergences across comparable tasks but, importantly, differences across industries. In fact, if the distribution of capital and worker types were identical across industries, wage premia would be the same. This highlights that industry-specific characteristics, particularly their production plans and the degree of complementarity or substitutability among factors of production, play a central role in explaining real wage differentials.

Finally, parameter θ is common across all industries, capturing the strength of sorting and segregation forces and serving as a proxy for economy-wide labour market concentration (Figure 2). It represents a single structural distortion: as sorting and segregation intensify, the workforce composition within each industry becomes less diverse, generating a systematic allocation of tasks toward selected industries.

Closing the model, profit maximization by firm-h in industry-s not only displays the optimal amounts of each type-a's wage rates, but also a choice on the optimal quantities of both ICT and non-ICT capital stocks:

$$k_h(j,s) \;:\; p_h(s)\, f_{k_h(j,s)} = \mathcal{M}^{\epsilon}\, R(j) \tag{11}$$

where $\mathcal{M}^{\epsilon} = \frac{\epsilon}{\epsilon-1}$ is the (constant) price mark-up, $f_{k_h(j,s)}$ being marginal products of both capital types from the production function specified in eq. (9), and $R(j)$ the capital-specific rental rate, for $j = \{phy, ict\}$.

According to the structure of the model, equilibrium conditions read as follows.

(Equilibrium) *An equilibrium for this economy is defined as an households' choice of job place, a combination of factors' prices $\{w(a,s), R(phy), R(ict)\}$, and a set of aggregate quantities $\Omega = \{Y, K(phy), K(ict), L(rt), L(nrt)\}$, such that: (a) each household picks the firm-industry tuple that maximizes eq. (5); (b) according to the occupational choice, each household maximizes its expected-utility version of the utility in eq. (5); (c) final and sectoral good producers maximize their revenues; (d) given the availability of workers in each job task as in eq. (7), optimal wages are determined by the equilibrium of labour demand and supply; (e) firms choose also capital bundles to maximize their profits; and (f) all markets clear, shaping Ω.*

Proof in Appendix B.

3. From Theory to Data

In this section I quantitatively evaluate the model and bring it to the data. To gauge the performance and the consistency of the model in addressing salient features of the inter-industry wage structure of the United States economy, I split the universe of industries into three groups (following the "contribution" result – Fact n. 1 – of the data motivating section), namely the top 25%, the middle 50% and the bottom 25% of industries according to the overall growth in their real log-wage. The time window considered is the usual, spanning from 2003 to 2022.

TABLE 4: INDUSTRY WAGE AND RELATIVE TASK SIZE

	$log(w(s))$		
	(1)	(2)	(3)
$g(rt/nrt, s)$	.016*** (.006)	.001 (.007)	-.020*** (.002)
$g(nrt/rt, s)$	.129*** (.036)	.250*** (.084)	.318*** (.043)
Industry FE	✓	✓	✗
Time FE	✗	✓	✗

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). Standard error in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries in 2003-2022 on* $N = 1240$ *observations. The Fixed Effects (FE) regressions are of the form* $y(log(w_t(s)) \mid \mathcal{X}_{i,t}, \mathcal{Z}_{j,t}) = \beta_c + \beta_i \mathcal{X}_{i,t} + \delta_j \mathcal{Z}_{j,t} + u_t$, *with* $\mathcal{X}_i$ *being the regressors, and* $\mathcal{Z}_j$ *a set of controls. All series are in logs. Constant not reported to save space. Source: BEA, BLS and own calculations.*

3.1. VALIDATING THE MODEL IMPLICATIONS

As a first step to validation, I quantify the relevance of the labour force composition on the industry-specific wage level. Beside the usual role of capital and labour types in both $\mathcal{V}(s)$ and $\mathcal{Q}(s)$, the industry-specific wage level in eq. (10) considers a sizeable role of the industry benefit of each task, namely $\mathcal{B}(a, s)\ \forall s$, which points directly to the substitution among job tasks at the industry layer. To offer a better understanding, I now quantify its role by providing reduced-form evidence on how it impacts industry wages.[24] A set of panel Fixed Effects (FE) regressions

$$y\Big(log(w_t(s)) \mid \mathcal{X}_{i,t}, \mathcal{Z}_{j,t}\Big) = \beta_c + \beta_i \mathcal{X}_{i,t} + \delta_{j,t} \mathcal{Z}_{j,t} + u_t$$

where $\mathcal{X}_i$ comprises the regressors – task ratio and its inverse, $g(\cdot)$ – and $\mathcal{Z}_j$ is a set of employment-related controls, is performed. Results are presented in Table 4. Column 1 presents the baseline estimation with industry FE only. A positive effect of increases in both ratios is observed, with the non-routine-to-routine worker ratio exerting a stronger role on industry real log-wages, consistent with stylized Fact 4 and the theoretical wage-setting equations. Column 3 examines cross-sectional variation by omitting industry FE. In this specification, an increase in the routine-to-non-routine ratio reduces wages, contrasting with the industry-aggregated prediction of equation (10). Nonetheless, the positive impact of the relative non-routine task ratio remains robust and quantitatively substantial compared with the baseline.

Employment and relative wages.– Another central hypothesis of the model concerns the labour market. Since the employment measure of a task in a given industry is defined as in equation (7), linking task-level employment to relative nominal task wages, the same regression procedure used for the wage equation can be applied. Results, reported in Table C.1, show strong empirical support: employment

[24] As a reminder, the model distinguishes between two types of workers, $a = \{rt, nrt\}$, so that $\mathcal{B}(rt,s) = \left[\ell(rt,s)/\ell(nrt,s)\right]^{-\varsigma}$ and $\mathcal{B}(nrt,s) = \left[\ell(nrt,s)/\ell(rt,s)\right]^{-\varsigma}$, where fractions are identified by $g(\cdot, s)$.

in both routine and non-routine tasks rises with the associated relative wage. The effect is substantial for routine workers and even larger for non-routine workers, indicating that industries offering higher wages for a given task relative to the rest of the economy attract more workers to that task.

3.2. Calibration

The vector of structural parameters to be calibrated is $\Theta = \left(\alpha_s,\, \epsilon,\, \lambda_s,\, \mu_s,\, \theta,\, \rho_s,\, \sigma_s\right)_{\forall s}$, and comprises features of both households and industries. A total of 17 parameters is computed, and the strategy can be summarized in three steps: *(i)* some parameters are calibrated directly from the data, and some are externally taken; *(ii)* the elasticities of substitution are directly and individually estimated via panel robust regressions at industry level; finally, *(iii)* other parameters are internally estimated by moment-matching of key features of the U.S. economy.

Data and external calibration.– The elasticity of substitution among different varieties (ϵ) is externally set to 6 (so to have an annual mark-up of 20%). Differently, the production function's share of non-ICT capital (α) is directly taken by manipulating data from BEA: as in Arvai and Mann (2022), for each group of industries, I compute one minus the share of labour in total output, adjusted by considering the weight of non-ICT capital in the total capital stock. This procedure reports $\alpha = \left\{0.263, 0.195, 0.514\right\}$, order from bottom to top groups.[25]

Elasticities of substitution.– The key model parameters, namely $\left(\rho_s, \sigma_s\right)_{\forall s}$, are estimated through the general equilibrium propoerties of the model as in Karabarbounis and Neiman (2014). For each industry group, labour, capital, and profits are expressed as income shares of output and blended with the industry-side F.O.C.s in eq. (11) to exploit the evolution of the capital rental rate (eq. 6). This allows the resulting condition to be expressed in differences between two periods. Using a linear approximation around a zero over-time trend, this approach yields two separate estimating equations that both relate the industry-level labour shares, $s_\ell\left(a, s\right)$ and $s_\ell\left(s\right)$, to technological capital:

$$\frac{s_{\ell,t}\left(nrt, s\right)}{1 - s_{\ell,t}\left(nrt, s\right)} \widehat{s}_\ell\left(nrt, s\right) = \beta_c + \left(\rho - 1\right) \widehat{\zeta}\left(s\right) + u_t \tag{12}$$

and

$$\frac{s_{\ell,t}\left(s\right)}{1 - s_{\ell,t}\left(s\right)} \widehat{s}_\ell\left(s\right) = \beta_c + \left(\sigma - 1\right) \widehat{\zeta}\left(s\right) + \beta_k \left(\widehat{\frac{\ell\left(nrt, s\right)}{k\left(ict, s\right)}}\right) + u_t \tag{13}$$

where *hatted* variables identify their own percent change between arbitrary t and $t-1$ periods, and ζ is the industry-related quantity of ICT capital relative to physical capital. The first equation identifies the elasticity of substitution between ICT capital and non-routine labor (ρ) by treating the ICT composite as a nested inter-

[25] Estimates are consistent with the argument against the usual "alpha equal one-third" rule – that the elasticity of output with respect to capital (*i.e.*, the capital share of output in a neoclassical production function) is 0.33 – by Vollrath (2024) for the U.S. economy in 1948-2018.

mediate input. By contrast, equation (13) estimates the elasticity of substitution between routine and non-routine workers (σ), derived from the full production structure in eq. (9). Importantly, the estimated coefficients from these regressions do not correspond to $\beta_{\zeta}^{(\rho,\sigma)} = \left([\rho,\sigma] - 1\right)$; instead, the specification directly identifies the elasticity parameters (ρ,σ), since the left-hand-side variable (the labour share trend) is augmented by the trend in the relative ICT capital stock, $\widehat{\zeta}\ (s)$.

The intuition behind the two equations is straightforward. A negative relationship between trends in labour shares and trends in relative capital quantities arises only when the estimated elasticities imply sufficient complementarity, that is, when $(\rho,\sigma) < 1$. Abstracting from other possible economic factors (such as output productivity, capital-or labour-augmenting technology, profit share's or mark-up's growth), an increase in the relative stock of ICT capital then leads to a consequential decline in the labour share of each occupational group.[26] Estimated overall correlations between trends in both labour shares and relative ICT stock among 3-digit U.S. 2017 NAICS industries are $corr_{\rho} = -0.76$ and $corr_{\sigma} = -0.75$ between years 2003 and 2022.

Given the relatively short time dimension, the estimation may be sensitive to outliers. A robust regression procedure is therefore employed, iteratively downweighting observations that lie far from the fitted regression line.[27] The resulting estimates of equations (12) and (13) are reported in Table 5. A key finding is that all industry groups except the middle one exhibit complementarity between ICT capital and non-routine tasks, as indicated by $\sigma_s > \rho_s$ for any $s = \{bot, top\}$. This pattern reflects a broadly similar adoption of technological change across industries, coupled with heterogeneous effects on labour force composition, as the magnitude of the (ρ,σ) gap varies across groups. Consistent with the evidence in Section 1, industries at the top of the distribution benefit the most from task reallocation driven by technological change. All estimated elasticities are statistically significant.[28]

Interpreting ρ, top industries exhibit the strongest complementarity between ICT capital and non-routine workers, followed by bottom and then middle industries. Values of $\rho \approx 1$ indicate greater gross substitutability, implying weaker complementarity. Accordingly, top industries sustain larger stocks of ICT capital while employing relatively more non-routine workers than other industry groups. Turning to σ, the

[26] This framework differs from Karabarbounis and Neiman (2014), who require elasticities greater than one so that a decline in the user cost of capital – driven by falling investment prices – reduces the labour share through substitution away from labour. Here, quantities rather than prices are used, as capital stocks are directly observable at the industry level in BEA data.

[27] Following Karabarbounis and Neiman (ibid.), this approach mitigates potential violations of OLS assumptions arising from small samples by reducing the influence of high-residual observations. At each iteration, observations with Cook's distance greater than one – indicative of high leverage – are removed, so that the weight of each observation is no longer uniform at $\frac{1}{n}$ in a data sample with n observations.

[28] Estimating the elasticity of substitution between capital and labour is inherently challenging, as it reflects both demand- and supply-side forces, and there is no consensus on its precise value. Existing aggregate and sectoral estimates range from 0.3 to 0.9 (Arrow et al. 1961, Klump et al. 2007, Herrendorf, Herrington, et al. 2015, Alvarez-Cuadrado et al. 2018, Oberfield and Raval 2021), from unity (Berndt 1976), to between 1.25 and 1.6 (Piketty 2013, Karabarbounis and Neiman 2014). Moreover, as emphasized by Oberfield and Raval (2021), it remains unclear whether time-series or micro-level data are more appropriate for identification.

TABLE 5: BASELINE ESTIMATION

	ρ	*Std.Err.*	*95% CI*	σ	*Std.Err.*	*95% CI*
bottom	.329	.04	[.250, .407]	.634	.08	[.482, .785]
middle	.420	.02	[.375, .466]	.400	.05	[.310, .491]
top	.249	.03	[.188, .310]	.766	.06	[.656, .877]

Estimation of the elasticities of substitution as given in eqs. (12) and (13), for 3-digit U.S. 2017 NAICS industries over the period 2003-2022. ρ refers to the estimate of the pair $(\ell\,(nrt, s)\,;\,k\,(ict, s)\,)$, and it exploits the degree of substitutability between non-routine workers and ICT capital; σ relates to the pair $(\ell\,(rt, s)\,;\,[\ell\,(nrt, s)\,,\,k\,(ict, s)]\,)$, and it is the degree of substitutability between routine workers and the joint combination of non-routine workers and ICT capital.

elasticity of substitution between routine workers and the ICT composite (or, by symmetry, between routine and non-routine tasks), values closer to one imply a lower propensity to retain routine workers as ICT capital and non-routine employment expand. This pattern is precisely what emerges from Table 5: top industries display the lowest tendency to employ routine workers at rates comparable to non-routine workers, followed by bottom industries, with middle industries showing the weakest response. Together, the estimated elasticities point to substantial heterogeneity in the degree of structural transformation across industries, implying uneven reallocation of economic activity toward selected sectors. This reallocation is driven by differences in industry-level prices and quantities of capital-labour combinations, as reflected in the observed trends in labour shares. Finally, the estimates align with the wage-setting mechanism in eq. (10): industries characterized by lower ρ and higher σ are those that command higher wage premia.

Matching moments.– To conclude the calibration, the remaining parameters are estimated by matching salient moments in the data using the Method of Simulated Moments (MSM), discussed in Mc Fadden (1989). These parameters include the industry-group-specific weights $(\mu_s, \lambda_s)_{\forall s}$ in the production function, as well as θ, which governs the dispersion of households' idiosyncratic productivities and, in turn, the strength of sorting and segregation effects. Given the parameters estimated thus far, the calibration leaves seven parameters to be identified by matching seven moments. Each λ_s is pinned down by the ICT capital share of the corresponding industry group, while each μ_s is identified using the group-specific share of routine workers, exploiting its theoretical definition in eq. (7). Finally, since θ captures productivity dispersion and the resulting degree of labour market polarization, it is calibrated to match the real log-wage premium of routine workers in the top industry group relative to the bottom group.[29]

To match key moments of the production structure and households' productivities specification, the method estimates the vector of parameters $\widetilde{\Theta} = \{\lambda_s\,,\,\mu_s\,,\,\theta\}$ for $s = \{bot, mid, top\}$, by minimizing the *loss function*

[29] In the model, heterogeneity in household productivity is intrinsically linked to sorting and segregation choices. Moreover, the dynamics of labour market concentration in aggregate employment are closely mirrored by those of routine workers at the 3-digit U.S. 2017 NAICS industry level; see Figure 4.

TABLE 6: SUMMARY OF CALIBRATION

	parameter	***value***				***source***
		bottom	*middle*	*top*	*global*	
α	*physical capital, share of* $y(s)$	0.263	0.195	0.514		*data*
ϵ	*demand elasticity across firms*				6	*external*
μ	*weight of routine workers in* $y(s)$	0.676	0.495	0.337		*MSM*
λ	*ICT capital share in* $Q(s)$	0.457	0.465	0.451		*MSM*
θ	*households' productivities dispersion*				11.3	*MSM*
ρ	*EoS, ICT capital and non-routine*	0.329	0.420	0.249		*estimation*
σ	*EoS, routine and ICT composite*	0.634	0.400	0.766		*estimation*

Set of estimated parameters of the model. "data" implies that the values are directly computed from data sources, while in "external" I choose standard calibrated values from the literature. "MSM" refers to the Methods of Simulated Moments as in Mc Fadden (1989). "estimation" refers to previously estimated values under a specific procedure; these values are taken from Table 5.

$$\mathcal{L}\left(\widetilde{\Theta}\right) = \left(\widehat{m}(\widetilde{\Theta}) - \widetilde{m}\right)' \mathbf{W}\left(\widehat{m}(\widetilde{\Theta}) - \widetilde{m}\right)$$

to reckon $\widetilde{\Theta}$ that is, to minimize the distance between the estimated moments $\widehat{m}(\widetilde{\Theta})$ and the data moments counterpart, $\widetilde{m}$. Element $\mathbf{W}$ is an *efficient weighting* matrix that allows to implement an efficient estimator.[30] Table C.2 summarizes the resulting values, and compares the targeted data moments (expressed as mean values throughout the period 2003-2022) with that in the model.

Of particular interest is the estimate of θ. As discussed, it can be interpreted as a measure of labour market concentration arising from sorting and segregation mechanisms operating through the dispersion of households' productivities, and thus as the (average) elasticity of labour supply faced by firms at the industry level. Since $\theta > 1$, a higher value indicates less heterogeneity in individual productivity and, consequently, stronger incentives for workers to sort into specific industries rather than move across them. Put differently, the farther θ lies from unity, the more polarized the labour market becomes in terms of workforce characteristics, reflecting stronger sorting and segregation forces. The estimated value, $\theta = 11.3$, points to a highly concentrated labour market (as in Figure 2), consistent with recent evidence for the U.S. labour market in Yeh et al. (2022).[31]

Comparative discussion of the calibration.– A summary of the parametrization of the model is in Table 6. The calibration highlights that top industries – those experiencing the largest increases in real wages over the sample period –

[30] Practically, I first set it to be an identity matrix whose diagonal elements are the mean values of each moment in the data, to then update these values using the estimated vector valued function with the distance between data and simulated moments.

[31] Authors document how concentration – due to employer market power, referred to as aggregate markdown (ratio between wages and marginal revenue product of labour) –, decreased between 1977 and 2002, but thereafter it has been sharply increased. Such turn is well captured even at industry level (refer to Figure 4).

TABLE 7: IMPLIED FIT OF WAGE VARIANCES

moment	***data***	***model***
routine wage variance, across industries	2.285	2.289
non-routine wage variance, across industries	2.314	2.311
between-industry wage variance	2.299	2.300

Comparison of the model fitting of the data for moments related to 3-digit U.S. 2017 NAICS between-industry real log-wage variance structure over the period 2003-2022; variances are computed according to eq. (1). The first two rows compute this measure for routine and non-routine tasks, while the last row directly reports the wage dispersion across industries.

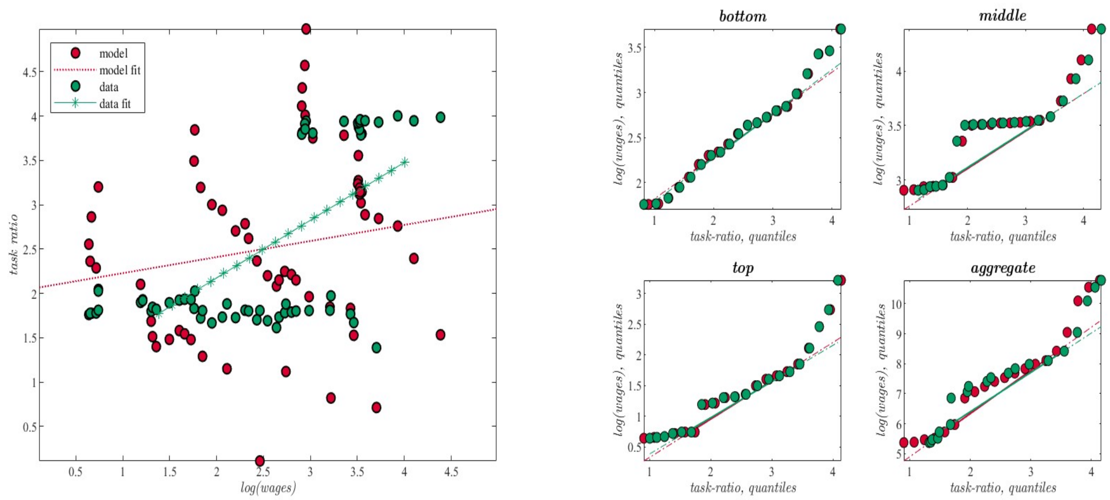
are characterized by the strongest complementarity between ICT capital and non-routine labour (lowest ρ) and, simultaneously, by a greater propensity to substitute routine workers with non-routine tasks (highest σ). By contrast, bottom industries also display meaningful ICT-non-routine complementarity, but this is accompanied by weaker substitutability between routine and non-routine tasks, limiting the re-allocation away from routine labour. These results closely align with the stylized facts documented in Section 1. While the non-routine-to-routine task ratio evolves primarily within industries (Fact 3), real wage growth is muted unless this shift is accompanied by an increase in the ICT-to-non-ICT capital ratio (Fact 4).

Moreover, top industries feature a substantial share of physical capital (α) in their production structure, alongside the lowest weight assigned to routine labor (μ). The prominent role of physical capital may partly reflect automation forces, whose effects differ across industry groups. For instance, bottom industries display an intermediate reliance on non-ICT capital but assign the largest weight to routine labour. Such heterogeneity suggests that automation operates unevenly across sectors.[32] A further latent indicator of automation emerges from the estimated ICT-capital weights in the ICT composite (λ), which indicate that industries with the strongest real log-wage growth place relatively less weight on ICT capital within the composite. Overall, the MSM calibration is internally consistent and mirrors the estimated elasticities of substitution across industry groups.

3.3. MODEL FIT

The primary scope of the model is to detect the industry's determinants that account for between-industry wage inequality, moment not directly targeted in the MSM parameters' estimation. Table 7 reports the fitting of real log-wage variance across industries for routine and non-routine workers, as well as that for industry-specific wage rates. Tasks' specific variances are computed according to eq. (1) given the analytical wage series sparking from eq. (10), while the between-industry wage variance is found by averaging these two dispersion indicators. Given the calibrated values

[32] Automation (industrial robots and artificial intelligence) tends to displace routine tasks while increasing the employment share of non-routine workers. At the industry level, however, the evidence on employment growth versus displacement remains mixed and context-dependent; see Filippi et al. (2023) for a detailed review. In this model, automation is therefore captured only indirectly through changes in $(\alpha_s, \mu_s, \lambda_s)_{\forall s}$.



(A) TASK RATIO VS. WAGES

(B) QUANTILES

FIGURE 3: MODEL AND DATA COMPARISON

Note: the figure shows the comparison between the model series and the related series in the data. Panel 3a represents the correlation in the baseline equilibrium between the task ratio (non-routine over routine tasks) with the empirical and model-implied real log-wage level, and the corresponding lines show the linear fit of the correlation; Panel 3b plots the parametric curves associated to quantiles of the considered HP-filtered series (model vs. data) one against each other. Series are scaled to be in the same range for graphical comparison. All plots consider industries to be grouped in terms of bottom, middle, and top industries' groups. Red circles refer to the model, while red ones to the data.

of parameters, the model built has the virtue of well targeting all the considered measures of inequality on real log-wages.

Another central hypothesis to validate in the cross-section is the model's ability to capture an increase in industry real log-wage following a rise in the task ratio, as discussed in Subsection 3.1. To this end, Panel 3a plots the correlation between the share of non-routine workers over the mass of routine workers and the industry wage as both in the data and in the model. Imposed model structure is able to successfully replicate the upward-sloping correlation implied by the data.

Finally, the validity of the calibration strategy can be assessed by checking how the model performs in addressing some other data moments, in particular those not targeted throughout the estimation procedure. I pick all the moments related to the wage structure in the economy, and evaluate the fitting also at the industry-group levels. Table C.3, Appendix C, reports the outcome. The direction of the untargeted moments in the data is captured by the model. An important observation is that almost all the appraised wage-moments have negative slope, thus linking my model to the secular decline in real wages (*e.g.*, Massenkoff and Wilmers 2023). The only increasing moments are the "top/bottom wage ratio" and the "aggregate task premium" (reflecting the wage premium of non-routine over routine tasks), making the model suitable for studying the evolution of this wage gap.

To conclude, the model performs well in replicating both targeted and untargeted moments related to the wage structure of the U.S. economy over the period 2003-2022: fitting is detected either at the aggregate and industry levels, and also for differences among routine and non-routine workers.

4. Counterfactual Analysis

The purpose of the following analysis is to understand whether heterogeneity in the industrial composition of the U.S. economy contributes to observed trends in wage inequality. To do so, a counterfactual analysis evaluates the role of variations in structural parameters on between-industry wage inequality. Specifically, the model's parameters are fully re-estimated over two equal periods, and the associated moments are computed by sequentially "switching on" changes in one or more parameters while holding the others fixed.[33] The primary goal is to identify the key drivers of wage variance across industries and quantify the share of observed inequality explained by the "structural" or by the "quantity" effects.

In the set of counterfactual exercises the core parameters are three: the elasticity of substitution between ICT capital and non-routine labour, namely ρ; the elasticity of substitution between routine and non-routine labour, namely σ; and the household productivity dispersion, namely θ, which captures sorting, segregation, and overall labour market concentration. Their contributions are evaluated both individually and jointly. Variations in industry-specific elasticities , $(\rho_s, \sigma_s)_{\forall s}$, isolate the role of structural transformations across industries, independent of changes in input quantities. Changes in θ reflect the degree of labour market concentration: with imperfectly elastic labour supplies, higher value implies stronger sorting and segregation tuple, making labour supply more industry-specific.

4.1. Selecting the parameters

Table 8 summarizes the evolution of the two substitution elasticities (ρ, σ) across two periods, 2003-2012 and 2013-2022. Top industries display a pronounced change in only one elasticity: the complementarity between ICT capital and non-routine workers declines, while the "gross substitution" between routine and non-routine workers shows only a modest increase. In the middle group, non-routine workers strengthen their complementarity with ICT capital, accompanied by a slight reduction in complementarity with routine workers. The bottom group experiences a simultaneous drop in both the substitutability across job types and the complementarity between ICT capital and non-routine labour. These patterns highlight heterogeneous structural adjustments across industries over time.

Figure 4 provides intuition on the evolution of labour market concentration by plotting the Herfindahl-Hirschman Index (HHI) for both worker types and aggregate employment.[34] Since θ is estimated via MSM, it reflects not only a cross-sectional measure but also the variation of sorting and segregation: higher θ implies that similar workers cluster more tightly within industries. A clear pattern emerges: labour market concentration rises steadily in the first half of the sample, then flattens in

[33] Tables 9 and C.8 report the results for overall between-industry wage inequality. In Appendix C, the same procedure is applied separately to routine (Table C.9) and non-routine (Table C.10) workers.

[34] Further details are provided in Appendix C.

TABLE 8: TIME-VARYING ESTIMATION

	2003-2012		2013-2022	
	ρ	σ	ρ	σ
bottom	.355 [.12]	.366 [.12]	.819 [.05]	.326 [.06]
middle	.431 [.04]	.429 [.08]	.345 [.04]	.438 [.09]
top	.408 [.04]	.367 [.12]	.508 [.06]	.358 [.05]

Estimation of the elasticities of substitution as given in eqs. (12) and (13) over different time span (2003-2012 and 2013-2022), in absolute values, for 3-digit U.S. 2017 NAICS industries. Standard errors in parenthesis, $[\cdot]$, and 95% confidence interval significant but not reported. ρ refers to the estimate of the pair $(\ell(nrt,s)\,;\,k(ict,s))$, and it exploits the degree of substitutability between non-routine workers and ICT capital; σ relates to the pair $(\ell(rt,s)\,;\,[\ell(nrt,s),\,k(ict,s)])$, and it is the degree of substitutability between routine workers and the joint combination of non-routine workers and ICT capital.

the second half (green line), driven by declining concentration among non-routine workers (blue line). Re-estimation over the two sub-periods yields $\theta_{2003\text{-}2012} = 7.26$ and $\theta_{2013\text{-}2022} = 7.79$, indicating that variation in industry-level worker concentration was stronger in the first period and more moderate in the second, yet overall concentration remains high.[35] Hence, tracking concentration of routine tasks suffices to capture the economy-wide pattern, consistent with using the top/bottom industry wage ratio for routine jobs to calibrate θ. Together, heterogeneous changes in (ρ, σ), coupled with variations in θ, effectively capture the reallocation of capital and labour types across industries, reflecting both structural transformations and rising labour market concentration for routine and non-routine tasks.

Key is to assess the role of differential effect of structural changes. As a preview of the results, heterogeneous shifts in both elasticities matter: changes in ρ and σ individually explain a substantial portion of real log-wage variance, but it is their joint evolution that accounts for the largest share of U.S. wage inequality. In addition, higher labour market concentration – reflected in stronger sorting and segregation effects – amplifies the inequality captured by these structural differences.

4.2. INSPECTING THE MECHANISM: MODEL RE-CALIBRATION

The core exercise consists in re-estimating the full set of model parameters over two equally sized subperiods (2003-2012 and 2013-2022) and quantifying the effects of allowing one – or a subset – of parameters to change at a time. The new estimates are reported in Table C.6. This approach provides a direct bridge between data-driven shifts in structural transformations across U.S. industries and the observed evolution of wage inequality, as disciplined by the model. The objective is to isolate the contribution of each structural parameter by computing counterfactual

[35] Recall that $\theta > 1$: as it approaches to 1, labour market concentration decreases. Hence, the two estimates are consistent both with the theory and the figure: the concentration of workers at industry level is higher in the second half ($\theta_{2013\text{-}2022}$) than in the first one ($\theta_{2003\text{-}2012}$).

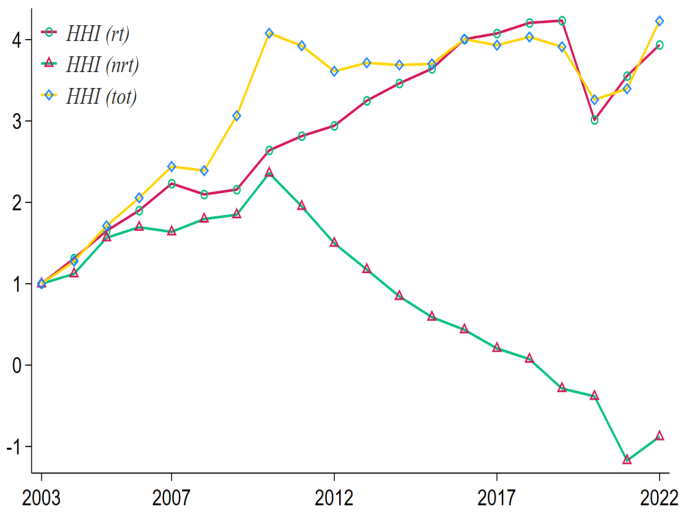


FIGURE 4: LABOUR MARKET CONCENTRATION AS A STRUCTURAL CHANGE

Note: the figure represents the evolution in labour market concentration as measured by the Herfindahl-Hirschman Index (HHI) – on the y-axis – by routine and non-routine (red and green lines, respectively) job tasks, and by total employment (yellow line). Series are normalized relative to their initial value in 2003, set to 1. *Source*: BLS and own calculations.

wage variance levels in which only a given parameter moves from its first-period to its second-period value, while all remaining parameters are held fixed at their initial levels. Formally, given the re-estimated parameters, consider the following model:

$$\Delta var\left(w(s)-\overline{w}\right)=f\left(\Phi_s\left(x,\tau_1\right),\Theta_s(p,\tau_1),\Phi_s\left(x,\tau_2\right)\mid\Delta\Theta_s\left(p_{\tau_2}^{\{\rho,\sigma,\theta\}},-p_{\tau_1}\right)\right)\quad\text{(Model A)}$$

where the change in between-industry wage variance is a function of the input factor series in both periods, $\Phi_s\left(x,\tau_1\right)$ and $\Phi_s\left(x,\tau_2\right)$, the whole set of parameters in period one, $\Theta_s(p,\tau_1)$, and a subset of parameters in the second period keeping fixed the others, $\Theta_s\left(p_{\tau_2},-p_{\tau_1}\right)$, for any industry-$s$. Practically, given the change in capital and labour types, wage levels and variances are estimated under two set of parameters: *(i)* full set related to period one, and *(ii)* full set related to period one with one (or a combination) related to its period two level; these parameters are those selected in Subsection 4.1.

Results are reported in Table 9. Column 2 displays the between-industry variance in the second period (τ_2) for both the data and the model specification described in column 1. To assess the contribution of individual structural parameters, column 3 reports the model-implied wage variance obtained by holding all parameters at their first-period values, except for the parameter(s) of interest, which are set at their second-period levels. The resulting counterfactuals highlight the central role of industry-heterogeneous changes in the substitutability of production factors in shaping U.S. between-industry wage inequality. In particular, the joint evolution of the two substitution elasticities – ρ, governing complementarity between ICT capital and non-routine labour, and σ, governing substitution between routine and non-routine workers – emerges as pivotal. Isolated changes in ρ alone would have generated a non-negligible increase in wage dispersion (.88), while changes in σ alone would

TABLE 9: MODEL COUNTERFACTUAL, CHANGE

industry wages	$var(w)_{\tau_2}$	Δ ***model*** $\mid \Delta\, m\Big(\Phi(x,\tau_2), \Theta = \{p_{\tau_2}, -p_{\tau_1}\}\Big)$ *level*	*share, model*	*share, data*
DATA	1.18			
MODEL	1.09			
$\Delta\sigma$		2.37	2.18	1.99
$\Delta\rho$		.88	.81	.74
$\Delta(\sigma,\rho)$		1.12	1.03	.94
$\Delta\theta$		1.34	1.23	1.13
$\Delta(\sigma,\rho,\theta)$		1.16	1.07	.98

Quantification of Model A. Model implied between-industry real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by the change in a specified parameter, $\frac{m\left[\Phi(x,\tau_2)\,\mid\,\Theta(p,\tau_2),\Theta(-p,\tau_1)\right]}{m\left[\Phi(x,\tau_2)\,\mid\,\Theta(p,\tau_2)\right]}$*, where* $\Phi(x,\tau_2)$ *identifies the series in the second period, and* Θ *the set of parameters where some of them,* (p,τ_2)*, are taken in the second period, while* $(-p,\tau_1)$ *reflects the set of all the parameters in the first period but those considered in the second period. Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

have implied an even larger surge in inequality (2.37). Strikingly, when both elasticities are allowed to change simultaneously, their opposing magnitudes partially offset each other, yielding an implied between-industry real log-wage variance (1.12) that closely aligns with both the full model calibration (1.09) and the observed data (1.18). This evidence underscores that it is the coordinated – and uneven – restructuring of task substitutability across industries, rather than movements in any single elasticity in isolation, that accounts for the observed evolution of wage inequality.

Column 5 quantifies the contribution of each parameter to the observed rise in U.S. wage inequality by reporting the share of the empirical real log-wage variance explained by the variance implied by parameter shifts in the model. Isolated changes in ρ account for about 74% of the observed between-industry wage dispersion, while heterogeneous and joint shifts in both elasticities explain as much as 94% of the data-implied variance (corresponding to 103% of the variance generated by the fully calibrated model for the 2013-2022 period). These results indicate that large wage differentials across industries are driven primarily by industry-specific differences in the substitutability between routine and non-routine workers, rather than by variation in the substitutability between ICT capital and non-routine labour alone. Crucially, it is the interaction of these two structural dimensions – how industries reorganize tasks internally and how they combine labour with technology – that accounts for the bulk of the observed increase in wage inequality.

These conclusions naturally extend to the interpretation of changes in θ, which rises from $\theta_{2003\text{-}2012} = 7.26$ to $\theta_{2013\text{-}2022} = 7.79$. Taken in isolation, this increase – signaling stronger worker concentration through intensified sorting and segregation –

would substantially amplify real log-wage dispersion (1.34). However, once these sorting-segregation forces interact with the industry-heterogeneous shifts in substitution elasticities documented in Table 8, their effect becomes more nuanced: labour market concentration slightly sharpens the economy-wide level of between-industry wage inequality, raising the model-implied variance (1.16). Put differently, when industries differ in how they reorganize tasks and combine labour with technology, a more concentrated labour market magnifies existing wage gaps rather than creating them on its own. When variations in industry-specific structural characteristics – captured by joint changes in both substitution elasticities – are considered together with changes in labour market concentration, the combined component $\Delta(\sigma, \rho, \theta)$ accounts for nearly the entire observed increase in U.S. between-industry real *log*-wage variance, explaining about 98% of its data-implied level.

Sensitivity.– So far, the analysis has focused on changes in the three core parameters of the model, $(\rho_s, \sigma_s, \theta)_{\forall s}$, while holding fixed the production function parameters $(\alpha_s, \mu_s, \lambda_s)_{\forall s}$. The results are further strengthened by the complementary, reverse exercise. In this case, the counterfactual asks what the level of between-industry wage variance would have been had one parameter remained fixed at its initial (period-1) value, while all other parameters were allowed to evolve from period one to period two. Formally, this amounts to the following experiment:

$$\Delta var\Big(w(s) - \overline{w}\Big) = f\Big(\Phi_s\left(x, \tau_1\right), \Theta_s(p, \tau_1), \Phi_s\left(x, \tau_2\right) \mid \Delta\Theta_s\left(p_{\tau_2}, -p_{\tau_1}^{\{\rho,\sigma,\theta\}}\right)\Big) \quad \text{(Model B)}$$

whose results are presented in Table C.8.[36] Absent any change in σ, thereby assigning a first-order role to the joint evolution of ρ, θ, and the industry-specific production weights and income shares, the model would explain about 79% of the observed rise in wage inequality. By contrast, fixing both elasticities simultaneously, $\left(\Delta\Theta_{|(\sigma,\rho)}\right)$, leads the model to overpredict dispersion: the implied variance (1.31) exceeds that observed in both the data and the baseline model. A similar overstatement emerges when all structural parameters are held fixed, leaving only changes in income shares to operate. Conversely, holding constant the productivity dispersion parameter θ while allowing both substitution elasticities and production weights to vary captures most of the observed inequality. In line with the stylized facts in Section 1, industry-heterogeneous shifts in substitution elasticities combined with changes in factor weights alone explain roughly 88% of the data-implied between-industry wage variance (96% in the model). This result underscores that wage inequality is driven primarily by uneven structural transformation across industries, rather than by changes in labour market concentration per se.

Investigations in this section highlight a central feature of the U.S. wage struc-

[36] This counterfactual exercise differs from that in Table 9 by allowing production function weights to vary as well. Specifically, it admits industry-specific changes not only in the structural parameters $(\rho_s, \sigma_s)_{\forall s}$, but also in the production parameters $(\alpha_s, \mu_s, \lambda_s)_{\forall s}$, alongside shifts in households' productivity dispersion captured by θ. In doing so, the analysis jointly accounts for structural transformation, changes in factor intensities, and evolving labour market concentration, providing a more comprehensive assessment of the forces shaping between-industry wage inequality.

ture: structural heterogeneity across industries is the dominant source of wage inequality. When the model is disciplined by the data, most of the observed dispersion is traced back to industry-specific differentials in the elasticities of substitution between capital and labour types (94%, Table 9), and this share remains large when combined with heterogeneity in factor weights within production (88%, Table C.8). Conditional on these structural differences, the increase in labour market concentration through stronger sorting and segregation across industries acts as an amplifier of inequality. Once both mechanisms are allowed to interact, the model explains up to 98% of the observed between-industry variance in real log wages over the past two decades (Table 9).[37] Overall, the evidence points to uneven structural transformation across industries – rather than quantity-based technological change – as the key force shaping the evolution of U.S. wage inequality.

The role of labour market power.– Model specification allows also to inspect whether monopsony power by the part of firms plays a significant role in between-industry wage inequality. Labour market failures due to employer market power in wage-setting is a widely discussed channel (*e.g.*, Prager and Schmitt 2021, Dodini et al. 2021), though Card, Rothstein, et al. (2024a) show that, in many manufacturing industries, industry-level wage premiums are not positively correlated with wage markdowns. In Appendix D, the previous exercises are replicated under a version of the model where monopolistically competitive firms (h, s) optimally exploit monopsonistic power. In this setting, eq. (10) would display an average wage markdown over the marginal revenue product of labour, $\mathcal{M}^{\theta} = \frac{\theta}{1+\theta} \in (0, 1)$, which is an increasing function of the labour supply elasticity θ (*e.g.*, Card, Cardoso, Heining, et al. 2018). As illustrated in Figure 2, when $\theta \to 1$ household productivities are more dispersed, labour supply is more elastic, and monopsony power – and the associated wage markdown – declines. Results for $\Delta\theta$ and $\Delta(\sigma, \rho, \theta)$ under monopsony remain largely unchanged in both direction and magnitude relative to the baseline, indicating that changes in between-industry wage inequality are not primarily driven by variations in aggregate wage markdowns.

4.3. Skill-Biased Technological Change

Finally, the analysis turns to the systematic impact of Skill-Biased Technological Change (SBTC). The model can evaluate how simultaneous changes in capital-labour series affect between-industry wage inequality when the evolution of production technology parameters $(\alpha, \mu, \lambda, \rho, \sigma)$ and labour market concentration (θ) is held fixed. Given the re-estimation procedure, consider the following counterfactual setup:

$$\Delta var\Big(w(s) - \overline{w}\Big) = f\Big(\Phi_s\left(x, \tau_1\right), \Delta\Phi_s\left(x_{\tau_2}, -x_{\tau_1}\right) \;\Big|\; \Theta_s\left(p, \tau_1\right)\Big) \qquad \text{(Model C)}$$

where the change in between-industry wage variance is a function of industry-

[37] These conclusions are robust when wage dispersion is examined separately for routine and non-routine tasks (Tables C.9 and C.10, Appendix C). Substitution elasticities exert a more balanced effect for routine workers, while sorting and segregation forces are relatively more important for non-routine workers.

specific factor series, $\Phi_s(x, \tau_1)$, and parameters, $\Theta_s(p, \tau_1)$, taken in period one, while considering one (or a combination of) series at the second period level while keeping fixed the others, $\Phi_s(x_{\tau_2}, -x_{\tau_1})$ to first period.

Table C.11 presents the results. Column (2) reports the between-industry variance in the second period (τ_2) for both the data and the model, while columns (3)-(5) show, for each series in column (1), the implied wage variance, its share relative to the "all-series" model, and the fraction of the observed variance explained. Differences in routine workers alone substantially inflate wage inequality (2.62), with ICT capital contributing to a lesser extent (1.69). Combinations of both labour types, and in combination with ICT capital, account for more than the actual real log-wage variance (112-113%). Compared to Table 9, these "changing-series" shares of data-driven wage variance are considerably higher than when also including substitution elasticities and sorting/segregation effects. This implies that the "quantity effect" of the SBTC, by itself, tends to overstate between-industry wage inequality. Changes in capital and labour series are necessary but not sufficient to fully explain observed real log-wage variance: the key drivers are the heterogeneous structural parameters, which vary unevenly across industries. The next section illustrates this more clearly by recomputing between-industry wage inequality while fully neutralizing structural differences, isolating the contribution of factor input variations alone.

5. Factor Quantities and Productivity Changes

As a concluding step in the analysis to disentangle the main channels underlying U.S. between-industry wage inequality, I examine the extent to which divergences in production inputs matter when industries are constrained to share the same structural parameters. Taking τ_0 as the initial year of the sample (2003),

$$var\Big(w(s) - \overline{w}\Big) = f\Big(\Delta\Phi_s(x), \Phi_s(-x, \tau_0) \;\Big|\; \Theta_{=\forall s}(p)\Big) \qquad \text{(Model D)}$$

Put differently, the exercise conjectures what the cross-industry variance in real log-wages would have been if only a selected set of inputs – capital types, labour types, or their combinations – had evolved over time, $\Delta\Phi_s(x)$, while all remaining inputs were held fixed at their initial levels, $\Phi_s(-x, \tau_0)$, and all industries were constrained to share common structural parameters, $\Theta_{=\forall s}(p)$. In the counterfactuals presented below, these parameters are set to their economy-wide means: $\mu = .503$, $\lambda =,458$, and $\theta = 11.3$, as estimated via the MSM procedure, together with a physical-capital income share of $\alpha = .324$ and a price-markup parameter fixed at $\epsilon = 6$. Treating all industries as a single group, the elasticities of substitution are re-estimated and set to $\rho = .333$ and $\sigma = .595$.[38]

Table 10 presents a model-based decomposition of Table 7, following the frame-

[38] These values drawn Figure C.1. Associated standard errors are $Std.Err._{(\rho)} = .02$ and $Std.Err._{(\sigma)} = .03$, with 95% confidence bands of $[.301, .366]_{(\rho)}$ and $[.545, .645]_{(\sigma)}$. Results in Table 10 are robust to re-estimating all parameters for the full economy (Table C.13), rather than imposing the mean calibration from Table 6.

TABLE 10: MODEL VS. DATA COUNTERFACTUAL, SERIES AND MEAN PARAMETERS

	data	**model** $\mid \Delta\,\Phi\,(x)$					$\Delta\,(tfp)$	
		$\Delta\ell\,(rt)$	$\Delta\ell\,(nrt)$	$\Delta\,(\ell)$	$\Delta k\,(ict)$	$\Delta\,(tech)$	*mean*	*baseline*
WAGES, VARIANCE								
routine	2.285	.35	.16	.22	.24	.15	.25	.22
non-routine	2.314	.32	.12	.17	.23	.13	.21	.23
industry	2.299	.33	.14	.19	.23	.14	.23	.23

Quantification of Model D. Changes in variances in real log-wages in the model induced by variations in one or more series, keeping fixed the others, and imposing the mean parameters to be homogeneous across industries. $\Delta\,(\ell)$ refers to joint variations in routine and non-routine series, $\Delta\,(tech)$ is associated to simultaneous changes in both ICT capital and non-routine workers, while changes in estimated industry-specific Hicks-neutral exogenous total factor productivity (TFP), given eq. (14), are captured by $\Delta\,(tfp)$; TFP series are taken under the same (mean) parameters' values, or given the baseline calibration (Table 3.2). In all the columns, the variation is computed throughout the period-by-period percentage differential thus identifying overall changes in empirical trends implied both by the data and the model; same values differ in terms of decimals.

work in Model D. The counterfactual analysis reveals that shifts in industry-specific routine workers explain the largest portion of the variation, while dynamics in non-routine employment, $\Delta\ell\,(nrt)$, contribute little – even when combined with changes in ICT capital, $\Delta\,(tech)$. Divergent ICT capital across industries has a non-negligible effect, but it remains quantitatively minor in terms of observed real log-wage variance. Overall, industry-level differences in ICT adoption and in the composition of routine and non-routine employment account for roughly 6-15% of observed wage inequality across industries and tasks.[39]

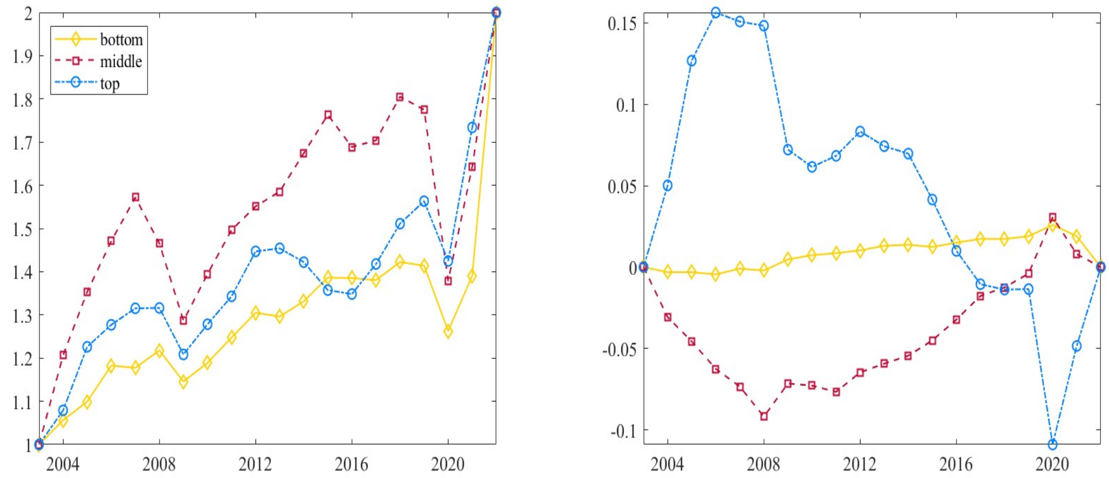
Productivity differentials.– So far all the analysis has been centred on excluding total factor productivity (TFP). Such patterns cannot be directly observed in the data, but sectoral series can be recovered from the theoretical model. In fact, Proposition 1 allows to write the industry-level counterpart of eq. (9) as

$$y\,(s) = z\,(s)\; f\Big(k\,(phy,s)\,,k\,(ict,s)\,,\ell\,(rt,s)\,,\ell\,(nrt,s)\;\Big|\;\Theta_s\Big)$$

where $z\,(s)$ is an exogenous measure of product-augmenting Hicks-neutral TFP for industry-s, with $k\,(j,s)$ and $\ell\,(a,s)$, for $j = \{phy, ict\}$ and $a = \{rt, nrt\}$, being its quantities of capital and labour types, and Θ_s identifying the vector of calibrated and estimated parameters.[40] Using industry-by-industry data on the annual, seasonally adjusted value added extracted from BEA to measure output, $y\,(s)$, it is possible to recover the year-by-year series for sectoral productivity from

[39] A similar exercise, weighting each capital and labour series according to heterogeneous structural parameters (Table C.7), produces comparable results: routine labour and ICT capital slightly dominate, explaining approximately 5-20% of the variance. Yet, even when all series are held constant over time, the model only marginally overestimates variance, indicating that structural parameters play a more important role in determining the overall level of wage inequality across industries and tasks.

[40] Neutrality in the sense of Hicks (1932) implies that the marginal rate of substitution between capital and labour inputs is not altered, and hence including productivity does not pose threads to the parameters' identification strategy of Subsection 3.2, which may be confounded by other forms of neutrality (labour-augmenting Harrod-neutrality, or capital-augmenting Solow-neutrality).



(A) UNIQUE CALIBRATION, MEAN

(B) DIFFERENCE WITH BASELINE

FIGURE 5: ESTIMATED PRODUCTIVITIES UNDER MEAN PARAMETERS

Note: the figure shows the estimated Hicks-neutral exogenous total factor productivity (TFP) measures estimated from the model. Panel 5a plots the series given a calibration where all the parameters are evenly set at the same *mean* values for all the industry-groups, while Panel 5b shows the difference between such series and the estimated TFP measures using the baseline calibration reported in Table 6. Series are scaled to be in the same range for graphical comparison.

$$log\ z_t\,(s) = log\ y_t\,(s) - \alpha_s\ log\ k_t(phy, s) - (1-\alpha_s)\ log\ v_t\,(s) \tag{14}$$

where $v_t\,(s)$ comprises the relation among ICT capital, routine and non-routine workers along the parameters governing their associated weights, the different elasticities of substitution, and the degree of labour market concentration.

Estimated series for $s = \{bot, mid, top\}$ are shown in Figure 5. Panel 5a reports the estimated series under uniform industry-level calibrated parameters, while Panel 5b shows the differences relative to the baseline calibration from Table 6. Across both estimations and all industry groups, TFP rises over time but exhibits clear drops during the 2008 Great Financial Crisis and the 2020 COVID-19 shock. Differences between the two TFP measures are minimal, though slightly more pronounced for top and middle industries. The right-hand side of Table 10 reports the magnitude of shifts in industry-specific productivity for both calibrations. As with factor quantities, isolated changes in TFP have little explanatory power for the observed cross-industry variance in real log wages.

Overall, this section highlights the contribution of industry-specific patterns in capital, labour, and productivity to wage inequality, holding all structural parameters constant. Notably, both individual and combined shifts in these series account for only 6-15% of the observed variance, reinforcing the conclusion from Section 4 that cross-industry wage inequality is predominantly driven by heterogeneous structural parameters, both across industries and over time.

Conclusions

Over the past decades, U.S. wage inequality has risen sharply, drawing renewed attention on its origins. Evidence increasingly points to differences across industries as a dominant force, with structural transformations serving as a key mechanism to interpret observed changes in U.S. labour market and in its wage structure. This paper formalizes this intuition within a general equilibrium framework, featuring heterogeneous substitution between capital and labour types and a labour market characterized by concentration, which amplifies sorting and segregation effects due to imperfect labour mobility.

A structural estimation of the model successfully replicates the U.S. wage structure across industries. Critically, inequality is largely driven by structural differences between industries: trend-heterogeneous substitution elasticities between routine and non-routine workers emerge as the main determinant of between-industry wage variance, whereas variation in ICT-non-routine substitutability plays a secondary role. Combined, these channels explain 94% of the observed real log-wage variance. By contrast, when differences in structural parameters are neutralized, only 6-15% of observed wage variance can be accounted for through changes in capital, labour inputs, or sectoral TFP alone. These findings strongly suggest that differences in structural parameters across industries serve as the predominant force governing the observed trend of wage inequality.

The model also illuminates broader labour market dynamics. Trends in sorting and segregation, captured through labour market concentration, have a marginal effect on wage dispersion once structural transformations are considered. Variations in substitution elasticities and labour concentration together explain nearly 98% of industry-specific wage inequality. When combined with heterogeneous weights of capital and labour inputs, the model accounts for 88% of the between-industry wage variance. These conclusions remain robust even when the analysis is disaggregated to examine wage inequality separately for routine and non-routine workers.

Overall, the analysis underscores that wage inequality is primarily shaped by the reallocation of economic activity across industries, driven by differences in the substitutability between routine and non-routine tasks. Shifts in capital-labour substitutability are secondary, suggesting that structural changes on the labour side – how tasks are organized and how workers are employed – are the main drivers of the secular rise in the U.S. wage inequality.

Two elements emerge as fundamental: *(i)* structural technology parameters, and *(ii)* the composition of capital and labour within industries. While both are essential for understanding wage dynamics, the persistent increase in inequality appears largely driven by an "indirect" effect of Skill Biased Technological Change – specifically, the evolving substitutability across tasks and the reallocation of employment within industries. Further explanations on what determines structural differences in technological parameters are called for.

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Appendix

(Outline) *In this appendix I report all the material complementary to the main text; it is made of additional tables and figures, and of discussions on further analytical results. Section A complements and enriches the exposition of the empirical findings in Section 1; Section B derives all the salient elements of the structural general equilibrium model outlined in Section 2, while Section C consolidates and further discusses the calibration strategy and the counterfactual analysis of the model of Sections 3-4. A replication is in Section D under both monopsony and monopolistically competitive power of firms.*

A. Motivating evidence: data, figures and tables

(Data details) *Data are for United States (US). For what concerns data from Bureau of Economic Analysis (BEA), private non-residential capital types are in net stocks evaluated at current costs by detailed industry dating back from 1925 to 2022.*[I] *Gross categories are "total equipment", "total structures", and "total intellectual property products", and each contains different types of assets according to their own National Income and Product Accounts (NIPA) asset type code; in total there are 96 different asset types. Data are provided for 74 industries at different layers.*[II] *Among all these types of capital assets I am going to classify them accordingly: digital equipment comprises all the electronic structures that are useful to process technological issues, that is "Mainframes", "PCs", "DASDs", "Printers", "Terminals", "Tape Drives", "Storage Devices", and "System Integrators"; the stock of intangible capital coincides with that of "Total Intellectual Property Products" (IPP);*[III] *while physical (or non-ICT) capital is the sum of all the remaining asset types. In addition to the aggregated capital types, I further define industry-specific ICT capital to be the combination of intangibles and digital equipment.*[IV]

BEA also collects data on wage levels of industries, together with their total employment. Herby, I extract also annual data on (seasonally adjusted) value added, wages and salaries, and of persons engaged in production (effective workforce) for

[I] Starting year is 1998 so that industries are classified with the latest available system, the U.S. 2017 NAICS; before, classification follows the U.S. Standard Industrial Classification (SIC) system. Assets' value is expressed in millions of U.S. dollars, and last update was in November 3, 2023.

[II] Classified according the BEA industry code system. Most are classified at 3-digit U.S. 2017 NAICS level (some 3-digit industries may fall in the same BEA code), four at 2-digit ("construction", "management of companies and enterprises", "educational services", and "other services, except government"), while five industries related to finance and insurance are at 4-digit level.

[III] A recent discussion analyses how the BEA collects data on IPP. Koh et al. (2020) argue that the effects of IPP capital on the U.S. labour share only emerged since 2013, when the BEA revised its methodology by re-classifying the notion of capital; now, the capital income comprises the rents arising from IPP investment. The authors compare the IPP effect on the labour share using both pre- and post-2013 data, finding out a negative effect of the rise in IPP on the U.S. labour share.

[IV] Similar classification in Arvai and Mann (2022), as well coherent with Eden and Gaggl (2018).

each industry. Complementing the analysis with real wages, data on annual (seasonally adjusted) Consumer Pirce Index (CPI) for all items (indexed at 2015 = 100) is taken from Federal Reserve Economic Data (FRED) database. Not exploiting BLS data throughout, the time window considered is 1998-2022. Both capital stocks and real wages are taken in industry per capita log-terms.

Production does not occur with capital only. A thorough examination of wage dispersion necessitates a careful consideration of the composition of the labour force within each industry. To this end, reference is made to the U.S. Bureau of Labor Statistics (BLS)' Occupational Employment and Wage Statistics (OEWS) programme, which provides comprehensive data covering over one hundred occupational categories. Regarding the data extracted from BLS, information are available for a total of more than 80 industries at 3-digit U.S. 2017 NAICS level starting from 2003 onwards.[V] *At workers' level, each industry displays information on a set of several and granular occupations classified according to the U.S. Office of Management and Budget's Standard Occupational Classification (SOC) system, i.e., occupations are categorized based on the type of job and on required skills, and employees are assigned to an occupation based on the work they perform and not on their education or training. However, since detailed occupations may vary across industries, for my purposes I classify occupations by considering their "major" group membership:*[VI] *I divide such major occupations in both routine and non-routine tasks. The latter group considers occupations such as "Management", "Business and Financial Operations", "Computer and Mathematical", "Architecture and Engineering", "Life, Physical, and Social Science", "Community and Social Service", "Legal", "Educational Training and Library", and "Arts, Design, Entertainment, Sports, and Media", while the left-outside occupations are comprised in the group of routine worker tasks.*

In the merging process (BEA and BLS), the 4-digit industries in the BEA data have been clustered into 3-digit category. The final dataset employed in the conducted analysis comprises information on a complete sample of 62 private-sector industries at the 3-digit U.S. 2017 NAICS system over an annual time window spanning from 2003 to 2022. This dataset forms the empirical foundation for the conduction of the main empirical analysis and subsequent investigation.

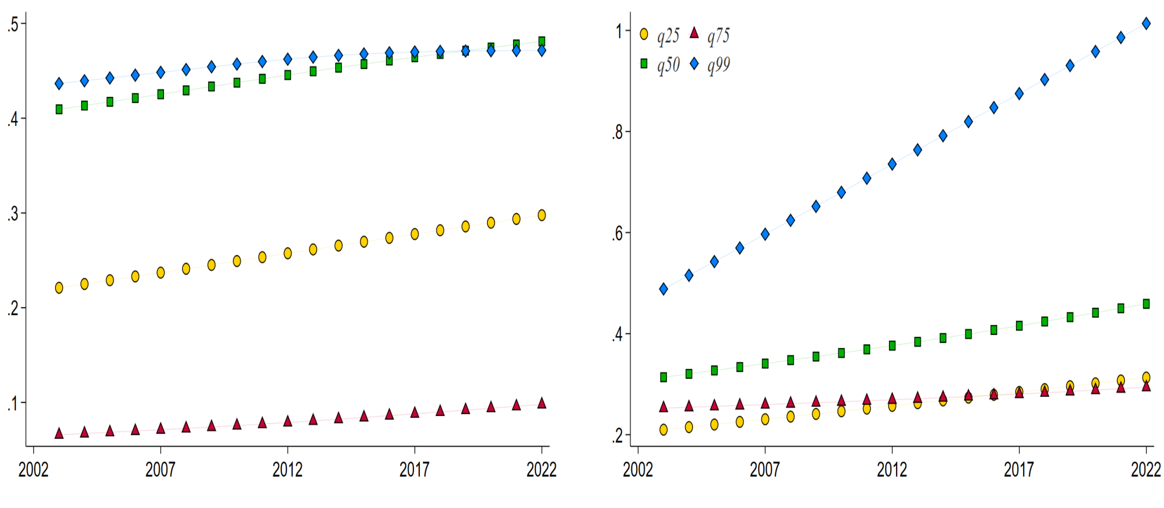
[V] The number of industries is variable year by year due to issues of data sampling. Moreover, prior 2003, these are classified according to the Standard Industrial Classification (SIC) system.

[VI] This choice arises from the fact that there are missing group definitions for some more detailed occupations. In fact, the number of occupations in each industry is not fixed: there are from 80 to 180 types of occupations for each industry, and these are clustered in common 22 major groups.

TABLE A.1: REGRESSIONS, CAPITAL STOCKS

	$log(w(s))$			
	(1)	(2)	(3)	(4)
$\beta_{k^{phy}}$	.134***	.122**	.164***	.114***
	(3.48)	(2.94)	(9.05)	(3.52)
$\beta_{k^{ict}}$	.052*			.049*
	(2.00)			(2.27)
$\beta_{k^{int}}$		.057*		
		(2.03)		
$\beta_{k^{deq}}$		-.003		
		(-.26)		
$\beta_{k^{int}} \times \beta_{k^{deq}}$			.009**	.009**
			(2.99)	(2.80)
R^2	.461	.491	.287	.497

*t-statistics in parentheses. * (p<0.05), ** (p<0.01), *** (p<0.001). Analysis at 3/4-digit U.S. 2017 NAICS industries over 1998-2022 on* $N = 1650$ *observations. Fixed Effects (FE) regressions are of the form* $y\left(\log\left(w_t(s)\right) \mid \mathcal{X}_{i,t}\right) = \beta_c + \beta_i \mathcal{X}_{i,t} + u_t$*, with* $\mathcal{X}_i$ *representing the different capital-labour ratios considered. Results are robust when controlling for the log size of industries, or when taking capital series directly in levels. Variables are in log format. Constant not reported to save space. Source: BEA and own calculations.*

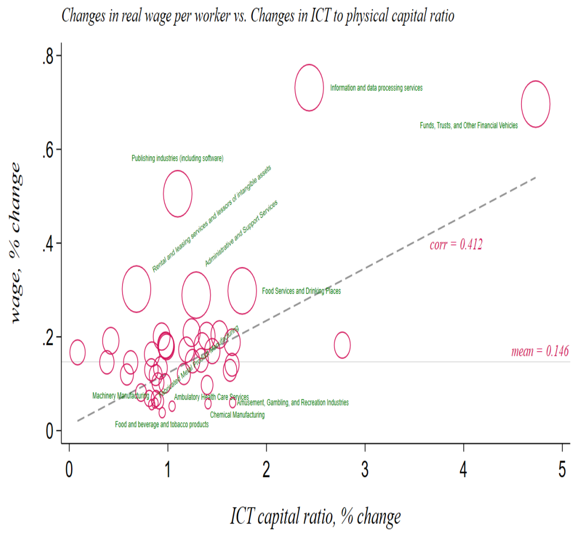


(A) ICT CAPITAL RATIO

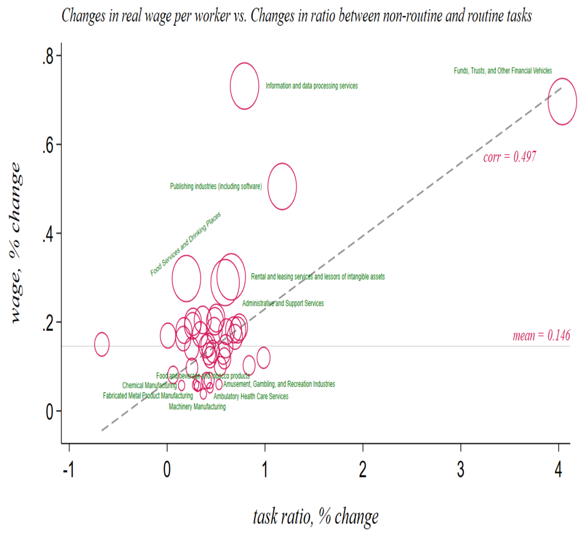
(B) TASK RATIO

FIGURE A.1: CHANGES BY PERCENTILES

Note: each subplot draws the HP-filtered trend in ICT capital ratio (ICT capital stock in physical capital quantity, Panel A.1a), and in task ratio (fraction of non-routine workers of routine ones, Panel A.1b). Series are divided according the growth in group-specific industry wage (*i.e.*, total $\Delta_{\%}$ in industry wage per worker). Industries are classified at 3-digit U.S. 2017 NAICS level. *Source*: BEA, BLS and own calculations.



(A) Real wages vs. ICT capital ratio



(B) Real wages vs. task ratio

Figure A.2: Industry correlations

Note: each subplot of the figure represents the correlation of overall percentage change in industry-specific real log-wage per capita with both ICT capital ratio (ICT over non-ICT capital) in Panel A.2a, and task ratio (non-routine over routine workers) in Panel A.2b. The oblique (dashed-grey) line represents the fitting curve with its associated degree of correlation, while the horizontal (solid-black) line identifies the mean value of all industries' overall percentage change in real log-wage per capita. For a better graphical visualization, the ICT ratio takes constant the initial level of physical capital. Each circle is referred to a specific industry, and I report the label only for more and less virtuous (top and bottom 10%) industries; circles' size captures different groups of industries, each expressing the group's position in the distribution of overall percentage change in their real log-wage. Plots are referred to 3-digit U.S. 2017 NAICS industries over the period 2003-2022. *Source*: BEA, BLS and own calculations.

Table A.2: Combined regressions, percentage changes

	$\Delta\, log(w(s))$		
	(1)	(2)	(3)
$\beta_{\Delta k}$	.054***		.009
	(.002)		(.008)
$\beta_{\Delta \ell}$	.035*		.035***
	(.018)		(.007)
$\beta_{\Delta k \times \Delta \ell}$		.982	.919***
		(.649)	(.152)
R^2	.030	.021	.037
N	1178	1178	1178

*t-statistics in parentheses. * (p<0.05), ** (p<0.01), *** (p<0.001). Analysis at 3-digit U.S. 2017 NAICS industries over 2003-2022. Fixed Effects (FE) regressions are of the form* $y\left(\Delta \log\left(w_t(s)\right) \mid \mathcal{X}_{i,t}\,,\, \mathcal{Z}_j\right) = \beta_c + \beta_i \mathcal{X}_{i,t} + \delta_j \mathcal{Z}_{j,t} + u_t$, *with* $\mathcal{X}_i$ *representing the percentage change in both ICT-to-physical capital and non-routine-over-routine workers ratios, and* $\mathcal{Z}_j$ *being a set of time-varying controls. Variables are in log format. Constant not reported to save space. Source: BEA, BLS and own calculations.*

TABLE A.3: COMBINED REGRESSIONS, LEVELS

	$log(w(s))$				
	Fe	*25th quantile*	*50th quantile*	*75th quantile*	*100th quantile*
β_k	.102**	.084***	.102***	.119***	.171*
	(.040)	(.029)	(.022)	(.027)	(.074)
β_ℓ	.299***	.269***	.300***	.328***	.416***
	(.078)	(.040)	(.030)	(.037)	(.104)
$\beta_{k\times\ell}$	.031	.029***	.031***	.034***	.041
	(1.89)	(.010)	(.008)	(.009)	(.025)

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). Standard error in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries in 2003-2022 on* $N = 1240$ *observations. The Fixed Effects (FE) regression is of the form* $y\left(log\left(w_t(s)\right) \mid \mathcal{X}_{i,t}, \mathcal{Z}_{j,t}\right) = \beta_c + \beta_i \mathcal{X}_{i,t} + \delta_j \mathcal{Z}_{j,t} + u_t$*, with* $i = k, \ell$*, and* $\mathcal{Z}_j$ *being a set of controls, and it has associated* $R^2 = .348$ *. Analogously, the conditional quantile regressions are then* $\mathcal{Q}_\omega\left(log\left(w_t(s)\right) \mid \mathcal{X}_{i,\omega t}, \mathcal{Z}_{j,\omega t}\right) = \beta_{i,\omega}\mathcal{X}_{i,\omega t} + \delta_{j,\omega}\mathcal{Z}_{j,\omega t} + u_{\omega t}$*, where* ω *represents each quantile (defined on the independent variable). Variables are all in log format. Constant not reported to save space, while quantile regressions do not have the constant term.* Source: *BEA, BLS and own calculations.*

TABLE A.4: COMBINED REGRESSIONS, STANDARD DEVIATIONS

	$sd\left[log(w(s))\right]$				R^2	
	(1)		(2)			
	$\beta_{sd(k)}$	β_ℓ	β_k	$\beta_{sd(\ell)}$	(1)	(2)
25th quantile	.256***	.070***	.008***	.403***		
	(.054)	(.015)	(.002)	(.010)		
50th quantile	.121***	.051***	.008***	.404***		
	(.034)	(.009)	(.002)	(.008)		
75th quantile	.031	.038***	.008***	.405***		
	(.039)	(.011)	(.002)	(.010)		
100th quantile	-.127	.015	.008	.408***		
	(.112)	(.024)	(.005)	(.021)		
Fe	.149***	.055***	.008***	.404***	.324	.793
	(.031)	(.010)	(.002)	(.007)		

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). Standard error in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries in 2003-2022 on* $N = 1240$ *observations. The Fixed Effects (FE) regression is of the form* $y\left(log\left(w_t(s)\right) \mid \mathcal{X}_{i,t}, \mathcal{Z}_{j,t}\right) = \beta_c + \beta_i \mathcal{X}_{i,t} + \delta_j \mathcal{Z}_{j,t} + u_t$*, with* $i = k, \ell$*, and* $\mathcal{Z}_j$ *being a set of controls. Analogously, the conditional quantile regressions are then* $\mathcal{Q}_\omega\left(log\left(w_t(s)\right) \mid \mathcal{X}_{i,\omega t}, \mathcal{Z}_{j,\omega t}\right) = \beta_{i,\omega}\mathcal{X}_{i,\omega t} + \delta_{j,\omega}\mathcal{Z}_{j,\omega t} + u_{\omega t}$*, where* ω *represents each quantile (defined on the independent variable). Variables are all in log or* sd *format Constant not reported to save space, while quantile regressions do not have the constant term. Source: BEA, BLS and own calculations.*

(Group decomposition) *Total change in wage inequality can be written also as*

$$
\begin{aligned}
\underbrace{\Delta\,\widetilde{var}\Big(w_t(s)-\overline{w}_t\Big)}_{\textit{total, wages}} = \; & \underbrace{\left(\frac{\ell_0(g)}{\ell_0}\right)\left[\Delta\,\widetilde{var}\Big(w_t(s\in g)-\overline{w}_t(g)\Big)\right]}_{\textit{within-group, wages}} \\
& + \underbrace{\sum_g \left[\Delta\,\widetilde{var}\Big(\ell_t(s)-\overline{\ell}_t(g)\Big)\right]\widetilde{var}\Big(w_0(s)-\overline{w}_0(g)\Big)}_{\textit{between-groups, employment}} \\
& + \underbrace{\sum_g \left[\Delta\,\widetilde{var}\Big(w_t(s)-\overline{w}_t(g)\Big)\right]\left[\Delta\,\widetilde{var}\Big(\ell_t(s)-\overline{\ell}_t(g)\Big)\right]}_{\textit{between-groups, interaction}} \\
& + \underbrace{\begin{aligned} & \underbrace{\sum_{\mathcal{G}_{/g}} \left(\frac{\ell_0\left(\mathcal{G}\right)}{\ell_0}\right)\left[\Delta\,\widetilde{var}\Big(w_t(s)-\overline{w}_t\left(\mathcal{G}\right)\Big)\right]}_{\textit{within-other groups, wages}} \\
+ \; & \underbrace{\sum_g \left[\Delta\left(\frac{\ell_t(g)}{\ell_t}\right)\widetilde{var}\Big(\overline{w}_t(g)-\overline{w}_t\Big)\right]}_{\textit{between groups, wages}} \end{aligned}}_{\textit{residual}}
\end{aligned}
\tag{A.1}
$$

where industries are partitioned in $g \in \mathcal{G}$ groups, and variances are employment-weighted. Subscript $t = 0$ indicates the initial level (starting year of the sample). Total change in variance can be decomposed in several components: (i) rise in variance for a particular group g; (ii) reallocation of employment across groups, keeping constant the variance of each group at its base level; (iii) cross-changes of wages and employment; (iv) rising variance within all other groups but g; and (v) rising variance between groups. Quantification of each component is in Table A.5; this decomposition is borrowed from Kleinmann (2023).

TABLE A.5: DECOMPOSITION OF THE RISE IN WAGE INEQUALITY

	industry group g				
share of the increase, wage variance	(1) *tails*	(2) *middle*	(3) *services*	(4) *manuf.*	(5) *other*
rising variance within the group	79%	32%	58%	11%	27%
employment reallocation across groups	34%	34%	17%	51%	10%
comovement (variance, employment)	7%	7%	3%	4%	5%
residual	-20%	27%	-22%	-34%	58%
total change across all industries	*100%*	*100%*	*100%*	*100%*	*100%*

Estimates of each component in eq. (A.1) for 3-digit U.S. 2017 NAICS industries between 2003 and 2022 related to $\log(w(s))$*. Operator* Δ *in the equation is* $x_t - x_{t-1}$*, and not a percentage change. The first row shows the share of total increase in variance due to rising variance in the group of industries; the second row shows the share due to changes in employment between that group and the other industries in the sample (employment reallocation), holding constant the change in variance in each group; the third row shows the share that is due to the cross-product of rising variance and rising employment share; the fourth row is so that the sum for each column is 100%. "tails" and "middle" are referred to overall percentage changes distribution in industry real wage per capita; "manuf." stands for manufacturing industries, while "other" does not consider services and manufacturing industries. Source: BEA and own calculations.*

(A) ROUTINE WORKERS

(B) NON-ROUTINE WORKERS

FIGURE A.3: LABOUR SHARES PATTERN

Note: the figure shows the fitted values and the associated linear trend for the labour share of both routine (Panel A.3a) and non-routine (Panel A.3b) workers. The fitted values are extracted from a Fixed Effects (FE) regression for both the labour share measures with year and industry fixed effects. *Source*: BLS and own calculations.

TABLE A.6: INDUSTRY CONTRIBUTION TO WAGE VARIANCE GROWTH

industry	***contribution share*** (%)	***group***
Administrative and Support Services	0.01	*top*
Agriculture	0.12	*top*
Ambulatory Health Care Services	0.23	*bottom*
Amusement, Gambling, and Recreation industries	0.62	*bottom*
Chemical Manufacturing	-0.26	*bottom*
Computer and Electronic Product Manufacturing	0.94	*top*
Fabricated Metal Product Manufacturing	0.14	*bottom*
Federal Reserve banks, Credit Intermediation, and related activities	0.47	*top*
Food Services and Drinking Places	2.36	*top*
Food and Beverage Stores	1.34	*bottom*
Food and Beverage and Tobacco Products	0.39	*bottom*
Funds, Trusts, and Other Financial Vehicles	0.10	*top*
General Merchandise Stores	1.29	*bottom*
Information and Data Processing Services	5.90	*top*
Machinery Manufacturing	-0.02	*bottom*
Management of Companies and Enterprises	2.71	*top*
Oil and Gas Extraction	0.26	*top*
Other Services, except government	0.43	*top*
Other Transportation and Support Activities	0.67	*bottom*
Paper Manufacturing	-0.02	*bottom*
Performing Arts, Spectator Sports, Museums, and related activities	-0.02	*top*
Printing and Related Support Activities	0.08	*bottom*
Professional, Scientific, and Technical Services	11.7	*bottom*
Publishing industries (including software)	2.23	*top*
Rail Transportation	-0.10	*bottom*
Real Estate	-0.04	*top*
Rental and Leasing Services and Lessors of Intangible Assets	-0.08	*top*
Securities, Commodity Contracts, and Other Financial Investments and Related Activities	8.59	*top*
Transportation Equipment Manufacturing	1.17	*bottom*
Warehousing and Storage	0.86	*bottom*

Contribution to between-industry wage variance growth of industries comprised in top and bottom (in terms of overall growth in real log-wage per capita, as reported by Table 1) groups over the period 2003-2022. Variance contribution follows the definition proposed in eq. (1). Industries are at 3-digit U.S. 2017 NAICS level. Source: BEA and own calculations.

B. Model derivation

(Household inter-temporal problem) *The household utility problem in eq. (5) can be rewritten inter-temporally, future-discounted by factor β, as*

$$\max_{\mathcal{C}_t^i,\, b_{t+1}^i,\, \{k_{t+1}^i(j)\}_{\forall j}} \mathcal{U}_{h,t}^i(a,s) = \sum_{t=0}^{\infty}\left[\beta^t \log\left(\mathcal{C}_t^i\right)\right] + \sum_{t=0}^{\infty}\left[\log \mathcal{B}_{h,t}(a,s)\right] + \wp_h^i(a,s)$$

$$\begin{aligned} s.t. \quad & \mathcal{C}_t^i + I_t^i(phy) + I_t^i(ict) + b_{t+1}^i - (1+r_t)\, b_t^i = \\ & = w_{h,t}(a,s)\ell_h^i(a,s) + R_t(phy)\, k_t^i(phy) + R_t(ict)\, k_t^i(ict) + \mathcal{D}_t^i \end{aligned}$$

$$with \quad k_{t+1}^i(phy) = \frac{I_t^i(phy)}{\zeta_t^i} + \left(1-\delta_{phy}\right) k_t^i(phy)$$

$$and \quad k_{t+1}^i(ict) = \frac{I_t^i(ict)}{\zeta_t^i} + (1-\delta_{ict})\, k_t^i(ict)$$

where the investment schedule comprises both assets, b_t^i (which are in zero net supply, thus exchanged only across households), and physical and ICT capital holdings, $k_t^i(j)$, with $j = \{phy, ict\}$. Household-specific productivity when working as type-a for firm-h in industry-s is denoted with $\wp_h^i(a,s)$. Each household inelastically supplies one unit of work, so that $\ell_h^i(a,s) = 1$. The firm benefit of household-a is defined to be $\mathcal{B}_{h,t}(a,s) = \left[g_{h,t}(a,s)\right]^{-\varsigma}$, where $g_{h,t}(a,s)$ reflects the relative share (i.e., the task ratio) of a given task, which is negatively scaled by the elasticity parameter ς, while $\mathcal{D}_t^i$ is the share of firms' profits that goes to household-i. Capital stock of household-i depreciates at a rate $\delta_{(j)}$ and accumulates over time by a law of motion which is function of $\zeta_{i,t}$, namely the quantity of ICT capital relative to that of non-ICT capital, that enters negatively in new capital investment, $I_t^i(j)$, under the idea that as the stock of capital owned by household-i becomes more sophisticated (larger ICT relative to non-ICT capital when ζ_t^i increases), it is required an higher rate of capital investment to keep constant the future stock of given capital type. Modelling the dynamics of capital in this way serves only to derive analytical solutions for the equations (12) and (13) estimating the pair (ρ, σ) of Section 3.

Utility maximization implies the Lagrangian function to be

$$\begin{aligned} \mathcal{L}_{\mathcal{C}_t^i,\, b_{t+1}^i,\, \{k_{t+1}^i(j)\}_{\forall j}} = & \sum_t \beta^t \left[\log \mathcal{C}_t^i\right] + \sum_t \left[\log \mathcal{B}_{h,t}(a,s)\right] + \wp^i(a,s) + \\ & - \sum_t \beta^t \psi^t \Big[\mathcal{C}_t^i + I_t^i(phy) + I_t^i(ict) + \\ & + b_{t+1}^i - (1+r_t)\, b_t^i - w_{h,t}(a,s) + \\ & - R_t(phy)\, k_t^i(phy) - R_t(ict)\, k_t^i(ict) - \mathcal{D}_t^i\Big] \end{aligned}$$

with ψ^t being the penalty multiplier. Optimality conditions are in order

$$\frac{1}{\mathcal{C}_t^i} = \beta^t \left(1 + r_{t+1}\right) \frac{1}{\mathcal{C}_{t+1}^i}$$
$$R_t = \left(1 + r_{t+1}\right) \zeta_t - (1-\delta)\zeta_{t+1}$$

The first is the usual Euler condition, which displays future path of consumption. The second implies that, aggregating across households, the path on interest rates on capital types, $R_t(phy) = R_t(ict) \equiv R_t$, *is linked to the path of aggregate relative quantity of ICT capital,* $\zeta_t = \int_i \zeta_t^i \,\mathrm{d}i$, *when* $\delta_{phy} = \delta_{ict} \equiv \delta$;[I] *in other words, given the constancy of both the discount factor and the capital depreciation rate(s), changes across states of the capital rental rate are determined by changes in the relative quantities across capital types. In the spirit of Karabarbounis and Neiman (2014), eq. (6) determines that investing in capital types is profitable as long as the marginal benefit of investment (the capital rental rate) is at least lower than its marginal cost (interest rate,* r_t *and depreciation rate,* δ*).*

□

(Labour supply derivation) *The probability of worker-*a *choosing firm-*$h = 1$ *in industry-*$h = 1$ *can be written as*

$$\left(\mathcal{P}_1(a,1) \equiv\right) \Longleftrightarrow \mathcal{P}(a,1) \qquad \textit{for notation purposes, abstract from firm subscript}$$

$$\begin{aligned}\mathcal{P}(a,1) &= Pr\Big[w\left(a,1\right)\mathcal{B}\left(a,1\right)\wp^i\left(a,1\right) > w(a,s)\mathcal{B}(a,s)\wp^i(a,s)\Big],\ \forall s \neq 1\\ &= Pr\left[\frac{w\left(a,1\right)\mathcal{B}\left(a,1\right)}{w(a,s)\mathcal{B}(a,s)}\wp^i\left(a,1\right) > \wp^i(a,s)\right],\ \forall s \neq 1\end{aligned}$$

For $s \in [2, S]$, *the partial derivative of* $\mathcal{P}(a,1)$ *with respect to* $\wp^i\left(a,1\right)$ *is*

$$\left\{\wp^i\left(a,1\right),\ \frac{w\left(a,1\right)\mathcal{B}\left(a,1\right)}{w\left(a,2\right)\mathcal{B}\left(a,2\right)}\wp^i\left(a,1\right),\ \ldots,\ \frac{w\left(a,1\right)\mathcal{B}\left(a,1\right)}{w(a,S)\mathcal{B}\left(a,S\right)}\wp^i\left(a,1\right)\right\}$$

so that $\mathcal{P}(a,1)$ *can be re-written as*

$$\begin{aligned}\mathcal{P}(a,1) &= F_{\wp^i(a,1)}\Big(\wp^i\left(a,1\right),\ \alpha_2\wp^i\left(a,1\right),\ \ldots,\ \alpha_s\wp^i\left(a,1\right)\Big),\ \forall\wp^i(a,s)\\ &= \int F_{\wp^i(a,1)}\Big(\wp^i\left(a,1\right),\ \alpha_2\wp^i\left(a,1\right),\ \ldots,\ \alpha_s\wp^i\left(a,1\right)\Big)\mathrm{d}\wp^i\left(a,1\right)\end{aligned}$$

for $\alpha_s = \frac{w(a,1)\mathcal{B}(a,1)}{w(a,s)\mathcal{B}(a,s)}$. *Now, recall that the parameter* $\wp^i(a,s)$ *is drawn from a multivariate Frechét-type cumulative distribution,*

[I] Alternatively, the common depreciation rate might be a weighted average of all the capital-types' depreciation rates.

$$F_i\Big(\wp^i_{h,\dots,H}\left(a,1\right),\dots,\wp^i_{h,\dots,H}(a,s),\dots,\wp^i_{h,\dots,H}\left(a,S\right)\Big) = exp\left[-\sum_s\left(\int_h \wp^i_h(a,s)\,\mathrm{d}h\right)^{-\theta}\right]$$

which becomes, in this framework (i.e., not considering firm's notation),

$$\begin{aligned} F\Big(\alpha_1\wp^i\left(a,1\right),\dots,\alpha_s\wp^i(a,s),\dots,\alpha_S\wp^i\left(a,S\right)\Big) &= exp\left[-\sum_s\alpha_s^{-\theta}\ \sum_s\wp^i(a,s)^{-\theta}\right] \\ &= exp\left[-\sum_s\Big(\alpha_s\ \wp^i(a,s)\Big)^{-\theta}\right] \end{aligned}$$

Taking its derivative with respect to $\wp^i(a,s)$ turns to write that

$$F_{\wp^i(a,1)}\Big(\wp^i\left(a,1\right),\ \alpha_2\wp^i\left(a,1\right),\ \dots,\ \alpha_s\wp^i\left(a,1\right)\Big) = -\left(-\theta\right)\wp^i(a,s)^{-\theta-1}exp\left[\overline{\alpha}\wp^i(a,s)^{-\theta}\right]$$

with $\overline{\alpha} = \sum_s\alpha_s^{-\theta}$. Evaluating the integral in $\mathcal{P}\left(a,1\right)$ yields to

$$\mathcal{P}\left(a,1\right) = \int\theta\ \underbrace{\wp^i(a,s)^{-\theta-1}}_{A}\ \underbrace{exp\left[\overline{\alpha}\wp^i(a,s)^{-\theta}\right]}_{B}\ \mathrm{d}\wp^i(a,s)$$

By multiplying and dividing by $\overline{\alpha}$ (so that A can be integrated with B), one gets

$$\begin{aligned} \mathcal{P}\left(a,1\right) &= \frac{\overline{\alpha}}{\overline{\alpha}}\int\theta\wp^i(a,s)^{-\theta-1}exp\left[\overline{\alpha}\wp^i(a,s)^{-\theta}\right]\mathrm{d}\wp^i(a,s) \\ &= \frac{1}{\overline{\alpha}}\int\overline{\alpha}\theta\wp^i(a,s)^{-\theta-1}exp\left[\overline{\alpha}\wp^i(a,s)^{-\theta}\right]\mathrm{d}\wp^i(a,s) \\ &= \frac{1}{\overline{\alpha}}\int\mathrm{d}F\Big(\wp^i\left(a,1\right),\dots,\wp^i(a,s),\dots,\wp^i(a,S)\Big) \\ &= \frac{1}{\overline{\alpha}} \\ &= \frac{1}{\sum_s\alpha_s^{-\theta}} \end{aligned}$$

By recalling that $\alpha_s = \frac{w(a,1)\mathcal{B}(a,1)}{w(a,s)\mathcal{B}(a,s)}$, it is possible to obtain that, $\forall s \neq 1$,

$$\mathcal{P}\left(a,1\right) = \frac{\Big(w\left(a,1\right)\mathcal{B}\left(a,1\right)\Big)^{\theta}}{\Big(\sum_s w(a,s)\mathcal{B}(a,s)\Big)^{\theta}}$$

where, including firm's subscript-h, it becomes

$$\mathcal{P}_1\left(a,1\right) = \frac{\Big(w_1\left(a,1\right)\mathcal{B}_1\left(a,1\right)\Big)^{\theta}}{\Big(\sum_{h,s} w_h(a,s)\mathcal{B}_h(a,s)\Big)^{\theta}}$$

Taking in general notation, $\forall h \in [1, H]$ *and* $\forall s \in [1, S]$*, it can be written as*

$$\mathcal{P}_h(a,s) = \frac{\Big(w_h(a,s)\mathcal{B}_h(a,s)\Big)^{\theta}}{\Big(\sum_{h,s} w_h(a,s)\mathcal{B}_h(a,s)\Big)^{\theta}} \quad , \quad \textit{with} \quad \mathcal{B}_h(a,s) = \big[g_h(a,s)\big]^{-\varsigma}$$

which denotes the fraction of type-a households choosing to work in firm-h, industry-s as in eq. (7).

To interpret the role of θ – the shape parameter of the Frechét distribution –, Figure B.1 plots the distribution of productivities under different values of θ, interpreted as being the degree of dispersion of households-specific efficiencies in working in firm-h in industry-s. Basically, the larger is the value of θ the lower is the variability of the productivity distribution (i.e., the lower is the dispersion of households' productivities), thus the higher is the degree of labour market concentration of workers across firms and industries.

Moreover, as explained in the main text to interpret eq. (7), "the number of workers of a given task-a willing to be employed in a certain firm-industry pair is determined by different optimal wage levels by firms in industries (wage premium), jointly with workforce composition in terms of both the firm-specific relative amount of workers of a certain type (sorting) and workers having the same efficiency in conducting that job task content (segregation). In this sense, workers in the same task are "highly rival factors" (Hicks 1932) since they can be freely substituted for one another, and this possibility ensures that high-wage workers do sort in high-wage firms and industries, and also that more efficient people would be preferred to get a job in high-wage workplaces than ones who are less productive".[II]

□

(Firm optimization) *Given $y_h(s)$ being the production function in eq. (9), and the firm's conditional demand given by eq. (8), the problem of a monopolistically competitive firm (h, s) choosing capital and labour endowments is*

[II] To build intuition, such idea can be formalized using an Horvath (2000)'s aggregator: the aggregate type-a workforce of industry-s can be taught as a CES aggregator of firm-specific labour measures of eq. (7),

$$\ell(a,s) = \left[\int_h \Big(\ell_h(a,s)\Big)^{\frac{1+\psi}{\psi}} \mathrm{d}h\right]^{\frac{\psi}{1+\psi}}$$

since each worker-i is endowed with $\ell_h^i(a,s) = 1$ unit inelastically supplied. Here, as $\psi \to \infty$, type-a workers become perfect substitutes; conversely, for any given value of parameter ψ comprised in $[0, \infty)$ workers won't be freely substituted across firms in the same industry. As a natural consequence, each worker would work in the firm (h, s) paying the highest wage according to its $\wp_h^i(a,s)$. More, at the intensive margin, firms in the same industry would pay the same wage level.

This is a crucial aspect to consider when moving from firm to industry dimension: free labour mobility within an industry – materialized when $\psi = \infty$ – allows to impose firms' optimal wages to be even, so to have a unique industry-s wage in equilibrium; refer to Proposition 1 and Remark 2.

Analogously, worker-a wage can be seen as $w(a,s) = \left[\int_h \Big(w_h(a,s)\Big)^{\frac{1+\psi}{\psi}} \mathrm{d}h\right]^{\frac{\psi}{1+\psi}}$ with $\psi = \infty$.

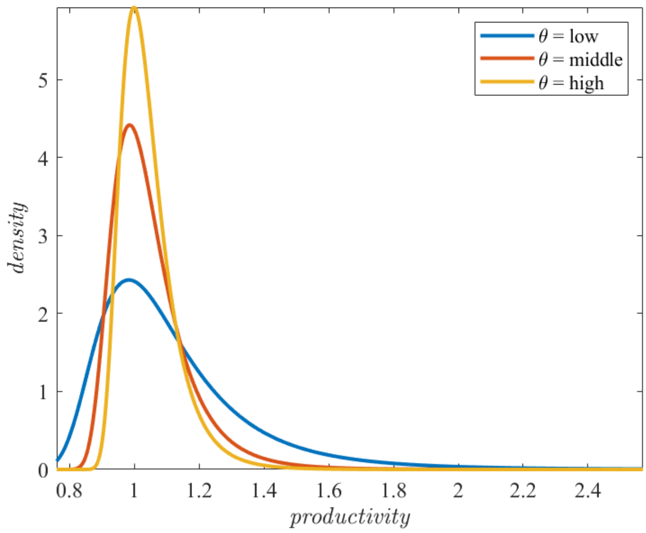


FIGURE B.1: VARIABILITY AND THE SHAPE PARAMETER

Note: the plot represents how the distribution of an arbitrary variable (*e.g.*, productivity) changes along different values of the shape parameter, keeping unchanged the scale parameter; lower shape parameter results in major variability.

$$\max_{p_h(s), \{k_h(j,s)\}_{\forall j}, \{\ell_h(a,s)\}_{\forall a}} p_h(s) y_h(s) - \left(\sum_j R(j)\, k_h(j,s) + \sum_a w_h(a,s)\, \ell_h(a,s) \right)$$

$$\text{s.t.} \quad y_h(s) = \left(\frac{p_h(s)}{p(s)} \right)^{\epsilon} y(s)$$

with $a = \{rt, nrt\}$ and $j = \{phy, ict\}$. The inter-temporal Lagrangian function for firm-h industry-s takes the form of

$$\begin{aligned} \mathcal{L}_{p_h(s), \{k_h(j,s), \ell_h(a,s)\}_{\forall j,a}} = p_h(s) y_h(s) - \Big(& R(phy)\, k_h(phy,s) + R(ict)\, k_h(ict,s) + \\ & + w_h(rt,s)\, \ell_h(rt,s) + w_h(nrt,s)\, \ell_h(nrt,s) \Big) + \\ & - \psi^t \Big[y_h(s) p_h(s)^{\epsilon} - y(s) \Big] \end{aligned}$$

with ψ^t being the penalty multiplier. Optimality conditions are in order

$$\frac{\partial \mathcal{L}}{\partial p_h(s)} \; : \; \psi = \frac{1}{\epsilon} p_h(s)^{1-\epsilon}$$

$$\frac{\partial \mathcal{L}}{\partial k_h(phy,s)} \; : \; p_h(s) f_{k_h(phy,s)} = \mathcal{M}R(phy)$$

$$\frac{\partial \mathcal{L}}{\partial k_h(ict,s)} \; : \; p_h(s) f_{k_h(ict,s)} = \mathcal{M}R(ict)$$

with the price mark-up being $\mathcal{M} = \frac{\epsilon}{\epsilon - 1}$, and first order conditions of firm-industry *specific output relative to capital-types are*

$$f_{k_h(phy,s)} = \alpha\left(k_h(phy,s)\right)^{\alpha-1}\left[y_h(s)\left(k_h(phy,s)\right)^{-\alpha}\right]$$

$$f_{k_h(ict,s)} = (1-\alpha)(1-\mu)\lambda\left[\left(k_h(phy,s)\right)^{\alpha}\mathcal{V}_h(s)^{\frac{1-\alpha-\varsigma}{\varsigma}}\mathcal{Q}_h(s)^{\frac{\varsigma-\varrho}{\varrho}}\right]\left(k_h(ict,s)\right)^{\varrho-1}$$

with

$$\mathcal{V}_h(s) = \mu\left(\ell_h(rt,s)\right)^{\varsigma} + (1-\mu)\,\mathcal{Q}_h(s)^{\frac{\varsigma}{\varrho}}$$

and

$$\mathcal{Q}_h(s) = \lambda\left(k_h(ict,s)\right)^{\varrho} + (1-\lambda)\left(\ell_h(nrt,s)\right)^{\varrho}$$

Note how aggregate capital rental rates, $R_t(j)$, are obtained by aggregating marginal product of capital types $\left(f_{k_h(j,s)}\right)$ across firms and industries.

Optimality conditions for wages are found by deriving directly for $\ell_h(a,s)$; this results in computing what follows.

$$\frac{\partial\mathcal{L}}{\partial\ell_h(rt,s)} : p_h(s)f_{\ell_h(rt,s)} = \mathcal{M}\,w_h(rt,s)$$

$$\frac{\partial\mathcal{L}}{\partial\ell_h(nrt,s)} : p_h(s)f_{\ell_h(nrt,s)} = \mathcal{M}\,w_h(nrt,s)$$

with the price mark-up still being $\mathcal{M} = \frac{\epsilon}{\epsilon-1}$, and first order conditions of firm (h,s)'s output relative to labour-types are

$$f_{\ell_h(rt,s)} = (1-\alpha)\mu\left(k_h(phy,s)\right)^{\alpha}\mathcal{V}_h(s)^{\frac{1-\alpha-\varsigma}{\varsigma}}\left(\ell_h(rt,s)\right)^{\varsigma-1}$$

$$f_{\ell_h(nrt,s)} = (1-\alpha)(1-\mu)(1-\lambda)\left(k_h(phy,s)\right)^{\alpha}\mathcal{V}_h(s)^{\frac{1-\alpha-\varsigma}{\varsigma}}\mathcal{Q}_h(s)^{\frac{\varsigma-\varrho}{\varrho}}\left(\ell_h(nrt,s)\right)^{\varrho-1}$$

with

$$\mathcal{V}_h(s) = \mu\left(\ell_h(rt,s)\right)^{\varsigma} + (1-\mu)\,\mathcal{Q}_h(s)^{\frac{\varsigma}{\varrho}}$$

and

$$\mathcal{Q}_h(s) = \lambda\left(k_h(ict,s)\right)^{\varrho} + (1-\lambda)\left(\ell_h(nrt,s)\right)^{\varrho}$$

Now, note that labour supplies for both types of tasks $a = \{rt, nrt\}$ are all determined by eq. (7), namely $\ell_h(a,s) = f\left(w_h(a,s),\, \mathcal{B}_h(a,s),\, \mathcal{W}_{\mathcal{H}}(a,\mathcal{S}),\, \mathcal{B}_{\mathcal{H}}(a,\mathcal{S})\right)$. Therefore, equating demand and supply of labour for each task-a results in deriving the associated optimal wage level in general equilibrium, that is

$$\begin{cases} p_h(s) f_{\ell_h(rt,s)} = \mathcal{M}\, w_h(rt,s) \\ \\ \ell_h(rt,s) = \left(\dfrac{w_h(rt,s)\mathcal{B}_h(rt,s)}{\sum_{h,s} w_h(rt,s)\mathcal{B}_h(rt,s)} \right)^{\theta} \end{cases}$$

and

$$\begin{cases} p_h(s) f_{\ell_h(nrt,s)} = \mathcal{M}\, w_h(nrt,s) \\ \\ \ell_h(nrt,s) = \left(\dfrac{w_h(nrt,s)\mathcal{B}_h(nrt,s)}{\sum_{h,s} w_h(nrt,s)\mathcal{B}_h(nrt,s)} \right)^{\theta} \end{cases}$$

By writing down the extensive forms for each derivative of $\ell_h(a,s)$ and exploiting the calculations to solve for $w_h(a,s)$, together with proposition 1, optimal wages are those reported in eq. (10). Note that $\mathcal{V}_h(s)$ expresses the substitutability between routine workers $(\ell_h(rt,s))$ and the ICT composite good, while $\mathcal{Q}_h(s)$ identifies the substitutability between ICT capital $(k_h(ict,s))$ and non-routine workers $(\ell_h(nrt,s))$, considering each firm-$h \in \mathcal{H}$ in industry-s.

Finally, firm (h,s) profits can be found by including equilibrium optimality conditions for both capital and wages in

$$\mathcal{D}_h(s) = p_h(s)y_h(s) - \left(\sum_j R(j)k_h(j,s) + \sum_a w_h(a,s)\ell_h(a,s) \right)$$

where $j = \{phy, ict\}$ identifies the types of capital in the economy.

□

(Proof of Proposition 1) *This heuristic proof is centred around the definition of the workers' measures as given by eq. (7) with no worker-a benefit, $\mathcal{B}_h(a,s) = 1$, $\forall a,h,s$. Imagine an economy in which there is only one industry-$h \in \mathcal{S} = 1$ populated by two firms, $\{h, h'\} \in \mathcal{H}$, and define $\mathcal{W}_{\mathcal{H}}(a,s) = w_h(a,s) + w_{h'}(a,s)$. Then:*

(a) when firms are assumed to be homogeneous in their size, i.e., when $\ell_h(a,s) \equiv \ell_{h'}(a,s)$, then they should set the same optimal wage level since

$$\left(\frac{w_h(a,s)}{\mathcal{W}_{\mathcal{H}}(a,s)} = \right) \ \ell_h(a,s) \equiv \ell_{h'}(a,s) \ \left(= \frac{w_{h'}(a,s)}{\mathcal{W}_{\mathcal{H}}(a,s)} \right)$$

holds only if $w_h(a,s) = w_{h'}(a,s)$.

(b) if firms are heterogeneous in size eq. (7) predicts how, in order to have $\ell_h(a,s) \neq \ell_{h'}(a,s)$, it is necessary that wage levels are different, $w_h(a,s) \neq w_{h'}(a,s)$. Assume that firm (h',s) sets a wage rate higher than firm (h,s), so that the former is larger than the latter. Assume further an increase in the wage chosen by firm (h',s), labelling the new level as $w_{h'}(a,s)'$, while the other wage remains unchanged. Henceforth, it must be the case that

$$w_{h'}(a,s) \to w_{h'}(a,s)', \quad \textit{so that} \quad \mathcal{W}_{\mathcal{H}}(a,s) < \mathcal{W}_{\mathcal{H}}(a,s)'.$$

For such firm, clearing the condition

$$\mathcal{W}_{\mathcal{H}}(a,s)' = \frac{w_{h'}(a,s)'}{\ell_{h'}(a,s)'} \tag{B.1}$$

means that there should be a related increase in its workforce, $\frac{\partial \ell_{h'}(a,s)}{\partial w_{h'}(a,s)} > 0$*, so that* $\ell_{h'}(a,s)' > \ell_{h'}(a,s)$*. Different scenario happens to the employees level of firm* (h,s)*: it turns out that* $\ell_h(a,s)' < \ell_h(a,s)$ *when* $w_{h'}(a,s) \to w_{h'}(a,s)'$ *since* $\frac{\partial \ell_h(a,s)}{\partial w_{h'}(a,s)} < 0$*. The implied mechanism is just*

$$\frac{\partial \ell_{h'}(a,s)}{\partial w_{h'}(a,s)} = -\frac{\partial \ell_h(a,s)}{\partial w_{h'}(a,s)}$$

However, the given change in the wage of firm (h',s) *not only has an impact on* $\ell_{h'}(a,s)$*, but it indirectly translates to the wage level of firm* (h,s) *due its negative effect on* $\ell_h(a,s)$*. The firm* (h,s) *version of the condition in eq. (B.1)*

$$\mathcal{W}_{\mathcal{H}}(a,s)' = \frac{w_h(a,s)'}{\ell_h(a,s)'} \tag{B.2}$$

implies that, after an increase in $\mathcal{W}_{\mathcal{H}}(a,s)$ *and a decrease in* $\ell_h(a,s)$ *due to a positive change in* $w_{h'}(a,s)$*, then the wage level of firm* (h,s) *must increase as well. It turns out that an increase in the wage level of firm* (h',s) *must determine an equal increase in the right-hand-side of both eqs. (B.1)-(B.2), which means*

$$\frac{w_h(a,s)'}{\ell_h(a,s)'} = \frac{w_{h'}(a,s)'}{\ell_{h'}(a,s)'}$$

so that $\frac{\partial w_h(a,s)}{\partial \ell_h(a,s)} = \frac{\partial w_{h'}(a,s)}{\partial \ell_{h'}(a,s)}$.

To sum up, when firms in a specific industry are of different size in terms of employed workers of type-a*, an increase in the wage level of a given firm causes: (i) an increase in the number of employed people in that firm,* $\frac{\partial \ell_{h'}(a,s)}{\partial w_{h'}(a,s)} > 0$*; (ii) a related decrease in the number of workers of the other firms,* $\frac{\partial \ell_h(a,s)}{\partial w_{h'}(a,s)} < 0$*, and thus an increase in other firms' wage levels,* $\frac{\partial w_h(a,s)}{\partial w_{h'}(a,s)} > 0$*. This is true as long as workers of each type-*a *are free to move across firms in the same industry at no cost and firms can hire new workers from the other firms;*

(c) by applying the same reasoning of point (b), if workers were perfectly mobile across industries, then it would have been the case that $\frac{w(a,s')}{\ell(a,s')} = \frac{w(a,s)}{\ell(a,s)}$*. This chance is ruled out by assuming a very high cost of transition from one industry to another such that no one among workers is keen to move.*

□

(Equilibrium characterization) *In equilibrium, the model should specify the clearing conditions of labour, capital, and goods markets. Starting from the labour market, since each household inelastically supplies one unit of labour, then it must hold that the total number of workers of type-a in firm (h, s) is $\ell_h(a, s) = \int_0^1 \ell_h^i(a, s)\mathrm{d}i$, so that the total labour supply is $L^S = \sum_a \sum_h \sum_s \ell_h(a, s)$. The measure of type-$a$ worker in firm (h, s) is given by eq. (7). Aggregating it across tasks and firms results in obtaining the industry-specific labour supply, $L(s) = \sum_{a,h} \ell_h(a, s)$. Analogously, aggregate labour supply of task-a is found by aggregating across firms and industries, $L(a) = \sum_{h,s} \ell_h(a, s)$. It follows that, considering $a = \{rt, nrt\}$, aggregate labour demand for this economy is just $L^D = \sum_a L(a) = L(rt) + L(nrt)$. Labour market clearing requires that $L^D = L^S$.*

For what concerns equilibrium in the capital market(s), total physical and ICT capital demands from industries are, respectively, $K^D(phy, s) = \sum_h k_h(phy, s)$ and $K^D(ict, s) = \sum_h k_h(ict, s)$, so that aggregate demands are simply determined: $K^D(phy) = \sum_s K^D(phy, s)$ and $K^D(ict) = \sum_s K^D(ict, s)$. By the part of supply, aggregating capital quantities over households would results in aggregate physical and ICT capital supplies: $K^S(phy) = \int_i k^i(phy)\mathrm{d}i$ and $K^S(ict) = \int_i k^i(ict)\mathrm{d}i$. Equilibrium in both markets requires $K(phy) \equiv K^D(phy) = K^S(phy)$ and $K(ict) \equiv K^D(ict) = K^S(ict)$, while market clearing in the capital market implies $K^D(phy) + K^D(ict) = K^S(phy) + K^S(ict)$.

Finally, aggregate profits to be given to households are $\mathcal{D}^D = \int_i \mathcal{D}_i \,\mathrm{d}i$, while $\mathcal{D}^S = \sum_s \mathcal{D}(s)$ are the total profits computed by aggregating industry-specific profits, $\mathcal{D}(s) = \sum_h \mathcal{D}_h(s)$. Equilibrium requires $\mathcal{D} \equiv \mathcal{D}^D = \mathcal{D}^S$. This implies that, by aggregating the households' inter-temporal budget constraints and imposing the clearing conditions so far, including also total quantities for

$$
\begin{aligned}
\mathcal{C}^i &= \int_i \mathcal{C}^i \,\mathrm{d}i \\
I(j) &= \int_i I^i(j) \,\mathrm{d}i\,, \quad \forall j \\
b &= \int_i b^i \,\mathrm{d}i \\
w &= \sum_a \sum_h \sum_s w_h(a, s) \\
\mathcal{B} &= \sum_a \sum_h \sum_s \mathcal{B}_h(a, s)
\end{aligned}
$$

where $\mathcal{B} = 1$ since it is the sum of relative quantities, the aggregate resource constraint for this economy at time t reads as

$$
\mathcal{C}_t + I_t(phy) + I_t(ict) + b_{t+1} - (1 + r_t)b_t = w_t \mathcal{B}_t L_t + R_t\Big(K_t(phy) + K_t(ict)\Big) + \mathcal{D}_t
$$

which equals the total output as defined by the final output CES aggregator, Y. Equilibrium conditions are described in Section 2.

□

C. MODEL ESTIMATION AND SIMULATION

(Estimating equations) *To derive the equations to estimate the elasticity of substitution between ICT capital and non-routine workers (ρ), and the elasticity of substitution between routine workers and ICT composite good (σ), i.e., that shown in eqs. (12) and (13), I apply the procedure implemented by Karabarbounis and Neiman (2014). The main steps to be implemented are:*[I]

1. *Define a CES production function, $y(\cdot)$ and compute the related F.O.C.s.; then, equate them to the aggregated (across firms and industries) F.O.C.s of the monopolistically competitive firms;*

2. *Define the following income shares. For a given labour force (ℓ), a given capital stock (k), and given profits $(\mathcal{D})$,*

$$s_\ell = \left(\frac{1}{\mathcal{M}}\right)\left(\frac{w(\ell)\ell}{w(\ell)\ell + Rk}\right) \quad , \quad s_k = \left(\frac{1}{\mathcal{M}}\right)\left(\frac{Rk}{w(\ell)\ell + Rk}\right) \quad , \quad s_{\mathcal{D}} = 1 - \frac{1}{\mathcal{M}}$$

3. *By combining the F.O.C. for capital (either for labour) with all the above shares, one gets an equation whose left-hand side is $1 - s_\ell\mathcal{M}$. Then, this should be written in changes between two arbitrary periods, whose resulting elements are labelled as $\widehat{x}$;*

4. *Use eq. (6) to substitute $\widehat{R}$;*

   $\rightarrow$ *use the Euler condition (from the households side) expressed in deviation between two arbitrary periods get $\widehat{(1+r)} = \frac{1}{\beta}$ so that, under constant β and δ, it holds that $\widehat{R} = \widehat{\zeta}$;*

5. *Once substituting out $\widehat{R}$, take a linear approximation of the resulting equation around $\widehat{\zeta} = 0$, thus obtaining the estimating equation.*

Apply this procedure for whatever CES functional form of the production function $y(\cdot)$. Note that, to carry out this procedure in the framework I propose it is necessary to assume equal marginal product of each type of capital. In fact, in eq. (6), trends in both capital rental rates are tied with trends in the quantity of ICT capital relative to non-ICT one, after aggregating optimal conditions in the firm problem. Here, the two capital rates are given by capitals' marginal productivity: thus, to have a unique capital rental rate, such that $R(phy) = R(ict) \equiv R$, equal marginal product of capital types are in order.

(Targeting moments for MSM) *Start from the weighting parameters, λ and μ. For each industry-$h \in \mathcal{S}$, the weight of ICT capital (λ) in the ICT composite is matched with the industry-specific ICT capital in the aggregate stock in the data.*

[I] Please refer to Karabarbounis and Neiman (2014) for further details and discussion.

Differently, the weight of routine workers (μ) in the production function is used to bridge the share of routine workers in the data with that predicted by the model, i.e., I implement the following identity using eq. (7): $\ell_{model}(a,s) = \left(\frac{w(a,s)}{\mathcal{W}(a,\mathcal{S})}\,\frac{\mathcal{B}(a,s)}{\mathcal{B}(a,\mathcal{S})}\right)^{\theta} \approx \ell_{data}(a,s)$. Of course, since the model's measures of employment are determined by relative wages, some slight differences in the estimated matched moments are in order.

Finally, I consider productivity dispersion parameter, θ, which directly relates to the between-industry wage difference for worker-a. Given $a = rt$, I use the wage premium of type-a working in top industry (s) relative to its counterpart in the bottom industry (s'):

$$\frac{w(rt,s)}{w(rt,s')} = \frac{\left[\Lambda(s)\,\chi(rt,s)\,\Big(k(phy,s)\Big)^{\alpha(s)}\,\mathcal{V}(s)^{\frac{1-\alpha(s)-\varsigma(s)}{\varsigma(s)}}\,\mathcal{B}(rt,s)^{\theta(\varsigma(s)-1)}\,\mathcal{WB}(rt,\mathcal{S})^{\theta(1-\varsigma(s))}\right]^{\frac{1}{1+\theta-\theta\varsigma(s)}}}{\left[\Lambda(s')\,\chi(rt,s')\,\Big(k(phy,s')\Big)^{\alpha(s')}\,\mathcal{V}(s')^{\frac{1-\alpha(s')-\varsigma(s')}{\varsigma(s')}}\,\mathcal{B}(rt,s')^{\theta(\varsigma(s')-1)}\,\mathcal{WB}(rt,\mathcal{S})^{\theta(1-\varsigma(s'))}\right]^{\frac{1}{1+\theta-\theta\varsigma(s')}}}$$

If considering changes over time, only $k(phy,\cdot)$, $\mathcal{V}(\cdot)$, $\mathcal{B}(\cdot)$, and $\mathcal{WB}(\cdot)$ are time-varying, with all the other parameters previously calibrated, targeted and fixed: this leaves θ as the only free parameter to match this moment. Note how I choose to pin down labour market concentration for routine workers in accordance with Figure 4.

TABLE C.1: EMPLOYMENT MEASURES AND TASKS' RELATIVE WAGES

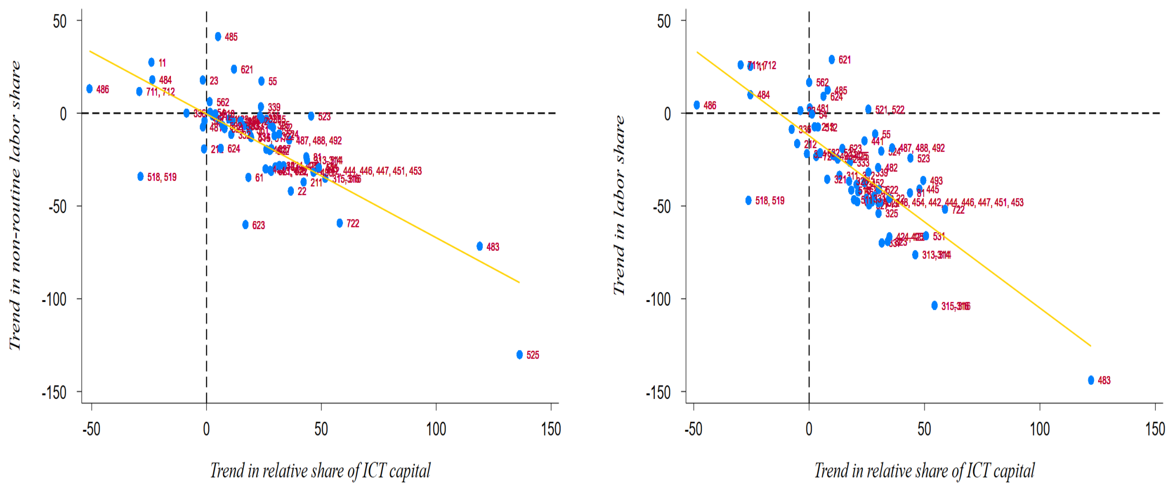
	$log(\ell(rt,s))$			$log(\ell(nrt,s))$		
	(1)	(2)	(3)	(1)	(2)	(3)
$\ell(rt,s\,\vert\,w\mathcal{B})$	.765*	.657*	3.55***			
	(.32)	(.28)	(.17)			
$\ell(nrt,s\,\vert\,w\mathcal{B})$				5.76***	3.71***	29.4***
				(1.9)	(1.1)	(2.2)
Industry FE	✓	✓	✗	✓	✓	✗
Time FE	✗	✓	✗	✗	✓	✗

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). Standard error in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries in 2003-2022 on N = 1240 observations. All the regressions are of the form $(y_t \mid \mathcal{X}_{i,t}, \mathcal{Z}_{j,t}) = \beta_c + \beta_i \mathcal{X}_{i,t} + \delta_j \mathcal{Z}_{j,t} + u_t$, with $\mathcal{X}_i$ being the regressors, and $\mathcal{Z}_j$ a set of controls. All series are in logs. Constant not reported to save space.* Source: *BLS and own calculations.*

TABLE C.2: METHOD OF SIMULATED MOMENTS

	parameter	*value*	***moment to match***	*fit*	
				data	*model*
μ_{bot}	weight of routines in $y(bot)$	0.6763	routine share, bottom	.0858	.0434
μ_{mid}	weight of routines in $y(mid)$	0.4953	routine share, middle	.1357	.0458
μ_{top}	weight of routines in $y(top)$	0.3366	routine share, top	.0985	.0199
λ_{bot}	weight of ICT in $\mathcal{Q}(bot)$	0.4565	ICT share, bottom	.3968	.3968
λ_{mid}	weight of ICT in $\mathcal{Q}(mid)$	0.4645	ICT share, middle	.3042	.3042
λ_{top}	weight of ICT in $\mathcal{Q}(top)$	0.4514	ICT share, top	.2990	.2990
θ	productivity dispersion	11.302	wage premium, $w\left(a, [s, s']\right)$	.9945	.9945

Estimated values and related matched moment using the Methods of Simulated Moments by Mc Fadden (1989).

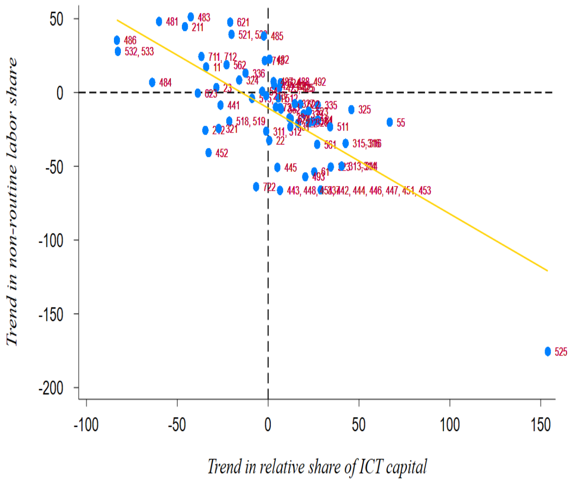


(A) ESTIMATES FOR ρ

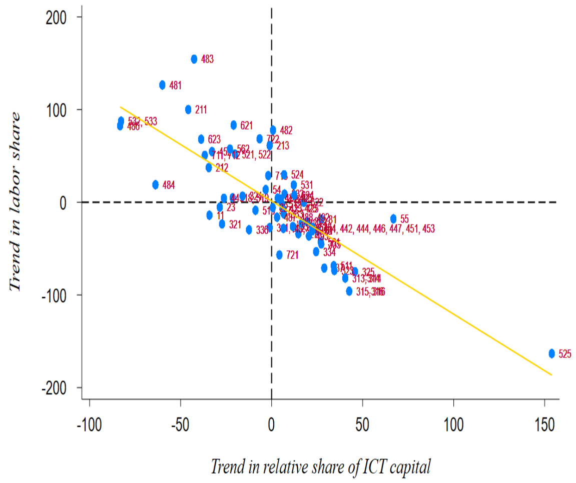
(B) ESTIMATES FOR σ

FIGURE C.1: CORRELATION BETWEEN ELASTICITIES AND RELATIVE ICT

Note: these scatterplots compute the correlation of trends in labour share with trends in the stock of ICT capital relative to physical capital, namely the left- and right-hand sides estimated through the elasticities of substitution as in eqs. (12) and (13), respectively. Each y-axis report the labour share, while common x-axis is referred to ICT capital. Panel C.1a refers the correlation of the elasticity of substitution between ICT capital and non routine workers, while Panel C.1b plots that of the entire labour share (considering both routine and non routine workers). Solid-gold negatively-sloped line is the estimated linear trend from robust regression. Labels to each point refer to the 3-digit U.S. 2017 NAICS code of the related industry. *Source*: BEA, BLS and own calculations.

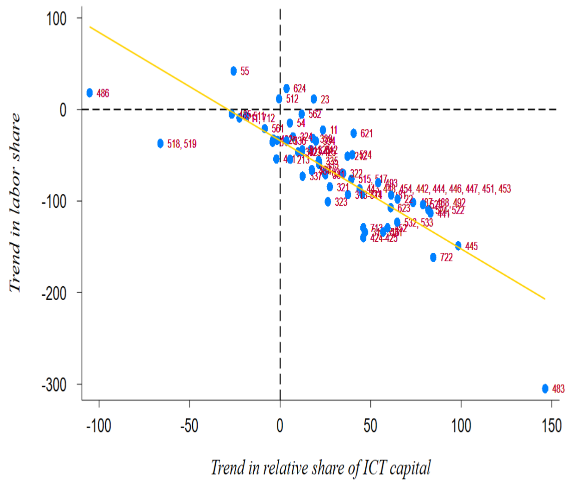


(A) ESTIMATES FOR ρ, 2003-2012



(B) ESTIMATES FOR σ, 2003-2012

(C) ESTIMATES FOR ρ, 2013-2022



(D) ESTIMATES FOR σ, 2013-2022

FIGURE C.2: CORRELATION BETWEEN IN-TIME ELASTICITIES AND RELATIVE ICT

Note: these scatterplots compute the correlation of trends in labour share with trends in the stock of ICT capital relative to physical capital, namely the left- and right-hand sides estimated through the elasticities of substitution as in eqs. (12) and (13), respectively; first row displays correlation in the period 2003-2012, while the second row that in the period 2013-2022. Each y-axis report the labour share, while common x-axis is referred to ICT capital. Panels C.2a and C.2c refer the correlation of the elasticity of substitution between ICT capital and non routine workers, while Panels C.2b and C.2d plot that of the entire labour share (considering both routine and non routine workers). Solid-gold negatively-sloped line is the estimated linear trend from robust regression. Labels to each point refer to the 3-digit U.S. 2017 NAICS code of the related industry. *Source*: BEA, BLS and own calculations.

TABLE C.3: MODEL FIT, UNTARGETED MOMENTS

moment	***fit***	
	data	*model*
aggregate task-premium	.001	.005
aggregate wage	-.008	-.073
routine wage, bottom	-.016	-.070
routine wage, middle	-.001	-.051
routine wage, top	-.007	-.079
non-routine wage, bottom	-.019	-.060
non-routine wage, middle	.003	-.074
non-routine wage, top	-.010	-.046

Untargeted moments to match to validate the calibration strategy. All moments, referred to real log-wages, are taken as percentage changes throughout the series.

(Market concentration) *To evaluate the pattern in concentration at industry level, the standard measure Herfindahl-Hirschman Index (HHI) is computed to account for market concentration of labour force for industries. To compute the dynamics in each year, labour market-level concentration for task-*a *is defined as*

$$HHI_{\ell(a)} = \sum_{s_{|g}} \left(\frac{\ell(a, s_{|g})}{\ell(a)} \right)^2 \tag{C.1}$$

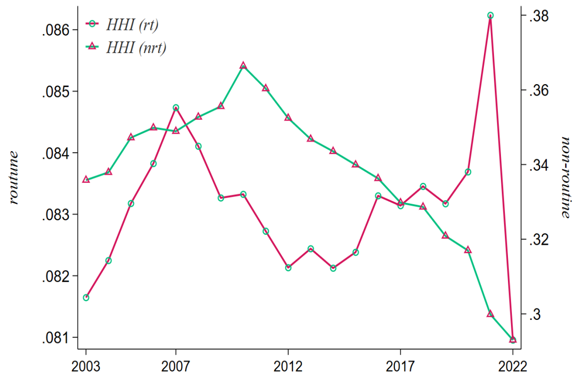
where the sum is over individual industries (s), or over a group of industries ($s_{|g}$*). Analogously, total employment concentration is defined by* $HHI_{\ell} = \sum_a HHI_{\ell(a)}$*. This measure is included in the range* $[0, 1]$*: a value of 1 identifies maximum market concentration, namely a single monopsonist in the labour market; conversely, a value of 0 results in a perfectly competitive environment. By definition, if industries have equal labour force size, the index would converge to the number of industries. An index below 0.15 points for an un-concentrated labour market, an index between 0.15 and 0.25 indicates a moderate concentration, while a value higher than 0.25 results in an highly concentrated labour market.*

Each subplot of Figure C.3 depicts the evolution in HHI score for both routine and non-routine workers. As a general observation, labour market concentration for both tasks has increased for top and bottom groups, but it has decreased for middle. With respect to bottom group, Panel C.3a indicates a joint increase in market concentration for both job tasks, with a very high concentration for non-routine workers $\left(HHI_{\ell(nrt)} > 0.25\right)$*; routine workers are approaching to a moderate concentration.*

In relation to middle group, after a steady increase, labour market concentration for non-routine workers constantly drops, even if concentration is still high since $HHI_{\ell(nrt)} > 0.25$*. The score for routine workers is low, but if follows cyclical fluctuations, with a steady increase before a recession, and thereafter a substantial drop.*

Finally, top group has seen a marked and steady increased in market concentration for non-routine workers, which is approaching to a moderate concentration; for the part of routine workers, concentration is high and, after a steady rise in the early years, then a flat pattern has occurred.

(A) BOTTOM



(B) MIDDLE

(C) TOP

FIGURE C.3: LABOUR MARKET CONCENTRATION BY GROUPS OF INDUSTRIES

Note: the figure represents the evolution in labour market concentration as measured by the Herfindahl-Hirschman Index (HHI). Each subplot measures the dynamics in each broadly defined group of industries by routine (red line, left axis) and non-routine (green line, right axis) tasks. *Source*: BLS and own calculations.

TABLE C.4: SUMMARY OF CALIBRATION, 2003-2012

	parameter	*value*				*source*
		bottom	*middle*	*top*	*global*	
α	*physical capital, share of $y(s)$*	0.131	0.091	0.254		*data*
ϵ	*demand elasticity across firms*				6	*external*
μ	*weight of routine workers in $y(s)$*	0.196	0.486	0.902		*MSM*
λ	*ICT capital share in $Q(s)$*	0.650	0.530	0.185		*MSM*
θ	*households' productivities dispersion*				7.26	*MSM*
ρ	*EoS, ICT capital and non-routine*	0.355	0.431	0.408		*estimation*
σ	*EoS, routine and ICT composite*	0.366	0.429	0.367		*estimation*

Set of estimated parameters of the model, first-half of the sample. "data" implies that the values are directly computed from data sources, while in "external" I choose standard calibrated values from the literature. "MSM" refers to the Methods of Simulated Moments as in Mc Fadden (1989). "estimation" refers to previously estimated values under a specific procedure; these values are taken from Table 8.

TABLE C.5: SUMMARY OF CALIBRATION, 2013-2022

	parameter	*value*				*source*
		bottom	*middle*	*top*	*global*	
α	*physical capital, share of $y(s)$*	0.131	0.104	0.260		*data*
ϵ	*demand elasticity across firms*				6	*external*
μ	*weight of routine workers in $y(s)$*	0.415	0.506	0.608		*MSM*
λ	*ICT capital share in $Q(s)$*	0.786	0.490	0.327		*MSM*
θ	*households' productivities dispersion*				7.79	*MSM*
ρ	*EoS, ICT capital and non-routine*	0.819	0.345	0.508		*estimation*
σ	*EoS, routine and ICT composite*	0.326	0.438	0.357		*estimation*

Set of estimated parameters of the model, second-half of the sample. "data" implies that the values are directly computed from data sources, while in "external" I choose standard calibrated values from the literature. "MSM" refers to the Methods of Simulated Moments as in Mc Fadden (1989). "estimation" refers to previously estimated values under a specific procedure; these values are taken from Table 8.

Table C.6: Method of Simulated Moments, sub-periods

	moment to match	2003-2012 value	data	model	2013-2022 value	data	model
μ_{bot}	routine share, bottom	0.196	0.087	0.097	0.415	0.085	0.051
μ_{mid}	routine share, middle	0.486	0.132	0.057	0.506	0.140	0.030
μ_{top}	routine share, top	0.902	0.100	0.013	0.608	0.097	0.083
λ_{bot}	ICT share, bottom	0.650	0.399	0.399	0.786	0.395	0.395
λ_{mid}	ICT share, middle	0.530	0.314	0.314	0.490	0.295	0.295
λ_{top}	ICT share, top	0.185	0.287	0.287	0.327	0.311	0.311
θ	wage premium, $w(a,[s,s'])$	7.259	0.994	0.994	7.787	0.995	0.995

Estimated values and related matched moment using the Methods of Simulated Moments by Mc Fadden (1989) for first and second half of the sample.

Table C.7: Variations in the series and structural heterogeneity

	data	model	model $\mid \Delta\Phi(x)$: $\Delta\ell(rt)$	$\Delta\ell(nrt)$	$\Delta(\ell)$	$\Delta k(ict)$	$\Delta(\ell, ict)$	$(all)_{\tau_0}$
Wages, variance								
routine	2.285	2.289	2.55	2.42	2.24	2.78	2.45	2.75
non-routine	2.314	2.311	2.78	2.44	2.37	2.78	2.39	2.83
industry	2.299	2.300	2.66	2.43	2.30	2.78	2.42	2.79

Changes in variances in real log-wages in the model induced by variations in one or more series, keeping fixed the others, and imposing the parameters to be that in the baseline calibration (i.e., quantification of Model D under industry-specific parameters). $\Delta(\ell)$ refers to joint variations in routine and non-routine series, while $\Delta(\ell, ict)$ is associated to simultaneous changes in both ICT capital and both types of workers. $(all)_{\tau_0}$ keeps fixed all the series (physical capital included), at their initial level, $\tau_0 = 2003$. In all the columns, the variation is computed throughout the period-by-period percentage differential thus identifying overall changes in empirical trends implied both by the data and the model; same values differ in terms of decimals.

(Further results of Subsection 4.2) *As in the main text, below I have performed an opposite exercise of Table 9 where the main structural parameters in the model, namely* $(\rho_s, \sigma_s, \theta)_{\forall s \in \{bot, mid, top\}}$ *are keeping fixed one or more at time, and let to change the non-ICT capital share and the weighting parameters in the production function, respectively* $(\alpha_s, \mu_s, \lambda_s)_{\forall s \in \{bot, mid, top\}}$*; the output of such exercise is shown in Table C.8.*

The two analysis performed are conducted by considering industries as a whole, that is, keeping the total employment in terms of both routine and non-routine workers, in order to address the determinants of between-industry wage inequality. However, as noticed, these two exercises can be performed also by distinguish the effects of changing parameters on separate job tasks: Table C.9 reports the two for routine workers, while Table C.10 the same but for non-routine workers.

Below it is possible to find a detailed comment for each employment group and, further below, a comparative comment that shows how the patterns holding for industry with aggregate employment hold also for routine and non-routine workers separately. Visually, these results are in Figures C.4a and C.5a for aggregate industries, and in Figures C.4b and C.5b by job task categories.

TABLE C.8: MODEL COUNTERFACTUAL, FIXING

		Δ ***model*** $\mid \Delta\, m\Big(\Phi(x,\tau_2), \Theta = \{p_{\tau_2}, -p_{\tau_1}\}\Big)$		
industry wages	$var(w)_{\tau_2}$	*level*	*share, model*	*share, data*
DATA	1.18			
MODEL	1.09			
$\Delta\Theta_{\vert\sigma}$		.94	.86	.79
$\Delta\Theta_{\vert\rho}$		2.21	2.04	1.87
$\Delta\Theta_{\vert(\sigma,\rho)}$		1.31	1.21	1.11
$\Delta\Theta_{\vert\theta}$		1.04	.96	.88
$\Delta\Theta_{\vert(\sigma,\rho,\theta)}$		1.30	1.20	1.10

Quantification of Model B. Model implied between-industry real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by fixing a specified parameter, $\frac{m\left[\Phi(x,\tau_2)\,\middle|\,\Theta(p,\tau_2),\Theta(-p^{\{\rho,\sigma,\theta\}},\tau_1)\right]}{m\left[\Phi(x,\tau_2)\,\middle|\,\Theta(p,\tau_2)\right]}$*, where* $\Phi(x,\tau_2)$ *identifies the series in the second period, and* Θ *the set of parameters where some of them,* (p,τ_2)*, are taken in the second period, while* $(-p,\tau_1)$ *reflects the set of all the parameters in the first period but those considered in the second period. Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

TABLE C.9: MODEL COUNTERFACTUALS FOR ROUTINE WORKERS

routine wages	$var(w)_{\tau_2}$	Δ ***model*** $\mid \Delta\, m\Big(\Phi(x,\tau_2), \Theta = \{p_{\tau_2}, -p_{\tau_1}\}\Big)$ *level*	*share, model*	*share, data*
		CHANGE		
DATA	1.14			
MODEL	.95			
$\Delta\sigma$		.99	1.04	.87
$\Delta\rho$		.35	.37	.30
$\Delta(\sigma,\rho)$		.66	.70	.58
$\Delta\theta$		.57	.60	.50
$\Delta(\sigma,\rho,\theta)$		.71	.76	.63
		FIXING		
DATA	1.14			
MODEL	.95			
$\Delta\Theta_{\mid\sigma}$		.71	.75	.62
$\Delta\Theta_{\mid\rho}$		1.15	1.22	1.01
$\Delta\Theta_{\mid(\sigma,\rho)}$		.84	.89	.74
$\Delta\Theta_{\mid\theta}$		.90	.95	.79
$\Delta\Theta_{\mid(\sigma,\rho,\theta)}$		.81	.86	.71

Quantification of Model A and Model B. Model implied industry-level between-routine workers real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by the change in a specified parameter, $\frac{m\left[\Phi(x,\tau_2) \mid \Theta(p,\tau_2),\Theta(-p,\tau_1)\right]}{m\left[\Phi(x,\tau_2) \mid \Theta(p,\tau_2)\right]}$, *where* $\Phi(x,\tau_2)$ *identifies the series in the second period, and* Θ *the set of parameters where some of them,* (p,τ_2), *are taken in the second period, while* $(-p,\tau_1)$ *reflects the set of all the parameters in the first period but those considered in the second period. Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

TABLE C.10: MODEL COUNTERFACTUALS FOR NON-ROUTINE WORKERS

non-routine wages	$var(w)_{\tau_2}$	Δ ***model*** $\mid \Delta\, m\Big(\Phi(x,\tau_2),\Theta=\{p_{\tau_2},-p_{\tau_1}\}\Big)$ *level*	*share, model*	*share, data*
		CHANGE		
DATA	1.21			
MODEL	1.22			
$\Delta\sigma$		3.74	3.06	3.09
$\Delta\rho$		1.41	1.15	1.16
$\Delta(\sigma,\rho)$		1.57	1.28	1.30
$\Delta\theta$		2.11	1.73	1.75
$\Delta(\sigma,\rho,\theta)$		1.61	1.31	1.33
		FIXING		
DATA	1.21			
MODEL	1.22			
$\Delta\Theta_{\mid\sigma}$		1.16	.95	.96
$\Delta\Theta_{\mid\rho}$		3.27	2.67	2.70
$\Delta\Theta_{\mid(\sigma,\rho)}$		1.78	1.45	1.47
$\Delta\Theta_{\mid\theta}$		1.18	.97	.98
$\Delta\Theta_{\mid(\sigma,\rho,\theta)}$		1.79	1.46	1.48

Quantification of Model A and Model B. Model implied industry-level between-non-routine workers real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by the change in a specified parameter, $\frac{m\left[\Phi(x,\tau_2)\,\middle|\,\Theta(p,\tau_2),\Theta(-p,\tau_1)\right]}{m\left[\Phi(x,\tau_2)\,\middle|\,\Theta(p,\tau_2)\right]}$*, where* $\Phi(x,\tau_2)$ *identifies the series in the second period, and* Θ *the set of parameters where some of them,* (p,τ_2)*, are taken in the second period, while* $(-p,\tau_1)$ *reflects the set of all the parameters in the first period but those considered in the second period. Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

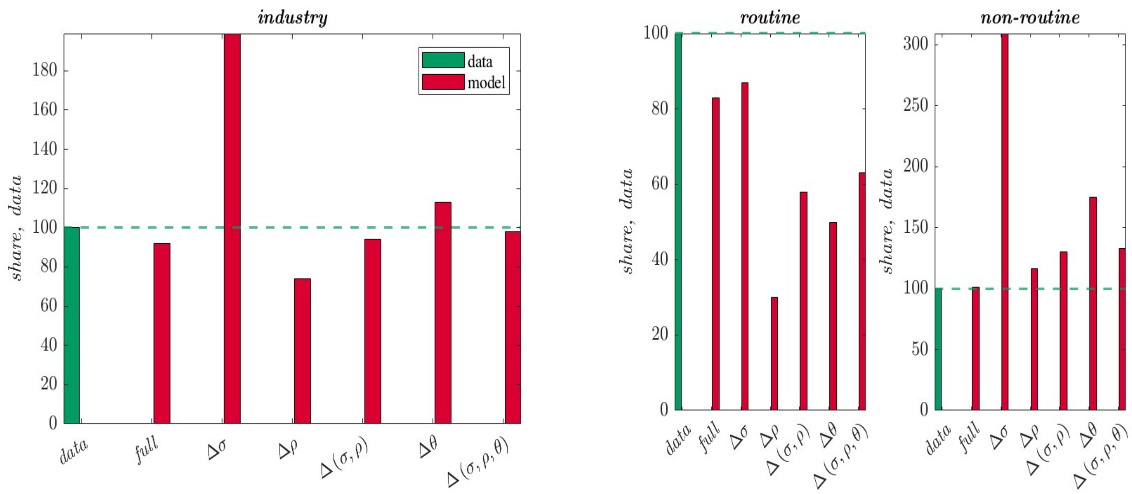
(Comparative comment) *Investigations on the contribution of structural parameters outline an important feature of the U.S. wage structure: observed structural differences among industries account for a sizeable fraction of wage inequality. All the effects can be summarized as follows:*

(a) ***routines****. Most of the share is accounted by trends in industry-heterogeneous differentials in elasticities of substitution among capital and worker types (58%, upper part of Table C.9), also taken in combination to different weights of factor inputs in production (79%, lower part of Table C.9). Considering these differences across industries, the rise in labour market concentration in terms of workers' complementarities (sorting and segregation effects) is an amplifier of U.S. wage inequality across routine job tasks in the last two decades, explaining 63% of the total between-industry routine real log-wage variance (upper part of Table C.9);*

(b) ***non-routines****. Most of the share is accounted by trends in industry-heterogeneous differentials in elasticities of substitution among capital and worker types (130%, upper part of Table C.10), also taken in combination to different weights of factor inputs in production (98%, lower part of Table C.10). Considering these differences across industries, the rise in labour market concentration in terms of workers' complementarities (sorting and segregation effects) is an amplifier of U.S. wage inequality across non-routine job tasks in the last two decades, explaining 133% of the total between-industry non-routine real log-wage variance (upper part of Table C.10);*

(c) ***industries****. Most of the share is accounted by trends in industry-heterogeneous differentials in elasticities of substitution among capital and worker types (94%, Table 9), also taken in combination to different weights of factor inputs in production (88%, Table C.8). Considering these differences across industries, the rise in labour market concentration in terms of workers' complementarities (sorting and segregation effects) is a amplifier of U.S. wage inequality across industries in the last two decades, explaining 98% of the total between-industry real log-wage variance (Table 9).*

Under a comparative perspective, routine workers wage inequality behaves similarly from that of non-routine and industries. All these variances are mostly explained by the substitution elasticity between routine and non-routine workers (σ) only and, jointly, the huge negative effect of σ on wage inequality is reduced due to changes in ρ. On the role of θ, again a clear division does not emerge: real wage dispersion increases after an increase in labour market concentration either for routine and non-routine workers and for industries. Henceforth, stronger workers' complementarities through stronger sorting and segregation effects result in negatively impacting all the considered categories, thus increasing real log-wage dispersion.

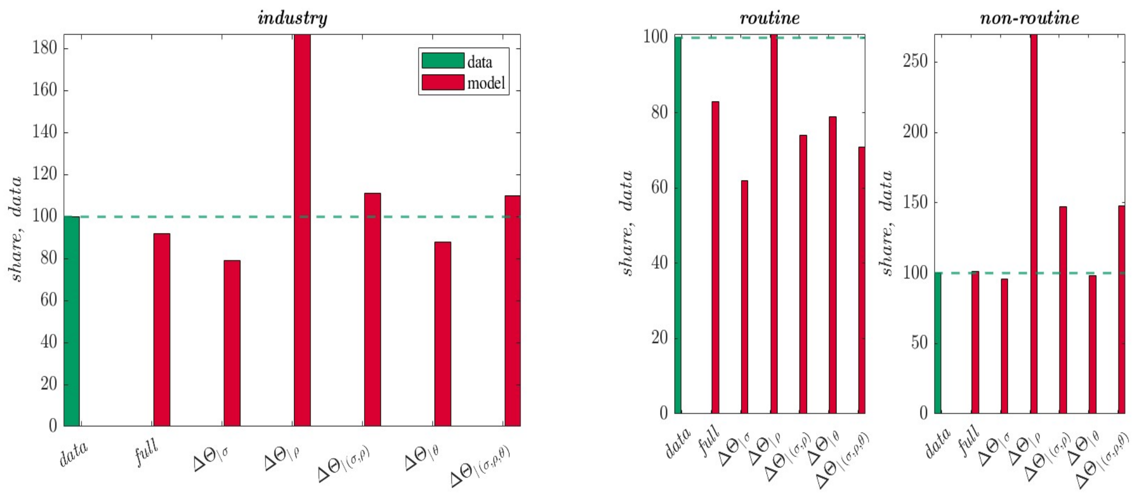


(A) INDUSTRY

(B) TASKS

FIGURE C.4: MODEL COUNTERFACTUAL, CHANGE

Note: these figures plot the second-period between-industry real log-wage variance, as a share of that in the data, implied by the model when changing one or more parameters at time as shown in Table 9 and in the upper part of Tables C.9 and C.10, for industry (total employment), routine and non-routine workers.



(A) INDUSTRY

(B) TASKS

FIGURE C.5: MODEL COUNTERFACTUAL, FIXING

Note: these figures plot the second-period between-industry real log-wage variance, as a share of that in the data, implied by the model when fixing one or more parameters at time as shown in Table C.8 and in the lower part of Tables C.9 and C.10, for industry (total employment), routine and non-routine workers.

(Comment for Skill-Biased Technological Change (SBTC) estimation) *As argued in the main text, Subsection 4.3, Skill-Biased Technological Change (SBTC) theory implies that wage differentials are shaped by different adoption rate of technology, so that dispersion in wages is mostly affected by how much an industry increases its technological capital and this, in turn, will increase its demand for high-skill or non-routine workers. Thus, below I have performed an exercise where the main structural parameters in the model, namely* $(\rho_s, \sigma_s, \theta)_{\forall s \in \{bot,mid,top\}}$*, and the production function weighting parameters, namely* $(\alpha_s, \mu_s, \lambda_s)_{\forall s \in \{bot,mid,top\}}$*, are keeping fixed at the first period, and let to change the series related to routine and non-routine workers, and that of ICT capital, as described in the main text by Model C; the output of such exercise is shown in Table C.11.*

The analysis performed is conducted by considering industries as a whole, that is, keeping the total employment in terms of both routine and non-routine workers, in order to address the determinants of between-industry wage inequality. As noticed, these two exercises can be performed also by distinguish the effects of changing parameters on separate job tasks; to this end, Table C.12 reports the SBTC analysis for both routine and non-routine workers. The impact of SBTC seems to have major effect on non-routine workers: in fact, while the variance explained by shifts in the series explain substantially less data-share compared to the case in which shifts in structural parameters are also considered, the wage variance of non-routine workers explodes. Therefore, as in the model, major impact of only the series is on non-routine workers.

TABLE C.11: MODEL COUNTERFACTUAL, SBTC

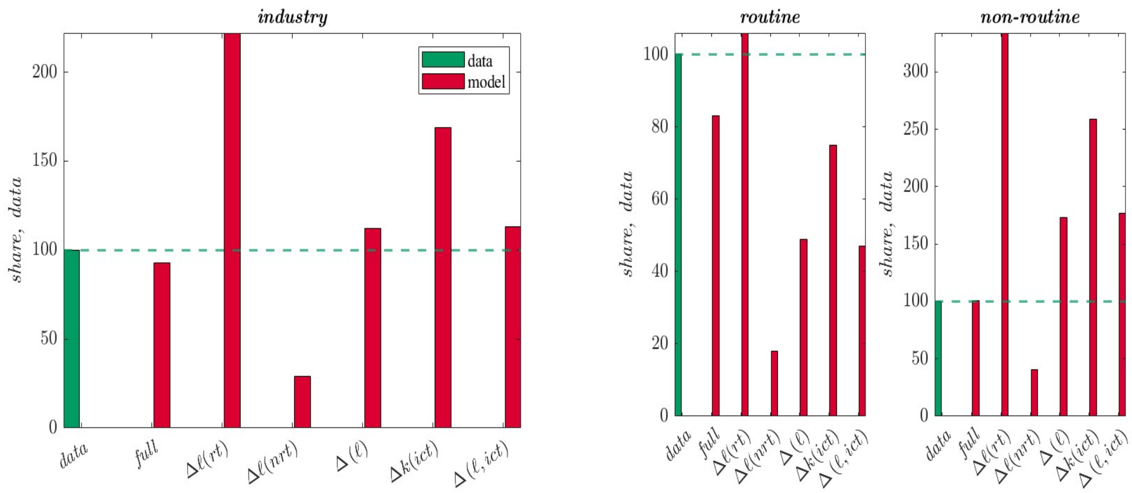
		Δ ***model*** $\mid \Delta\, m\Big(\Phi = \{x_{\tau_2}, -x_{\tau_1}\} \,\Big|\, \Theta(p, \tau_1)\Big)$		
industry wages	$var(w)_{\tau_2}$	*level*	*share, model*	*share, data*
DATA	1.18			
MODEL	1.09			
$\Delta\ell(rt)$		2.62	2.42	2.22
$\Delta\ell(nrt)$		.35	.32	.29
$\Delta\ell$		1.32	1.22	1.12
$\Delta k(ict)$		1.99	1.84	1.69
$\Delta(\ell, ict)$		1.34	1.23	1.13

Quantification of Model C. Model implied between-industry real log-wage variance changes between two time spans uniformly calibrated, with changes according to variations in some series; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by the change in a specified parameter, $\frac{m\left[\Phi(x,\tau_2),\Phi(-x,\tau_1)\,\middle|\,\Theta(p,\tau_1)\right]}{m\left[\Phi(x,\tau_2)\,\middle|\,\Theta(p,\tau_2)\right]}$*, where* $\Theta(x, \tau_1)$ *identifies the set of parameters in the first period, and* Φ *the set of capital and labour series where some of them,* (x, τ_2)*, are taken in the second period, while* $(-x, \tau_1)$ *reflects the set of all the series in the first period but those considered in the second period. Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

TABLE C.12: MODEL COUNTERFACTUAL, SBTC BY TASKS

		Δ ***model*** $\mid \Delta\, m\Big(\Phi = \{x_{\tau_2}, -x_{\tau_1}\} \,\Big|\, \Theta(p,\tau_1)\Big)$		
wages	$var(w)_{\tau_2}$	*level*	*share, model*	*share, data*
		ROUTINE		
DATA	1.14			
MODEL	.95			
$\Delta\ell(rt)$		1.21	1.28	1.06
$\Delta\ell(nrt)$		.20	.21	.18
$\Delta\ell$		.56	.59	.49
$\Delta k(ict)$		.86	.91	.75
$\Delta(\ell, ict)$		.53	.56	.47
		NON-ROUTINE		
DATA	1.21			
MODEL	1.22			
$\Delta\ell(rt)$		4.04	3.30	3.34
$\Delta\ell(nrt)$		.50	.40	.41
$\Delta\ell$		2.09	1.70	1.73
$\Delta k(ict)$		3.13	2.56	2.59
$\Delta(\ell, ict)$		2.14	1.75	1.77

Quantification of Model C for routine and non-routine workers separately. Model implied between-industry real log-wage variance changes between two time spans uniformly calibrated, with changes according to variations in some series; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by the change in a specified parameter, $\frac{m\left[\Phi(x,\tau_2),\Phi(-x,\tau_1)\,\middle|\,\Theta(p,\tau_1)\right]}{m\left[\Phi(x,\tau_2)\,\middle|\,\Theta(p,\tau_2)\right]}$*, where* $\Theta(x,\tau_1)$ *identifies the set of parameters in the first period, and* Φ *the set of capital and labour series where some of them,* (x,τ_2)*, are taken in the second period, while* $(-x,\tau_1)$ *reflects the set of all the series in the first period but those considered in the second period. Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*



(A) INDUSTRY (B) TASKS

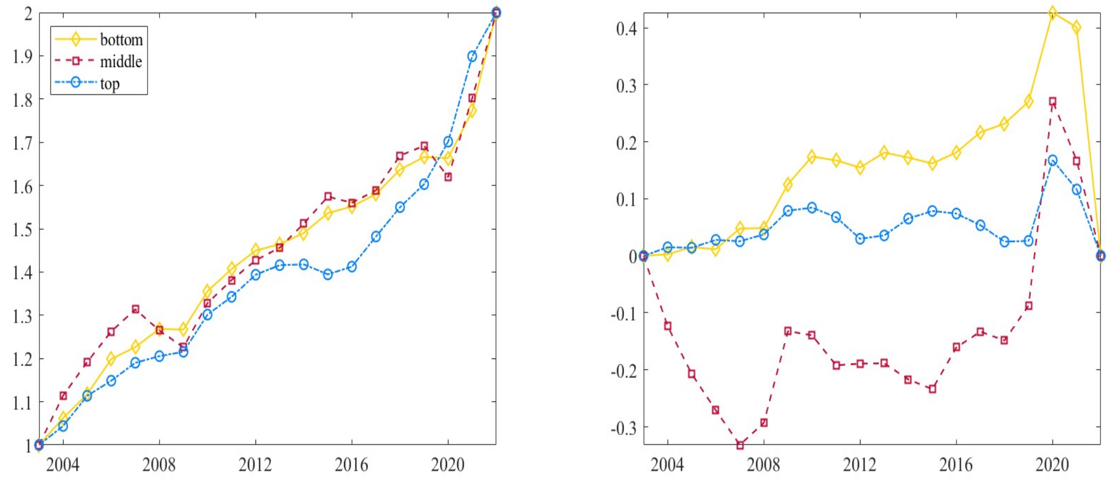
FIGURE C.6: MODEL COUNTERFACTUAL, SBTC

Note: these figures plot the second-period between-industry real log-wage variance, as a share of that in the data, implied by the model when changing one or more parameters at time while keeping fixed the all the parameters, as shown in Tables C.11 and C.12, for industry (total employment), routine and non-routine workers.

TABLE C.13: MODEL VS. DATA COUNTERFACTUAL, SERIES AND NEW PARAMETERS

		model $\mid \Delta\,\Phi(x)$						
							$\Delta\,(tfp)$	
	data	$\Delta\ell(rt)$	$\Delta\ell(nrt)$	$\Delta\ell$	$\Delta k(ict)$	$\Delta\,(\ell, ict)$	*newly*	*baseline*
WAGES, VARIANCE								
routine	2.285	.62	.41	.46	.54	.40	.57	.34
non-routine	2.314	.07	.02	.02	.04	.03	.02	.09
industry	2.299	.35	.22	.24	.29	.21	.30	.22

Quantification of Model D. Changes in variances in real log-wages in the model induced by variations in one or more series, keeping fixed the others, and imposing the newly estimated parameters to be homogeneous across industries. $\Delta\,(\ell)$ refers to joint variations in routine and non-routine series, $\Delta\,(tech)$ is associated to simultaneous changes in both ICT capital and non-routine workers, while changes in estimated industry-specific Hicks-neutral exogenous total factor productivity (TFP), given eq. (14), are captured by $\Delta\,(tfp)$; TFP series are taken under the same (newly estimated) parameters' values, or given the baseline calibration (Table 3.2). In all the columns, the variation is computed throughout the period-by-period percentage differential thus identifying overall changes in empirical trends implied both by the data and the model; same values differ in terms of decimals.

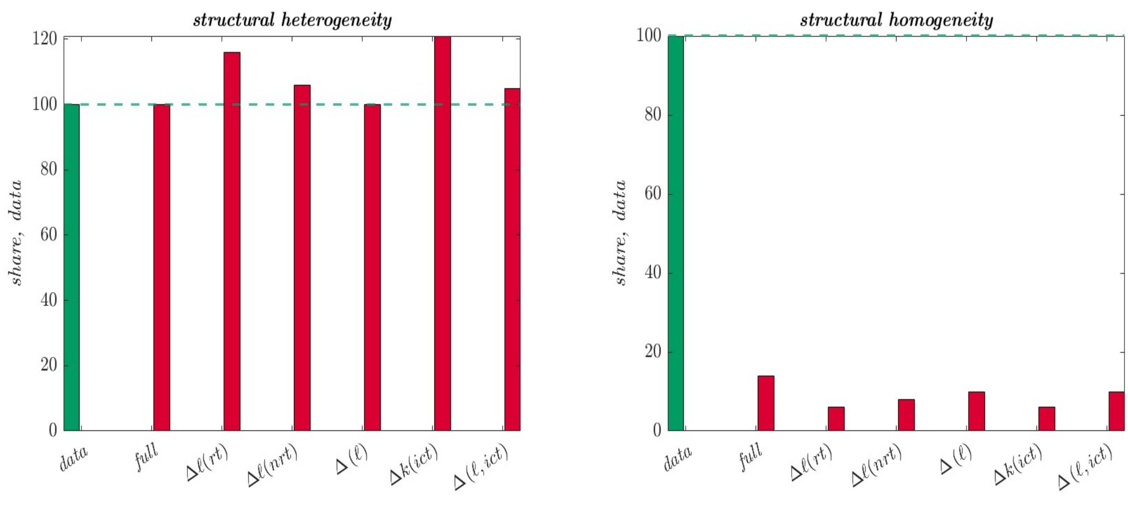


(A) UNIQUE CALIBRATION, NEWLY

(B) DIFFERENCE WITH BASELINE

FIGURE C.7: ESTIMATED PRODUCTIVITIES UNDER NEW PARAMETERS

Note: the figure shows the estimated Hicks-neutral exogenous total factor productivity (TFP) measures estimated from the model. Panel C.7a plots the series given a calibration where all the parameters are evenly set at the same *newly estimated* values for all the industry-groups, while Panel C.7b shows the difference between such series and the estimated TFP measures using the baseline calibration reported in Table 6. Series are scaled to be in the same range for graphical comparison.



(A) STRUCTURAL HETEROGENEITY

(B) HOMOGENEOUS PARAMETER

FIGURE C.8: MODEL COUNTERFACTUAL, SERIES

Note: these figures plot the between-industry real log-wage variance, as a share of the data, implied by the model when one or more series at time are changing, keeping fixed the parameters. Panel C.8a reports the result of Table C.7, where structural parameters are heterogeneous across industries. By contrast, Panel C.8b reports the outcome from Table 10, under industry-homogeneous structural parameters; note that a similar graph would appear if considering Table C.13.

D. Case under monopsony power

(Discussion) *In this appendix I am going to replicate all the analysis in the main text (calibration, estimation and counterfactuals from Model A and Model B) under the case in which firms are assumed to be monopolistically competitive and take the labour supply of each task-a as given. In this case, frictions due to a concentration in the labour market suggest that wages' formation is decided at firm level so that, taking labour supplies and industry variables as given, the profit-maximization problem for firm-h in industry-s becomes*

$$\max_{p_h(s),k_h(j,s),w_h(a,s)} \left[\mathcal{D}_h(s) \mid y_h(s), \ell_h(a,s)\right]$$

where firms have wage-setting power over employees. In other words, the profit maximization implies that the Cobb Douglas-nested CES production function in eq. (9) is subject to the labour supply curves specified by eq. (7), and to the conditional firm demand in eq. (8), thus exploiting both monopolistic and monopsonistic power, respectively, in combination. Whether one decides to assume or not a monopsony power by the part of firms (i.e., maximizing taking $\ell_h(a,s)$ as a constraint) does not change substantially the results and the conclusion of the main text. In fact, the only difference between optimal wages for routine and non-routine workers for a monopolistically competitive firm only, as in eq. (10) and in Appendix B, and that of a monopsonistic-monopolistically competitive firm is just the wage-markdown element $\mathcal{M}^\theta = \frac{\theta}{1+\theta}$ in both the composite parameters $\chi(a,s)$, a feature of firms when considering also a monopsonistic environment.

Since θ, measuring the degree of sorting and segregation effects in the economy, is the same for all industries, changes in variances due to changes in θ have an even effect in each industries, thus not changing the conclusion of the main text. This is because, as already noticed, the specification of the households' sorting choices comprises in itself the definition of monopsony power since the measure of workers of type-a in firm-h, industry-s is mostly determined by relative wages as in eq. (7).

TABLE D.1: SUMMARY OF CALIBRATION UNDER MONOPSONY

	parameter	***value*** bottom	*middle*	*top*	*global*	***source***
α	*physical capital, share of* $y(s)$	0.263	0.195	0.514		*data*
ϵ	*demand elasticity across firms*				6	*external*
μ	*weight of routine workers in* $y(s)$	0.676	0.490	0.333		*MSM*
λ	*ICT capital share in* $Q(s)$	0.455	0.456	0.439		*MSM*
θ	*households' productivities dispersion*				11.4	*MSM*
ρ	*EoS, ICT capital and non-routine*	0.329	0.420	0.249		*estimation*
σ	*EoS, routine and ICT composite*	0.634	0.400	0.766		*estimation*

Set of estimated parameters of the model. "data" implies that the values are directly computed from data sources, while in "external" I choose standard calibrated values from the literature. "MSM" refers to the Methods of Simulated Moments as in Mc Fadden (1989). "estimation" refers to previously estimated values under a specific procedure; these values are taken from Table 5.

TABLE D.2: METHOD OF SIMULATED MOMENTS UNDER MONOPSONY

	parameter	*value*	***moment to match***	***fit*** *data*	*model*
μ_{bot}	weight of routines in $y(bot)$	0.6760	routine share, bottom	.0858	.1420
μ_{mid}	weight of routines in $y(mid)$	0.4902	routine share, middle	.1357	.1596
μ_{top}	weight of routines in $y(top)$	0.3331	routine share, top	.0985	.0637
λ_{bot}	weight of ICT in $Q(bot)$	0.4552	ICT share, bottom	.3968	.3968
λ_{mid}	weight of ICT in $Q(mid)$	0.4585	ICT share, middle	.3042	.3042
λ_{top}	weight of ICT in $Q(top)$	0.4398	ICT share, top	.2990	.2990
θ	productivity dispersion	11.376	wage premium, $w\left(a, [s, s']\right)$	.9945	.9945

Estimated values and related matched moment using the Methods of Simulated Moments by Mc Fadden (1989).

TABLE D.3: MODEL FIT UNDER MONOPSONY, UNTARGETED MOMENTS

	fit	
moment	*data*	*model*
aggregate task-premium	.001	.005
aggregate wage	-.008	-.073
routine wage, bottom	-.016	-.070
routine wage, middle	-.001	-.051
routine wage, top	-.007	-.080
non-routine wage, bottom	-.019	-.060
non-routine wage, middle	.003	-.075
non-routine wage, top	-.010	-.046

Untargeted moments to match to validate the calibration strategy. All moments, referred to real log-wages, are taken as percentage changes throughout the series.

TABLE D.4: METHOD OF SIMULATED MOMENTS UNDER MONOPSONY, SUB-PERIODS

		2003-2012			**2013-2022**		
	moment to match	*value*	*data*	*model*	*value*	*data*	*model*
μ_{bot}	routine share, bottom	0.225	0.087	0.103	0.415	0.085	0.051
μ_{mid}	routine share, middle	0.492	0.132	0.061	0.506	0.140	0.030
μ_{top}	routine share, top	0.906	0.100	0.013	0.608	0.097	0.083
λ_{bot}	ICT share, bottom	0.647	0.399	0.399	0.786	0.395	0.395
λ_{mid}	ICT share, middle	0.518	0.314	0.314	0.490	0.295	0.295
λ_{top}	ICT share, top	0.170	0.287	0.287	0.326	0.311	0.311
θ	wage premium, $w(a,[s,s'])$	7.255	0.994	0.994	7.787	0.995	0.995

Estimated values and related matched moment using the Methods of Simulated Moments by Mc Fadden (1989).

Table D.5: Summary of calibration under monopsony, 2003-2012

	parameter	***value***				***source***
		bottom	*middle*	*top*	*global*	
α	*physical capital, share of* $y(s)$	0.131	0.091	0.254		*data*
ϵ	*demand elasticity across firms*				6	*external*
μ	*weight of routine workers in* $y(s)$	0.225	0.492	0.906		*MSM*
λ	*ICT capital share in* $Q(s)$	0.647	0.518	0.170		*MSM*
θ	*households' productivities dispersion*				7.26	*MSM*
ρ	*EoS, ICT capital and non-routine*	0.355	0.431	0.408		*estimation*
σ	*EoS, routine and ICT composite*	0.366	0.429	0.367		*estimation*

Set of estimated parameters of the model, first-half of the sample. "data" implies that the values are directly computed from data sources, while in "external" I choose standard calibrated values from the literature. "MSM" refers to the Methods of Simulated Moments as in Mc Fadden (1989). "estimation" refers to previously estimated values under a specific procedure; these values are taken from Table 8.

Table D.6: Summary of calibration under monopsony, 2013-2022

	parameter	***value***				***source***
		bottom	*middle*	*top*	*global*	
α	*physical capital, share of* $y(s)$	0.131	0.104	0.260		*data*
ϵ	*demand elasticity across firms*				6	*external*
μ	*weight of routine workers in* $y(s)$	0.415	0.506	0.608		*MSM*
λ	*ICT capital share in* $Q(s)$	0.786	0.490	0.326		*MSM*
θ	*households' productivities dispersion*				7.79	*MSM*
ρ	*EoS, ICT capital and non-routine*	0.819	0.345	0.508		*estimation*
σ	*EoS, routine and ICT composite*	0.326	0.438	0.357		*estimation*

Set of estimated parameters of the model, second-half of the sample. "data" implies that the values are directly computed from data sources, while in "external" I choose standard calibrated values from the literature. "MSM" refers to the Methods of Simulated Moments as in Mc Fadden (1989). "estimation" refers to previously estimated values under a specific procedure; these values are taken from Table 8.

Table D.7: Model counterfactuals, monopsony

industry wages	$var(w)_{\tau_2}$	Δ ***model*** $\mid \Delta\, m\Big(\Phi(x,\tau_2), \Theta = \{p_{\tau_2}, -p_{\tau_1}\}\Big)$ *level*	*share, model*	*share, data*
		CHANGE		
Data	1.18			
Model	1.08			
$\Delta\sigma$		2.36	2.19	1.99
$\Delta\rho$		.88	.82	.75
$\Delta(\sigma,\rho)$		1.12	1.03	.94
$\Delta\theta$		1.34	1.24	1.13
$\Delta(\sigma,\rho,\theta)$		1.16	1.07	.98
		FIXING		
Data	1.18			
Model	1.08			
$\Delta\Theta_{\mid\sigma}$		.94	.87	.79
$\Delta\Theta_{\mid\rho}$		2.21	2.04	1.87
$\Delta\Theta_{\mid(\sigma,\rho)}$		1.31	1.21	1.11
$\Delta\Theta_{\mid\theta}$		1.04	.96	.88
$\Delta\Theta_{\mid(\sigma,\rho,\theta)}$		1.30	1.20	1.10

Quantification of Model A and Model B given both monopsonistic and monopolistic power by the part of firms. Model implied between-industry real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by the change in a specified parameter, $\frac{m\left[\Phi(x,\tau_2)\,\middle|\,\Theta(p,\tau_2),\Theta(-p,\tau_1)\right]}{m\left[\Phi(x,\tau_2)\,\middle|\,\Theta(p,\tau_2)\right]}$, *where* $\Phi(x,\tau_2)$ *identifies the series in the second period, and* Θ *the set of parameters where some of them,* (p,τ_2), *are taken in the second period, while* $(-p,\tau_1)$ *reflects the set of all the parameters in the first period but those considered in the second period. Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

TABLE D.8: MODEL COUNTERFACTUALS FOR ROUTINE WORKERS, MONOPSONY

routine wages	$var(w)_{\tau_2}$	Δ ***model*** $\mid \Delta\, m\Big(\Phi(x,\tau_2), \Theta = \{p_{\tau_2}, -p_{\tau_1}\}\Big)$ *level*	*share, model*	*share, data*
		CHANGE		
DATA	1.14			
MODEL	.94			
$\Delta\sigma$		.97	1.03	.85
$\Delta\rho$		.33	.35	.29
$\Delta(\sigma,\rho)$		.64	.68	.57
$\Delta\theta$		.55	.59	.49
$\Delta(\sigma,\rho,\theta)$		.70	.74	.61
		FIXING		
DATA	1.14			
MODEL	.94			
$\Delta\Theta_{\mid\sigma}$		.71	.75	.62
$\Delta\Theta_{\mid\rho}$		1.15	1.22	1.01
$\Delta\Theta_{\mid(\sigma,\rho)}$		.84	.89	.73
$\Delta\Theta_{\mid\theta}$		.89	.95	.78
$\Delta\Theta_{\mid(\sigma,\rho,\theta)}$		.81	.86	.71

Quantification of Model A and Model B given both monopsonistic and monopolistic power by the part of firms. Model implied industry-level between-routine workers real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by the change in a specified parameter, $\frac{m\left[\Phi(x,\tau_2)\,\middle|\,\Theta(p,\tau_2),\Theta(-p,\tau_1)\right]}{m\left[\Phi(x,\tau_2)\,\middle|\,\Theta(p,\tau_2)\right]}$*, where $\Phi(x,\tau_2)$ identifies the series in the second period, and Θ the set of parameters where some of them, (p,τ_2), are taken in the second period, while $(-p,\tau_1)$ reflects the set of all the parameters in the first period but those considered in the second period. Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

Table D.9: Model counterfactuals for non-routine workers, monopsony

non-routine wages	$var(w)_{\tau_2}$	Δ ***model*** $\mid \Delta\, m\Big(\Phi(x,\tau_2), \Theta = \{p_{\tau_2}, -p_{\tau_1}\}\Big)$ *level*	*share, model*	*share, data*
		CHANGE		
DATA	1.21			
MODEL	1.22			
$\Delta\sigma$		3.76	3.08	3.10
$\Delta\rho$		1.44	1.18	1.19
$\Delta(\sigma,\rho)$		1.59	1.30	1.31
$\Delta\theta$		2.12	1.74	1.75
$\Delta(\sigma,\rho,\theta)$		1.63	1.33	1.34
		FIXING		
DATA	1.21			
MODEL	1.22			
$\Delta\Theta_{\mid\sigma}$		1.17	.96	.97
$\Delta\Theta_{\mid\rho}$		3.27	2.68	2.70
$\Delta\Theta_{\mid(\sigma,\rho)}$		1.78	1.46	1.47
$\Delta\Theta_{\mid\theta}$		1.18	.97	.97
$\Delta\Theta_{\mid(\sigma,\rho,\theta)}$		1.79	1.47	1.48

Quantification of Model A and Model B given both monopsonistic and monopolistic power by the part of firms. Model implied industry-level between-non-routine workers real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to variance levels considering bottom, middle, and top industries. Column 2 shows the level in the second period of the between-industry variance both in the data and in the period-two fully calibrated model. For the model specified in column 1: column 3 represents the variance level in the second period as implied by the change in the parameter(s), while column 4 computes the second period variance share of the period-two full model which is accounted by the change in a specified parameter, $\frac{m\left[\Phi(x,\tau_2)\,\middle|\,\Theta(p,\tau_2),\Theta(-p,\tau_1)\right]}{m\left[\Phi(x,\tau_2)\,\middle|\,\Theta(p,\tau_2)\right]}$, *where* $\Phi(x,\tau_2)$ *identifies the series in the second period, and* Θ *the set of parameters where some of them,* (p,τ_2)*, are taken in the second period, while* $(-p,\tau_1)$ *reflects the set of all the parameters in the first period but those considered in the second period. Column 5 reports the fraction of variance explained by changes in parameter(s) in the observed data-driven variance.*

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